# Representing Signals

1.1: Classification of signals

# Introduction

- **The** concept and theory of signals and systems are needed in almost all electrical engineering fields and in many other engineering and scientific disciplines as well. In this chapter we introduce the mathematical description and representation of signals and systems and their classifications.
- **Signals** Convey information or massage regarding to physical quantity or variable. Example: Voice, light intensity, Video…etc.
- **Signals** are represented mathematically as functions of one or more independent variables. For example, a speech signal can be represented mathematically by acoustic pressure as a function of time, and a picture can be represented by brightness as a function of two spatial variables. Now, we focus our attention on signals involving a single independent variable.
- Electrical Signals are easy to measure, transfer, and provide a suitable means to represent physical variable.

# Introduction

- In this course we are going to talk about Electrical Signals. Transducer are used to convert physical quantity into an electrical signal. Example: Microphone, camera, Sensors …etc.
- **A signal** is a function representing a physical quantity or variable, and typically it contains information about the behavior or nature of the phenomenon. For instance, in a RC circuit the signal may represent the voltage across the capacitor or the current flowing in the resistor. Mathematically, a signal is represented as a function of an independent variable t. Usually t represents time. Thus, a signal is denoted by



# Objectives:

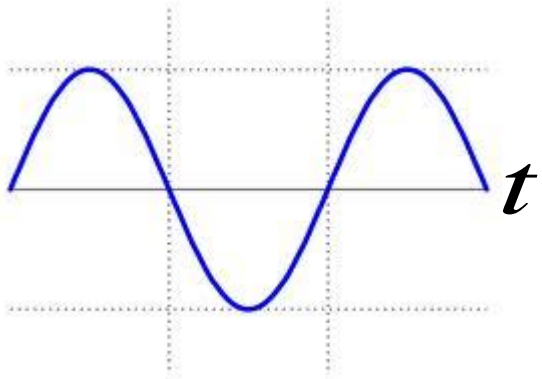
After completing this session, you will be able to:

1. To know the classification of the signals;
   - Continuous/ Discrete-time Signals.
   - Even/ Odd Signals.
   - Periodic/ Aperiodic Signals.
   - Energy/ Power Signals.
   - Deterministic/Random Signals
2. To know how to classify the signals graphically.
3. To know how to classify the signals mathematically.
4. To be aware why we classify the signals.

# 1– Continuous & Discrete time Signals

- **Continuous–time Signal** A signal $x(t)$ is a continuous–time signal if $t$ is a continuous variable. If t is a discrete variable, that is, $x(t)$ is defined at discrete times, then $x(t)$ is a discrete–time signal. Since a discrete–time signal is defined at discrete times, a discrete–time signal is often identified as a sequence of numbers, denoted by $\{x_n\}$ or $x[n]$, where $n$ = integer. Illustrations of a continuous–time signal $x(t)$ and of a discrete–time signal $x[n]$ are shown in Fig. 1–1.

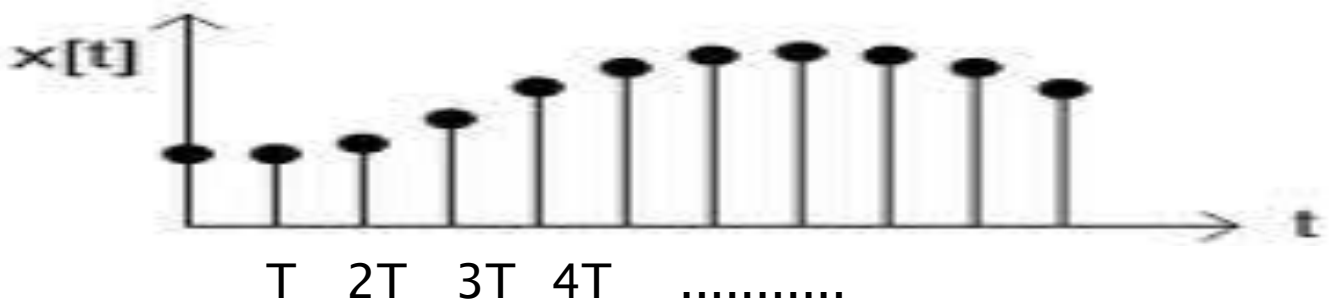
e.g. $x(t)$=sin(t)




which usually called "*Analog Signal* ".

# Continue..

- **Discrete- time Signals** will take values at selected time values, this time defined by $(T)$.

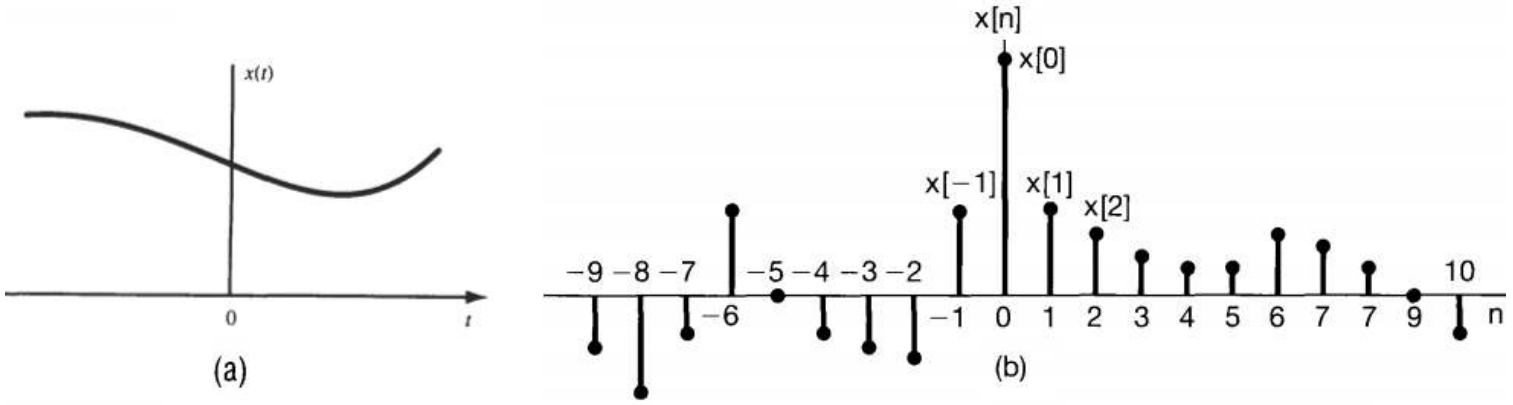



- we can represent this mathematically by:

$x(kT) = \sin(kT)$

Where $k = 0,1,2 \ldots etc$

# Fig. 1-1 Graphical representation of (a) continuous-time and (b) discrete-time signals



# 1– Continuous & Discrete time Signals

- The time (T) is called the sampling time; because the origin of any signal is analog then we take samples from it every T second to become a discrete signal.
- A digital signal is a discrete-time signal for which not only the time but also the amplitude has been made discrete.
- The process of converting a continuous-valued to discrete-time signal then to a digital (discrete-valued discrete-time) signal is known as *analog-to-digital conversion*.

# Continue..

- A discrete-time signal $x[n]$ may represent a phenomenon for which the independent variable is inherently discrete.
- For instance, the daily closing stock market average is by its nature a signal that evolves at discrete points in time (that is, at the close of each day).
- On the other hand a discrete-time signal $x[n]$ may be obtained by sampling a continuous-time signal

# Continue..

$x(t)$ such as $x(t_0), x(t_1), \ldots, x(t_n), \ldots$

$x[0], x[1], \ldots, x[n], \ldots$

or in a shorter form as $x_0, x_1, \ldots, x_n, \ldots$

so, we can understand that: $x_n = x[n] = x(t_n)$

and $x_n$ 's are called samples and the time interval between them is called the *sampling interval*. When the sampling intervals are equal (uniform sampling), then $x_n=x[n]=x(nT_s)$ where the constant $T$, is the sampling interval.

A discrete-time signal $\boldsymbol{x[n]}$ can be defined in two ways:

# Continue..

- ✓ We can specify a rule for calculating the *nth* value of the sequence. For example,

$$x[n] = x_n = \begin{cases} \left(\frac{1}{2}\right)^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Or,

$$\{x_n\} = \left\{1, \tfrac{1}{2}, \tfrac{1}{4}, \ldots, \left(\tfrac{1}{2}\right)^n, \ldots\right\}$$

- ✓ We can also explicitly list the values of the sequence. For example, the sequence shown in Fig. l-l(b) can be written as

$$\{x_n\} = \{\ldots, 0, 0, 1, 2, 2, 1, 0, 1, 0, 2, 0, 0, \ldots\}$$
$$\uparrow$$

Or,

$$\{x_n\} = \{1, 2, 2, 1, 0, 1, 0, 2\}$$
$$\uparrow$$

# Continue…

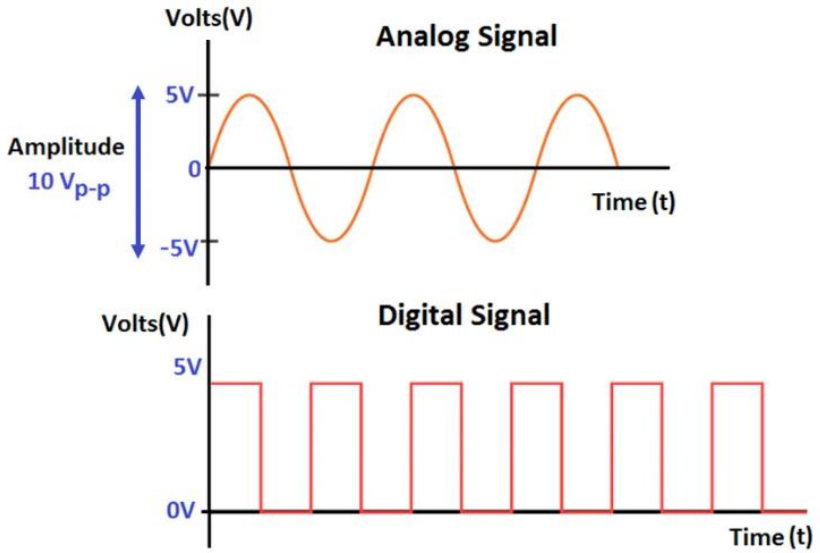
Note: We use the arrow to denote the $n=0$ term. We shall use the convention that if no arrow is indicated, then the first term corresponds to $n=0$ and all the values of the sequence are zero for $n < 0$.

The sum and product of two sequences as follows

$$\{c_n\} = \{a_n\} + \{b_n\} \rightarrow c_n = a_n + b_n$$
$$\{c_n\} = \{a_n\}\{b_n\} \rightarrow c_n = a_n b_n$$
$$\{c_n\} = \alpha\{a_n\} \rightarrow c_n = \alpha a_n \qquad \alpha = \text{constant}$$

# 2- Analog and Digital Signals:

- If a continuous-time signal $x(t)$ can take on any value in the continuous interval $(a, b)$, where a maybe $-\infty$ and b may be $+\infty$, then the continuous-time signal $x(t)$ is called an analog signal. If a discrete-time signal $x[n]$ can take on only a finite number of distinct values, then we call this signal a digital signal.



# 3– Real and Complex Signals:

- A signal $x(t)$ is a real signal if its value is a real number, and a signal $x(t)$ is a complex signal if its value is a complex number. A general complex signal $x(t)$ is a function of the form
- $x(t) = x^*(-t)$

$$x(t) = x_1(t) + jx_2(t), \text{where}, j = \sqrt{-1} = \text{or } 1\angle 90^o$$

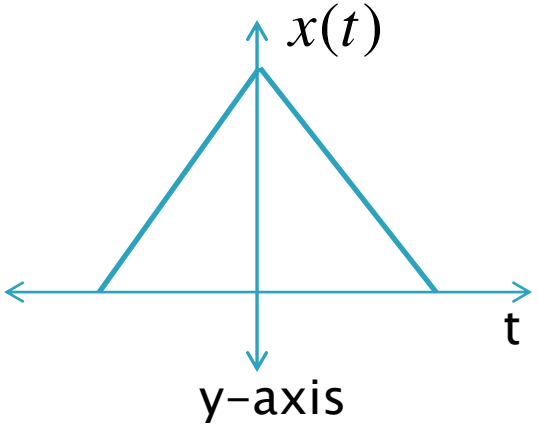
- $x^*(t) = x_1(t) - jx_2(t)$

# 4-Deterministic and Random Signals:

- Deterministic signals are those signals whose values are completely specified for any given time. Thus, a deterministic signal can be modeled by a known function of time $t$ . Random signals are those signals that take random values at any given time and must be characterized statistically.

# 2) Even & Odd Signals

- **<u>Even Signals</u>**: if the signal $x(t)$ has symmetry about the **y-axis**, then the signal is even signal.




- Mathematically:

$$x(t) = x(-t)$$

- e.g.:

$$x(t) = \cos(\omega t)$$

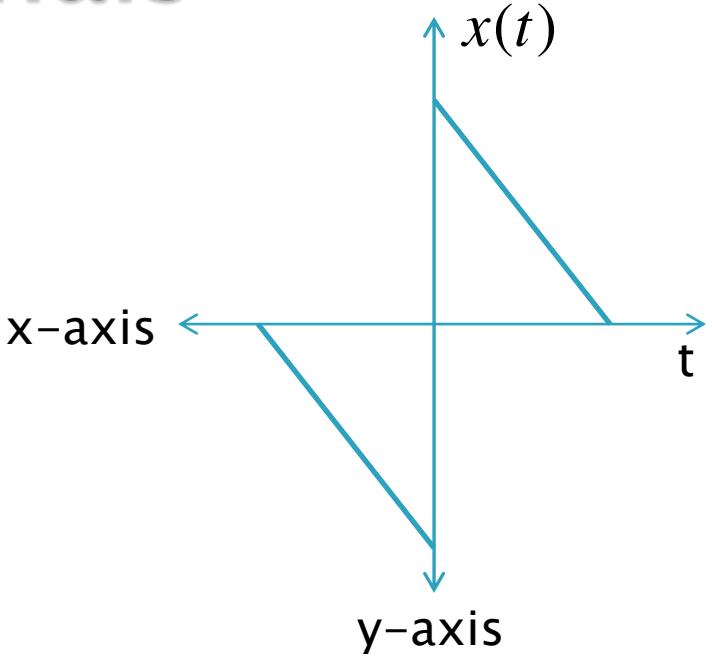
# 2) Even & Odd Signals

- That means if we reflect the signal $x(t)$ about the **y-axis** it will be the same signal.
- That means if we reflect the signal $x(t)$ about the **y-axis** then we reflect it again about the **x-axis** it will be the same signal.

# 2) Even & Odd Signals

- **Odd Signals** if:

$$x(t) = -x(-t)$$




- e.g.:

$$x(t) = \sin(\omega t)$$

# Example from Simon Haykin

**EXAMPLE 1.1** Develop the even/odd decomposition of a general signal $x(t)$ by applying th definitions of Eqs. (1.2) and (1.3).

***Solution:*** Let the signal $x(t)$ be expressed as the sum of two components $x_e(t)$ and $x_o(t)$ a follows:

$$x(t) = x_e(t) + x_o(t)$$

Define $x_e(t)$ to be even and $x_o(t)$ to be odd; that is,

$$x_e(-t) = x_e(t)$$

and

$$x_o(-t) = -x_o(t)$$

Putting $t = -t$ in the expression for $x(t)$, we may then write

$$\begin{aligned} x(-t) &= x_e(-t) + x_o(-t) \\ &= x_e(t) - x_o(t) \end{aligned}$$

# Continue..

Solving for $x_e(t)$ and $x_o(t)$, we thus obtain

$$x_e(t) = \frac{1}{2}\Big(x(t) + x(-t)\Big)$$

and

$$x_o(t) = \frac{1}{2}\Big(x(t) - x(-t)\Big)$$

# 2) Even & Odd Signals

- If the above two conditions are not applicable on a certain signal; then this signal is Neither odd nor even signal. In other word, the signal in this case has no symmetry.
- Classifying the signals to even/odd Signals are important in many areas of mathematical analysis, especially the theory of power series and Fourier series.

# 2) Even & Odd Signals

- Any Signal can be Expressed as the sum of an even part and an odd part; that is,

$$x(t) = x_e(t) + x_o(t)$$

- Where $x_e(t)$ is even and $x_o(t)$ is odd with:

$$x_e(t) = \tfrac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \tfrac{1}{2}[x(t) - x(-t)]$$

# 2) Even & Odd Signals

- Example 1: write and draw $x_e(t)$ and $x_o(t)$ for:

$$x(t) = e^{+t} \qquad -\infty < t < \infty$$

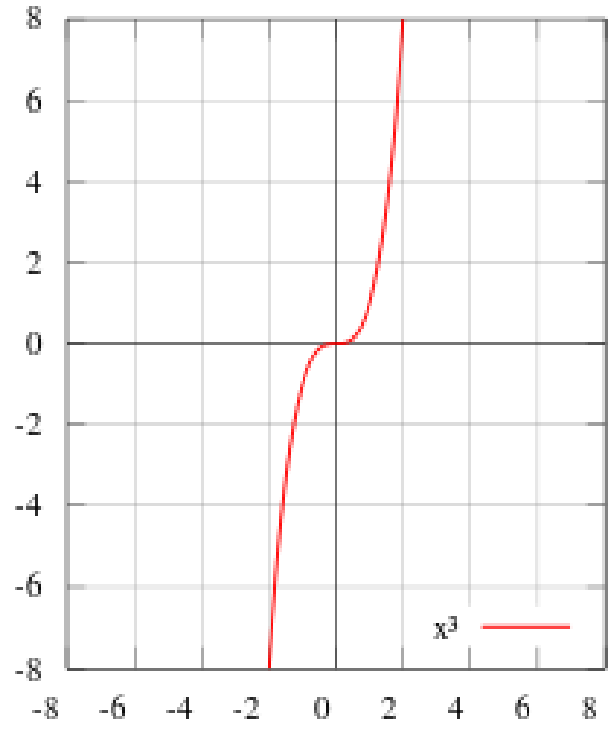
Sol:

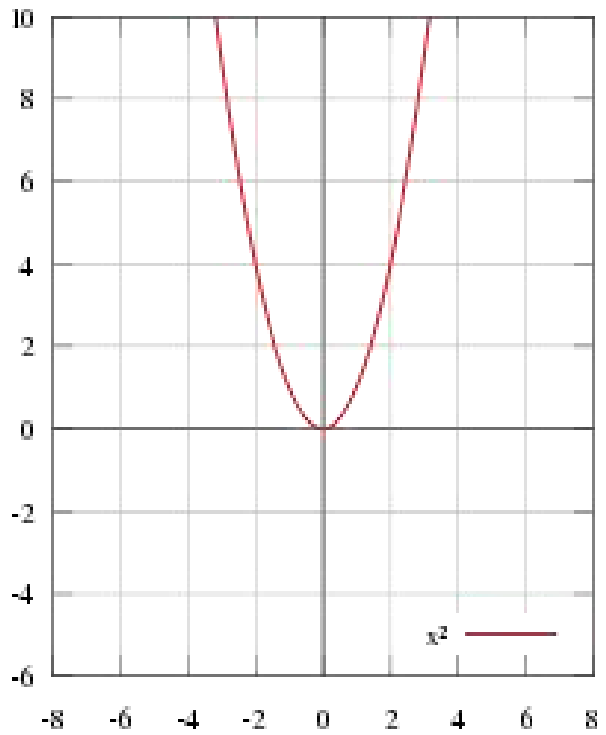
$$x_e(t) = \frac{[e^t + e^{-t}]}{2}$$

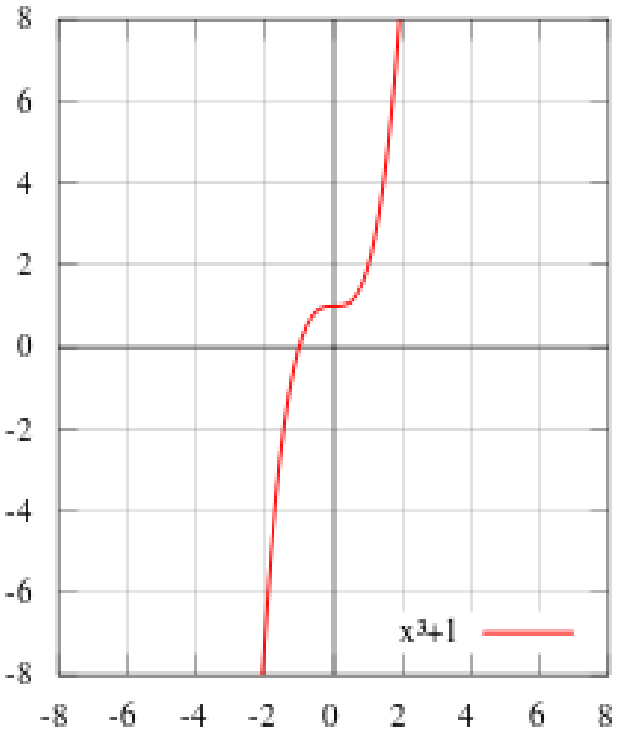
$$x_o(t) = \frac{[e^t - e^{-t}]}{2}$$

# Exercise 1:

Determine the type of symmetry of each of the following signals:



▶ **Drill Problem 1.2** The signals $x_1(t)$ and $x_2(t)$ shown in Figs. 1.12(a) and (b) constitute the real and imaginary parts of a complex-valued signal $x(t)$. What form of symmetry does $x(t)$ have?

***Answer:*** $x(t)$ is conjugate symmetric. ◀

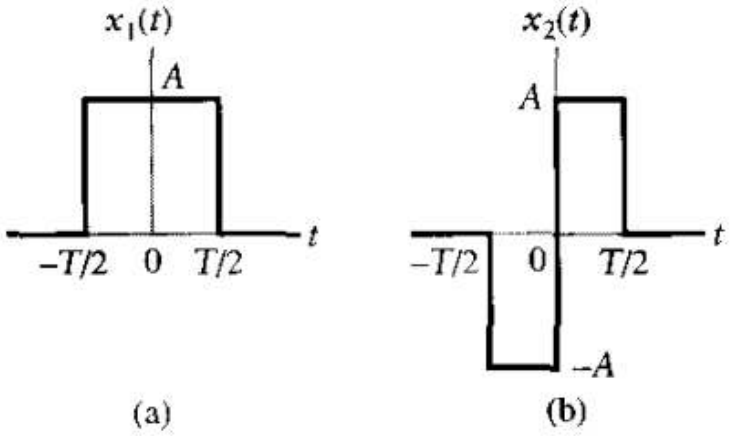
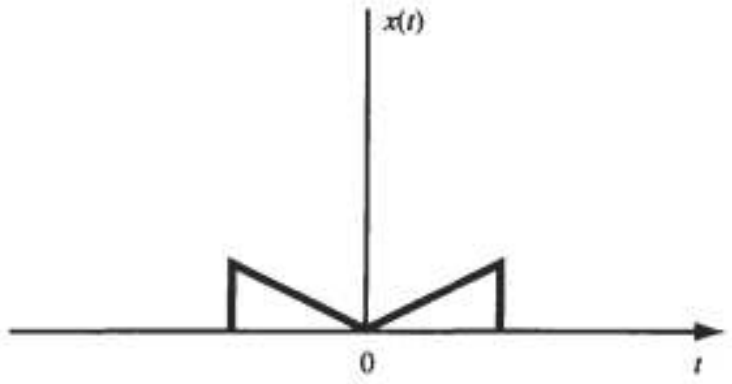
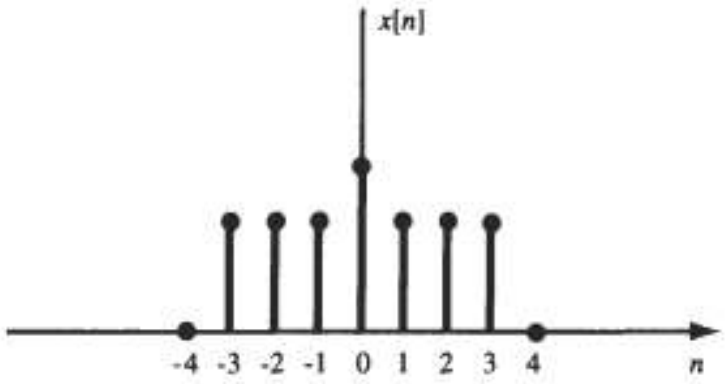


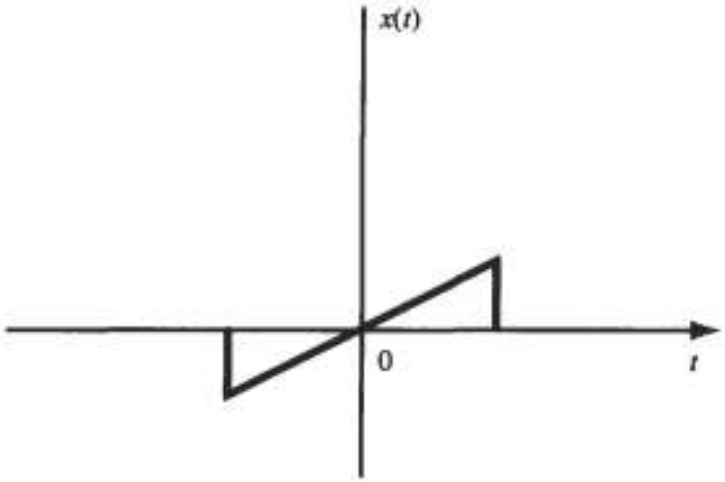

**FIGURE 1.12** (a) One example of continuous-time signal. (b) Another example of continuous-time signal.



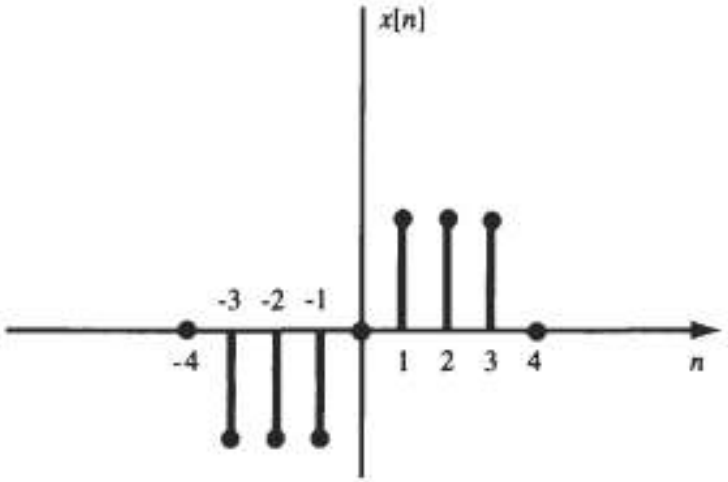

(a)




(b)




(c)




(d)

Any signal $x(t)$ or $x[n]$ can be expressed as a sum of two signals, one of which is even and one of which is odd. That is,

$$x(t) = x_e(t) + x_o(t)$$

$$x[n] = x_e[n] + x_o[n]$$

where

$$x_e(t) = \tfrac{1}{2}\{x(t) + x(-t)\} \qquad \text{even part of } x(t)$$

$$x_e[n] = \tfrac{1}{2}\{x[n] + x[-n]\} \qquad \text{even part of } x[n]$$

$$x_o(t) = \tfrac{1}{2}\{x(t) - x(-t)\} \qquad \text{odd part of } x(t)$$

$$x_o[n] = \tfrac{1}{2}\{x[n] - x[-n]\} \qquad \text{odd part of } x[n]$$

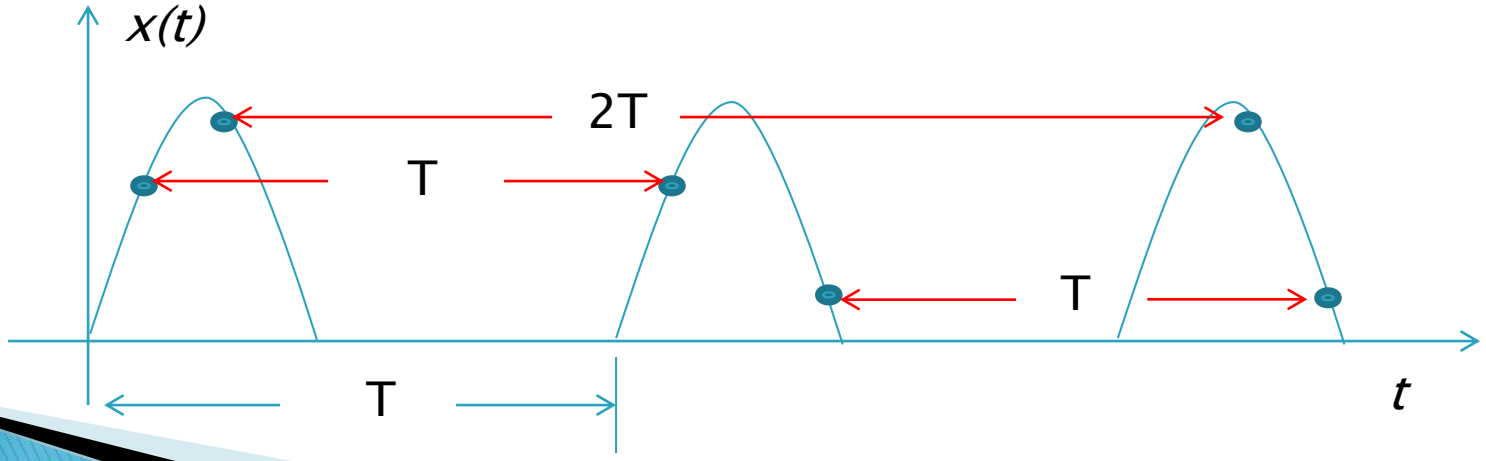
Note that the product of two even signals or of two odd signals is an even signal and that the product of an even signal and an odd signal is an odd signal

# 3) Periodic & Aperiodic Signal

- $x(t)$ is periodic with period $T > 0$ if:

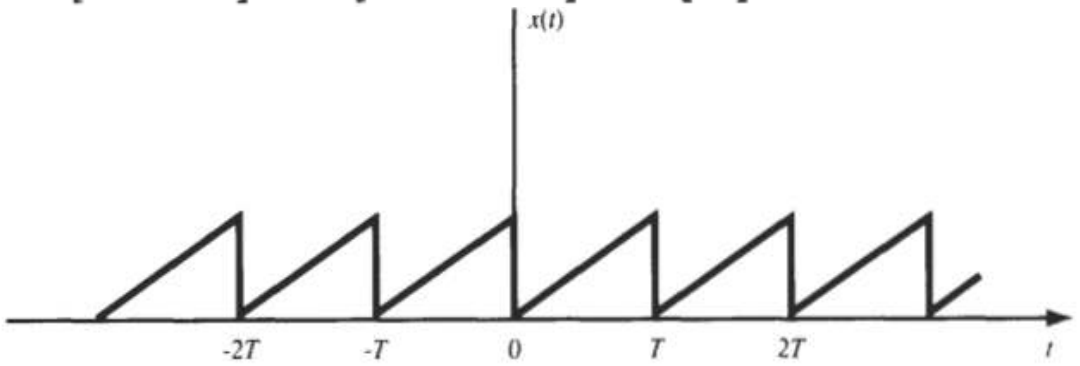
$$x(t) = x(t + nT) \quad where\ n = 1, 2, 3, ...etc.$$

- The minimum value of the period $T > 0$ that satisfies the above equation is called the **fundamental period.**

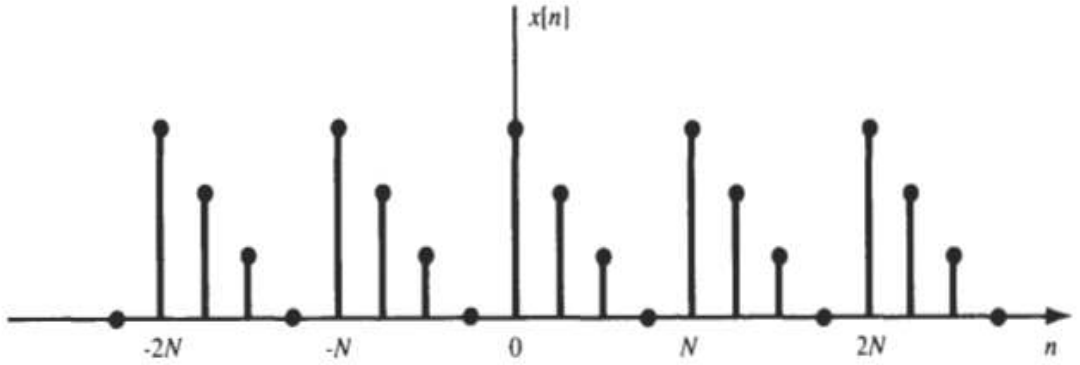


Periodic discrete-time signals are defined analogously. A sequence (discrete-time signal) x[n] is periodic with period N if there is a positive integer N for which

$$x[n+N] = x[n+mN] = x[n] \qquad \text{all } n$$




(a)



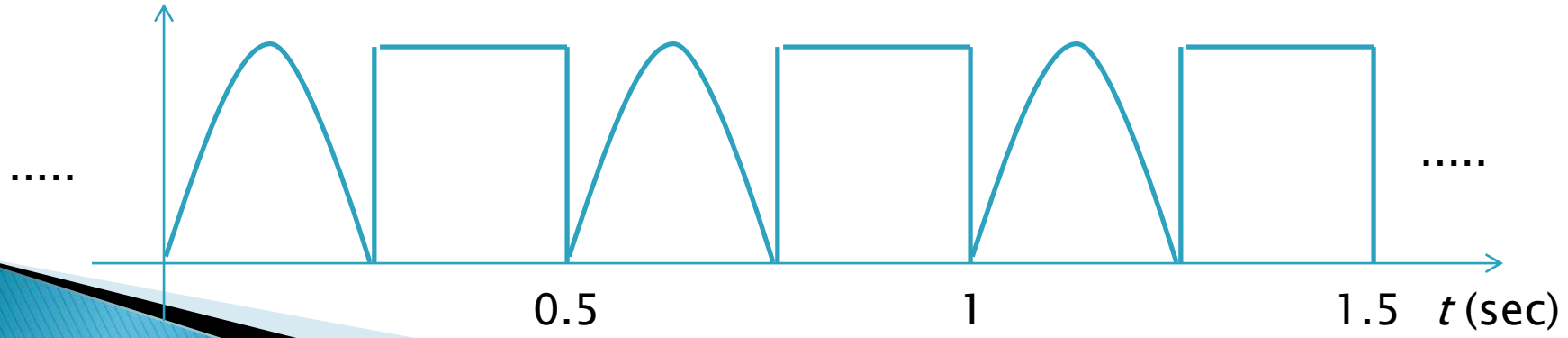
# 3) Periodic & Aperiodic Signal

- The number of times the signal repeat it self per second is called the frequency.

$$f = \frac{1}{T} \quad (Hertz)$$

$$\& \qquad \omega = 2\pi f \quad (rad/\sec)$$

- e.g.: the frequency of the following signal is 2Hz.

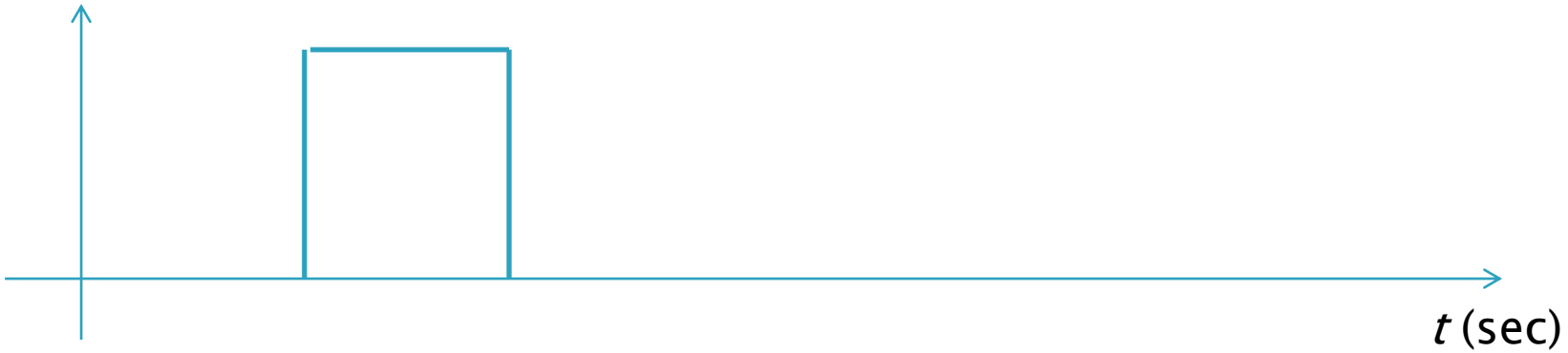


# 3) Periodic & Aperiodic Signal

- If there is no T that satisfies the above equation, x(t) is called Aperiodic. Where:

$$T \to \infty$$

- e.g.



# 3) Periodic & Aperiodic Signal Summation of two Periodic Signals:

- Sum of two periodic signals can be periodic only if the ratio of the respective periods of each signal can be expressed as a rational number.
- Some of the irrational numbers:

$$\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, ....$$

$$\pi = \frac{22}{7}$$

# 3) Periodic & Aperiodic Signal
# Summation of two Periodic Signals:

- Note that if the summation of to periodic signals produce periodic signal; the new period is **equal or larger than** the largest period.
- The new period is equal to the lowest common denominator (LCD) of the two periods.
- Non-linear operations (e.g. multiplication) of periodic signals produce periodic signal with ***different fundamental frequency***.

# *Example*

- Consider the following signals:

$$x_1(t) = \sin(10\pi t) \quad x_2(t) = \cos(25\pi t) \quad x_3(t) = \cos(20t)$$

calculate the new period of the follow:

I. $x_1(t) + x_2(t)$.

II. $x_1(t) + x_3(t)$.

III. $x_1(t) * x_3(t)$.

# *Continue…*

- By applying the condition of the periodic signals we find out that the three signals are periodic.

$$T_1 = 1/5 = 0.2$$

$$T_2 = 1/12.5 = 0.08$$

$$T_3 = \pi/10$$

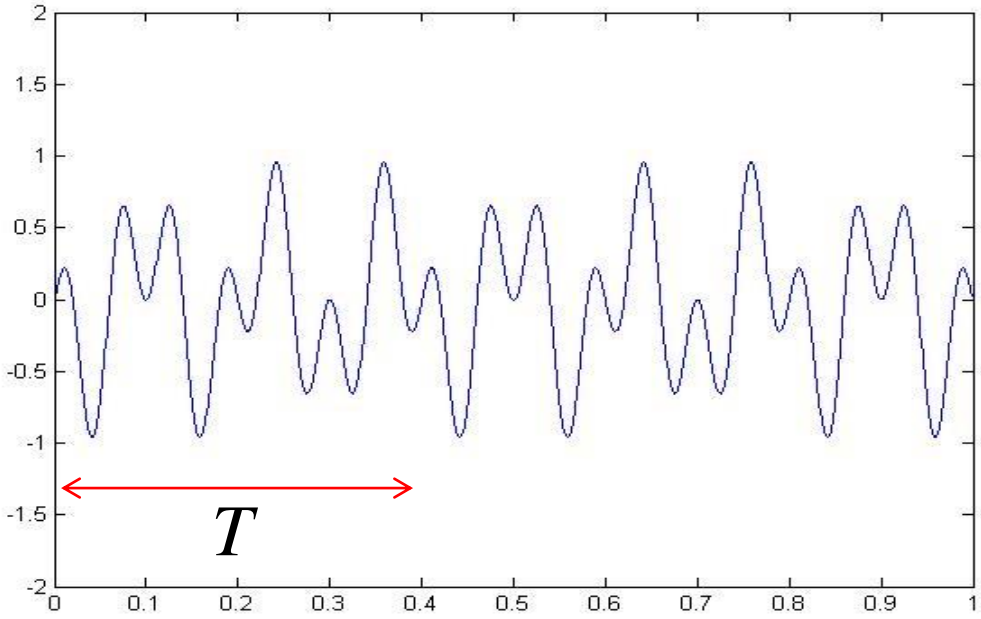
# *Continue…*

I. The ratio of $T_1/T_2$ or $T_2/T_1$ gives rational number. So, the new period using the LCD is:

$$T = 0.4s$$

Using MATLAB to draw $x_1(t) + x_2(t)$ signal:

```
>> t=0:0.001:10;
>> x1=sin(10*pi*t);
>> x2=cos(25*pi*t);
>> x=x1+x2;
>> plot(t,x)
>> axis([0 1 -2 2])
```



# *Continue…*

II. The ratio of $T_1/T_3$or $T_3/T_1$ gives irrational. For that reason there is no LCD for $T_1$ & $T_3$. So, the produced signal is not periodic.

III. The result of the multiplication of the two signals can be obtained by calculating the trigonometric identities first then the LCD.

$$x_1(t) * x_3(t)$$

$$\sin(10\pi t) * \cos(20t) = \tfrac{1}{2}[\sin(10\pi + 20)t + \sin(10\pi - 20)t]$$

$$= \tfrac{1}{2}[\sin(51.416t) + \sin(11.416t)]$$

now the period can be calculated by LCD.

# Example:

- Show that the complex exponential signal $x(t) = e^{jw_o t}$ is periodic and that its fundamental period is $\frac{2\pi}{w_o}$.

  **Sol**: $x(t)$ will be periodic if
- $x(t+T) = e^{jw_o(t+T)} = e^{jw_o t}$
- Then: $e^{jw_o(t+T)} = e^{jw_o t}\ e^{jw_o T}$

  we must have : $e^{jw_o T} = 1$
- If $w_o$=0, then $x(t) = 1$, this means its periodic for any period T, but if $w_o \neq 0$ so that $w_o T = 2\pi m$ or $T = \frac{2\pi m}{w_o}$
- Thus, the fundamental period $T_o$ the smallest positive T x(t) is given by $\frac{2\pi}{w_o}$

# Example

Let $x(t)$ be the complex exponential signal

$$x(t) = e^{j\omega_0 t}$$

with radian frequency $\omega_0$ and fundamental period $T_0 = 2\pi/\omega_0$. Consider the discrete-time sequence $x[n]$ obtained by uniform sampling of $x(t)$ with sampling interval $T_s$. That is,

$$x[n] = x(nT_s) = e^{j\omega_0 nT_s}$$

Find the condition on the value of $T_s$ so that $x[n]$ is periodic.

If $x[n]$ is periodic with fundamental period $N_0$, then

$$e^{j\omega_0(n+N_0)T_s} = e^{j\omega_0 nT_s} e^{j\omega_0 N_0 T_s} = e^{j\omega_0 nT_s}$$

Thus, we must have

$$e^{j\omega_0 N_0 T_s} = 1 \Rightarrow \omega_0 N_0 T_s = \frac{2\pi}{T_0} N_0 T_s = m2\pi \qquad m = \text{positive integer}$$

or

$$\frac{T_s}{T_0} = \frac{m}{N_0} = \text{rational number}$$

Thus $x[n]$ is periodic if the ratio $T_s/T_0$ of the sampling interval and the fundamental period of $x(t)$ is a rational number.

# Example

Consider the sinusoidal signal $x(t) = cos15t$ finds

a) Find the value of sampling interval $T_s$ such that $x[n] = x(nT_s)$ is periodic sequence.

b) Find the fundamental period of $x(nT_s)$ if $T_s = 0.1\pi$ seconds

(*a*) The fundamental period of $x(t)$ is $T_0 = 2\pi/\omega_0 = 2\pi/15$. $x[n] = x(nT_s)$ is periodic if

$$\frac{T_s}{T_0} = \frac{T_s}{2\pi/15} = \frac{m}{N_0}$$

where $m$ and $N_0$ are positive integers. Thus, the required value of $T_s$ is given by

$$T_s = \frac{m}{N_0} T_0 = \frac{m}{N_0} \frac{2\pi}{15}$$

(*b*) Substituting $T_s = 0.1\pi = \pi/10$ we have

$$\frac{T_s}{T_0} = \frac{\pi/10}{2\pi/15} = \frac{15}{20} = \frac{3}{4}$$

Thus, $x[n] = x(nT_s)$ is periodic.

$$N_0 = m\frac{T_0}{T_s} = m\frac{4}{3}$$

The smallest positive integer $N_0$ is obtained with $m = 3$. Thus, the fundamental period of $x[n] = x(0.1\pi n)$ is $N_0 = 4$.

# Exercise 2

which of the following functions (signals) is periodic?

1. $$x_1(t) = t\sin(t)$$

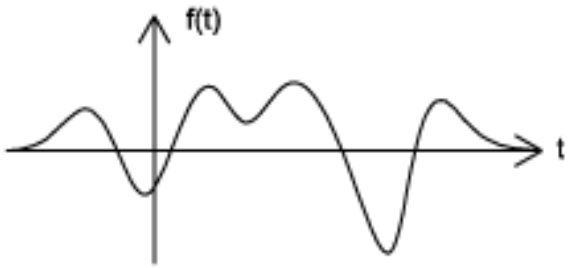
2. $$x_2(t) = e^{\sin(t)}$$

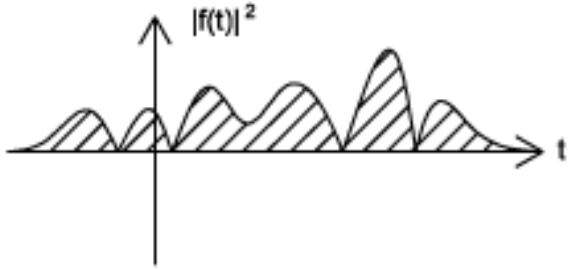
# 4) Energy & Power Signal

- **Energy Signal** : A signal is said to be energy signal when it has finite energy.
- Since we often think of signal as a function of varying amplitude through time, it seems to reason that a good measurement of the strength of a signal would be the area under the curve (The Energy of the signal).
- However, this area may have a negative part. This negative part does not have less strength than a positive signal of the same size. This suggests either squaring the signal or taking its absolute value, then finding the area under that curve.

# 4.1- Energy Signal

- It turns out that what we call the *Energy* of a signal is the area under the squared signal.




$$E_f = \int_{-\infty}^{\infty} (|f(t)|)^2 .dt$$

$$\text{Energy } E = \int_{-\infty}^{\infty} x^2(t)dt$$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

# For discrete signals

- Similarly, for a discrete-time signal $x[n]$, the normalized energy content $\boldsymbol{E}$ of $x[n]$ is defined as

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- The normalized average power $P$ of $x[n]$ is defined as

$$P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

# The following classes of signals are defined:

i. $x(t)$ $(or\ x[n])$ is said to be an energy signal (or sequence) if and only if $0 < E < m$, and $P = 0$.

ii. $x(t)$ $(or\ x[n])$ is said to be a power signal (or sequence) if and only if $0 < P < m$, thus implying that $E = m$

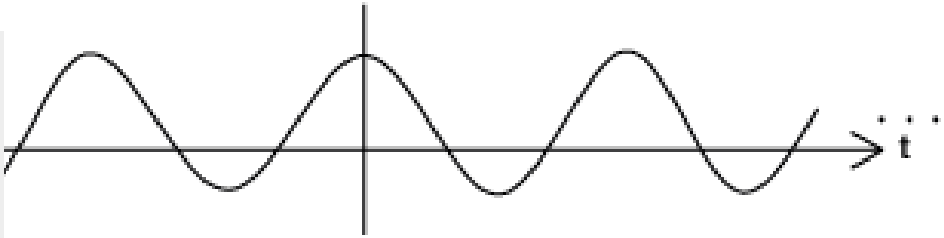
iii. Signals that satisfy neither property are referred to as neither energy signals nor power signals

Note that : a periodic signal is a power signal if its energy content per period is finite, and then the average power of this signal need only be calculated over a period.

# 4.2-Power Signal

- **Power Signal:** A signal is said to be power signal when it has finite power.
- Our definition of energy seems reasonable, and it is. However, what if the signal does not decay? In this case we have infinite energy for any such signal.
- Does this mean that a sixty hertz sine wave feeding into your headphones is as strong as the sixty hertz sine wave coming out of your outlet? Obviously not. This is what leads us to the idea of *signal power*.

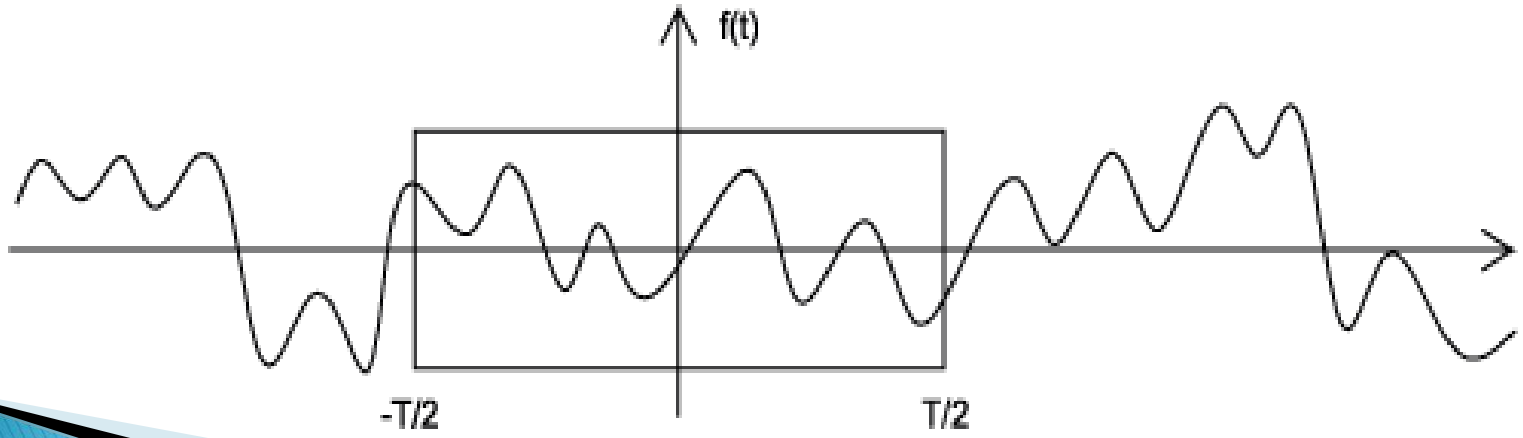
$$\text{Power } P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} x^2(t)dt$$



# 4.2-Power Signal

- Power is a time average of energy (energy per unit time). This is useful when the energy of the signal goes to infinity.

$$P_f = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} (|f(t)|)^2 .dt$$



# 4.2-Power Signal

- Note:

1. Aperiodic Signals Could be Energy signal if $E$ is finite & $P$=0.
2. Periodic Signals are always called power signals because if:

$$E = \infty \quad \& \quad P \neq 0$$

3. If $E$ is infinite, $P$ is not finite. So, $f(t)$ is neither energy nor power signal.

A signal can be categorized into energy signal or power signal: An **energy signal** has a finite energy, $0 < E < \infty$. In other words, energy signals have values only in the limited time duration. For example, a signal having only one square pulse is energy signal. A signal that decays exponentially has finite energy, so, it is also an energy signal. The power of an energy signal is 0, because of dividing finite energy by infinite time (or length).

On the contrary, the **power signal** is not limited in time. It always exists from beginning to end and it never ends. For example, sine wave in infinite length is power signal. Since the energy of a power signal is infinite, it has no meaning to us.

Thus, we use power (energy per given time) for power signal, because the power of power signal is finite, $0 < P < \infty$.

# Example 3:

- Given an exponential signal as defined below, find its energy.

$$f(t) = A.\exp(-k.t) \quad for \;\; 0 \le t$$

***Solution:***

- The solution is expressed by the equation:

$$E_f = \int_{-\infty}^{\infty} |f(t)|^2 .dt = \int_{0}^{\infty} [A.\exp(-k.t)]^2 .dt = \frac{A^2}{2k}$$

- An exponential signal is an *energy signal*, since its energy is a finite, non-zero value.

# Example 4:

- Given a sinusoidal signal as defined by the equation:

$$x(t) = A.\sin \omega t$$

Find its power.

***Solution:***

- A sinusoidal signal is a power signal, since the its power over a cycle is a finite, non-zero value. The energy associated with the sinusoidal signal is infinite.

$$P_x = \frac{1}{T}\int_0^T |x(t)|^2 .dt = \frac{1}{T}\int_0^T (A.\sin(\omega.t))^2 .dt = \frac{A^2}{2}$$

$$\frac{1}{T}\int_0^T (A.\sin(\omega.t))^2 .dt = 0.5(\sin^* \cos)^2 .dt$$

# Example 5:

- Given a square-wave signal as defined by the equation:

$$f(t) = \begin{cases} -A & -T/2 < t < 0 \\ A & 0 < t < T/2 \end{cases}$$

find its power.

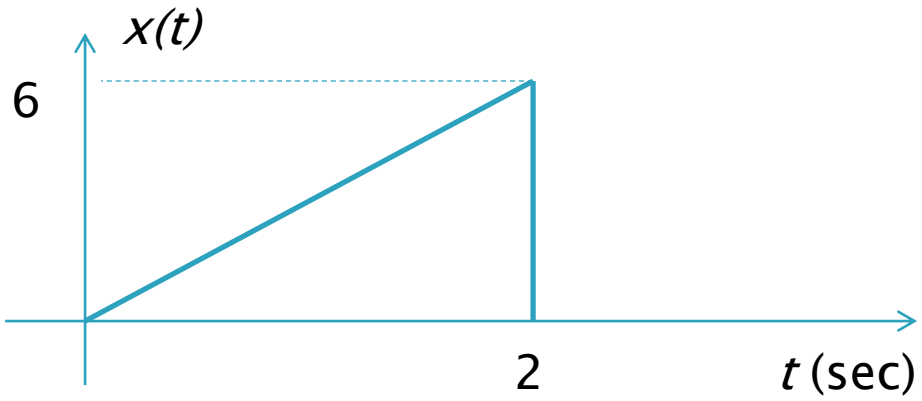
***Solution:***

- A periodic square-wave signal is a power signal, since the its power over a cycle is a finite, non-zero value. The energy associated with the square-wave signal is infinite.

$$P_f = \frac{1}{T}\int_{-T/2}^{T/2} |f(t)|^2 .dt = \frac{1}{T}\int_{-T/2}^{T/2} A^2 .dt = A^2$$

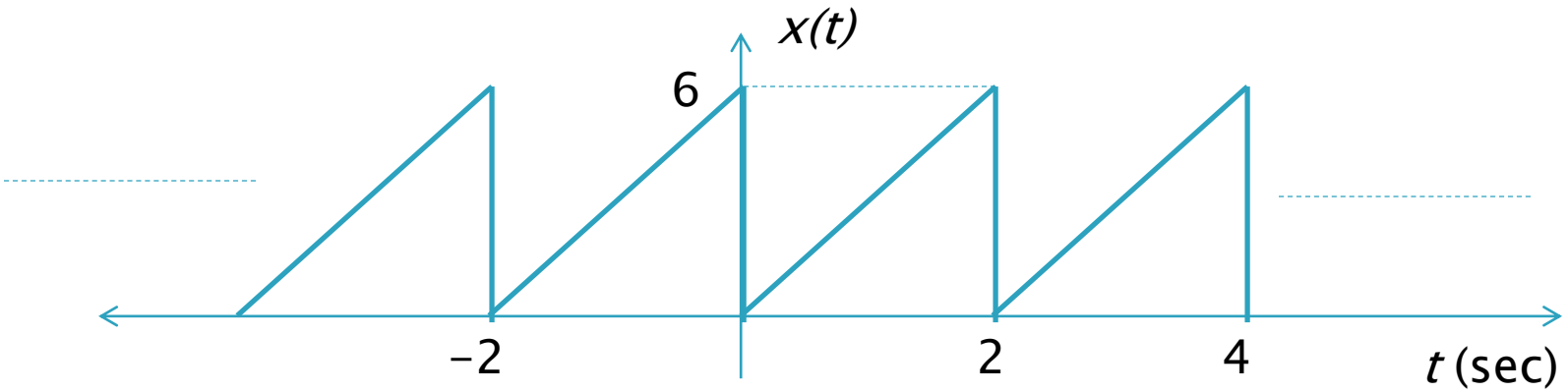
# Exercise 3:

- Consider *x(t)* shown below, determine if the signal is Energy or power Signal.



# Exercise 4:

- Consider *x(t)* shown below, determine if the signal is Energy or power Signal.



# Exercise 5:

- For $x(t) = t \sin \omega t$ check if the signal is:
  A. Even or odd.
  B. Discrete or continuous.
  C. Periodic or Aperiodic.
  D. Power or Energy Signal.

# Exercise 6:

- Is $x(t)=3cos(3\pi t)+sin(4\pi t)+5$ periodic? If yes, what is the Period of this signal? Then compute the power.

# Representing Signals

Transformations of continuous- Time Signal :1.2

# :Introduction

- A number of important operations are often performed on Signals, time shifting, time scaling, and time reflection. These operations are very common components to the real world systems.
- Operations on signals amplitude are easier to understand and students might already familiar with it but with different names. Such as, shifting on the amplitude called offset, such as DC offset in the circuit analysis. Scaling on the amplitude called Amplification or attenuation.

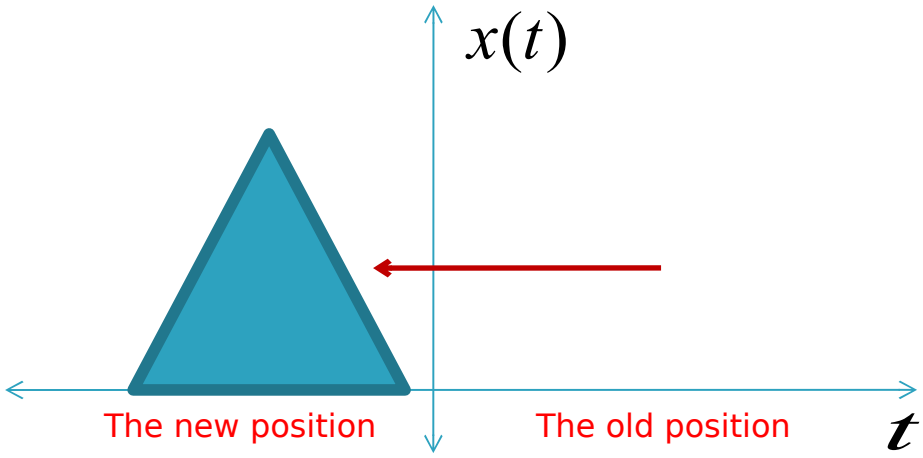
# :Objectives

- After completing this session, you will be able to:

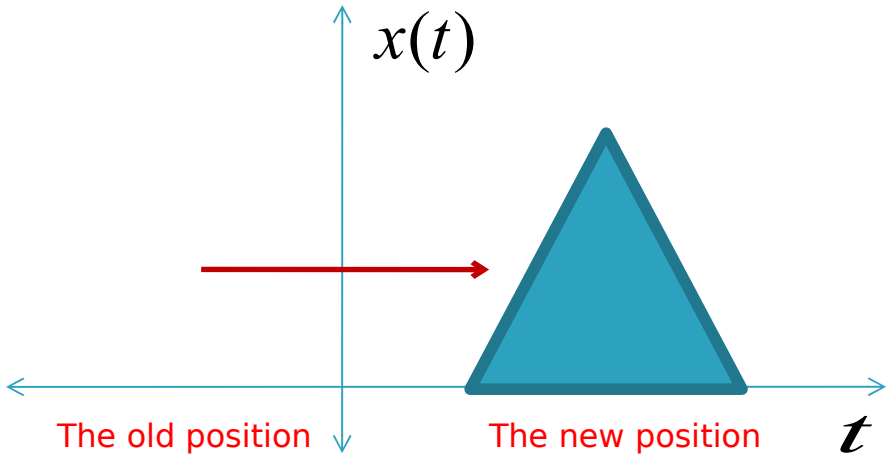
1. To know the type operations can be performed on the signals.
   I. Time shifting.
   II. Time scaling.
   III. Time reflections.
2. To perform time shifting, time scaling, and time reflection mathematically and graphically.
3. To perform combination of the above operations.
4. To know some applications on these operations.
5. To perform the mentioned operations on the amplitude of the signals.

# :Time- shifting (1

- Time-shifting on signal is the process that move the signal to right or to the left on the diagram.




Left Shifting Operation




Right Shifting Operation

# :Time- shifting (1

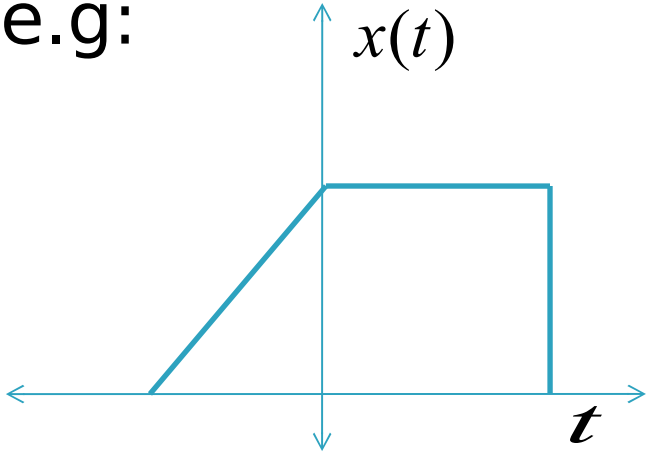
- Shifting the signal to the right is commonly known by the "*delay*" process.
- The shifted version of is represented mathematically by
- To perform time-shifting on $x(t)$ by a time $t_o$ the:

a) Right: subtract a fixed positive quantity "$t_o$" of the shift to the time variable in the function ***x(t)*** to get where $t_o>0$.

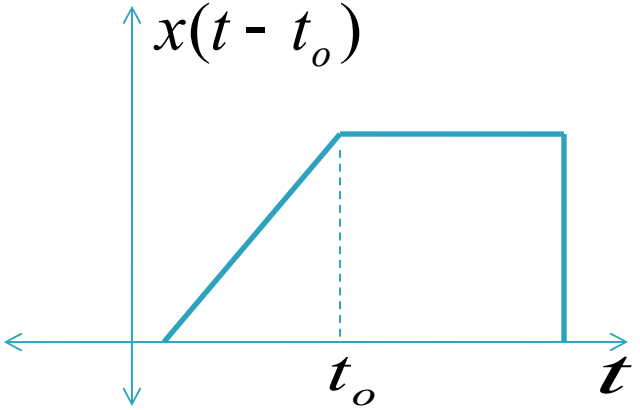
b) Left: Add a fixed positive quantity "$t_o$" of the shift to the time variable in the function ***x(t)*** to get ***)*** where .

# :Time- shifting (1

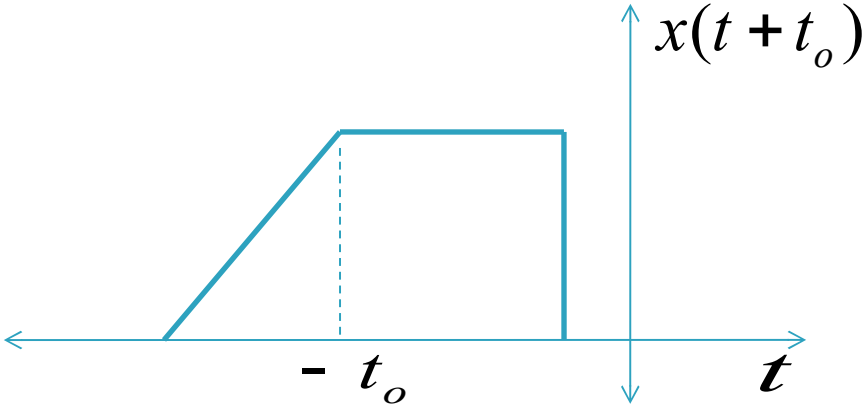
e.g:




The signal x(t) without shift




x(t) with shift to the right by $t_o$




x(t) with shift to the left by $t_o$

# :Example 1

- If $x(t) = t^2 - 5$ , write down the new version of x(t) after performing time shifting to:

A. x(t) to the right by 2 units.

B. x(t) to the left by 4 units.

Sol: Subtract the positive quantity "2" to the variable "t" as follow:

C. Add the positive quantity "4" to the variable "t" as follow:

$$x(t+4) = (t+4)^2 - 5 = t^2 + 8t + 11$$

# :Example 2

- If $f(t) = \sin(\pi t - 3)$, write the new version of f(t) if we:

A. Shifted t to the right by 3 units.

B. Shifted t to the left by $3/\pi$ units.

Sol:

C. $f(t - 3) = \sin(\pi(t - 3) - 3)$

D. $f(t + 3/\pi) = \sin(\pi(t + 3/\pi) - 3). = \sin(\pi t)$

# :Example 3

- shift $x(t) = \cos(t)$ to the right by $\frac{\pi}{2}$ .

Sol:

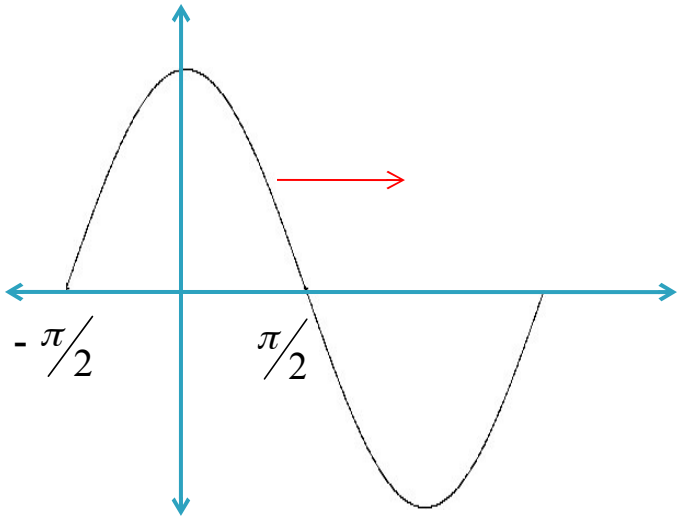
$$x(t - \frac{\pi}{2}) = \cos(t - \frac{\pi}{2})$$

From the trigonometric identities

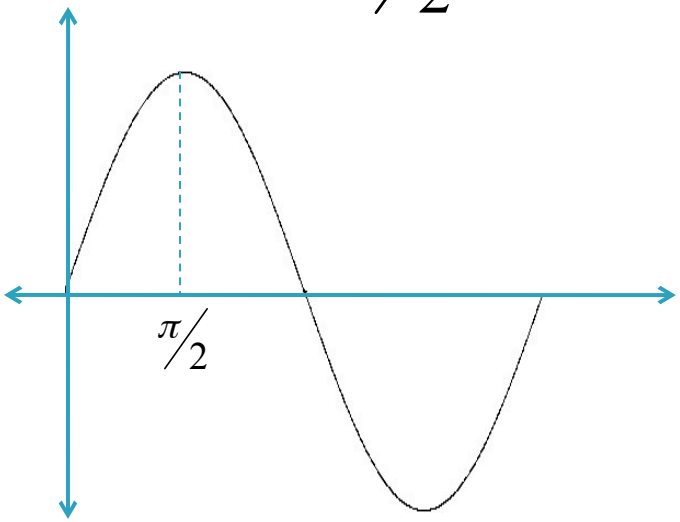
$$\cos(t - \frac{\pi}{2}) = \sin t$$

# Example 3

- The same result can be obtained graphically if we shift the cosine to the right by $\pi/2$ .




Cosine wave

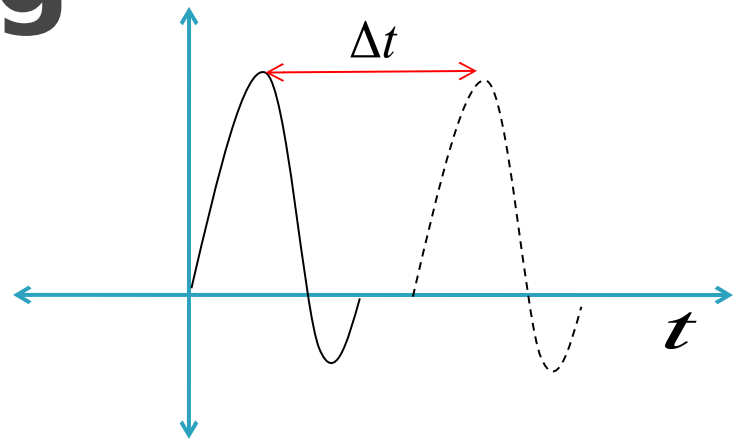



Sine wave

# Applications on time Shifting

- RADAR: is a system that uses pulses of electromagnetic waves to determine the position of a distant object. RADAR operates by sending out a signal, and then listening for echo's. If there is an object in range, the signal will bounce off that object and return to the RADAR station. The echo signal is a shifted version of the sent signal, by measuring the delayed time (the shifted time) as shown in the figure below, then multiply it by the speed of the light, you get the distance as follow:

# Applications on time Shifting

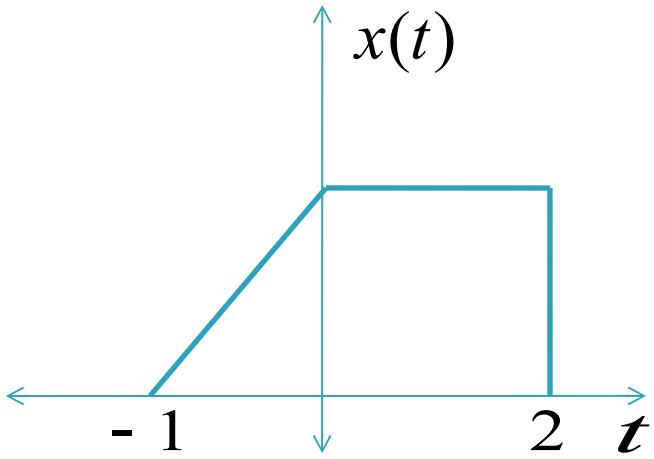



$$2D = C * \Delta t$$

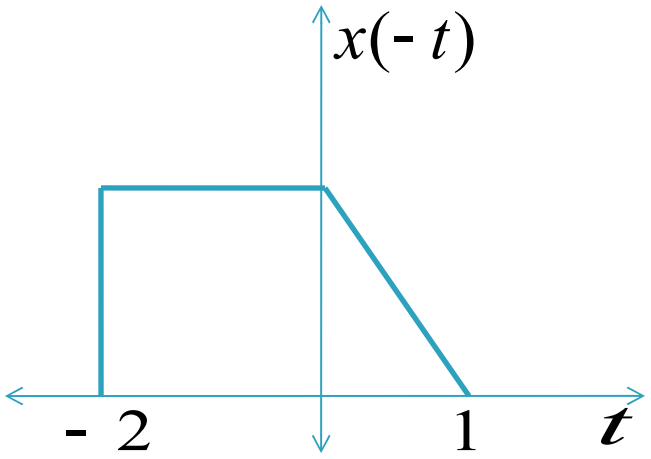
- Where D is the distance and C is the light speed.

# Time- Reflection (2

- Reflection on time is the process that rotate the signal around the y-axis.
- x(-t) is the reflected version of x(t).




The signal x(t) without reflection

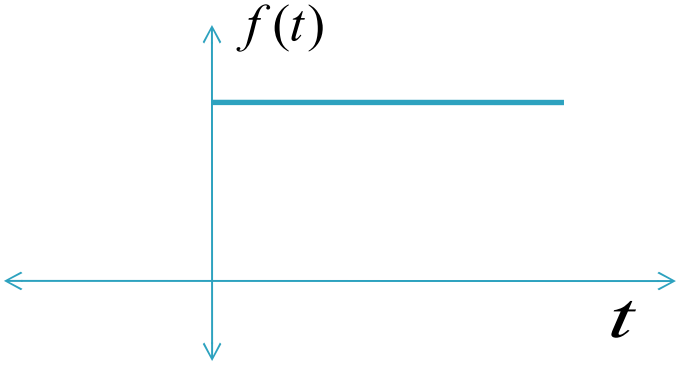



The reflected version of x(t)

# :Example 4

- Reflect the signal:

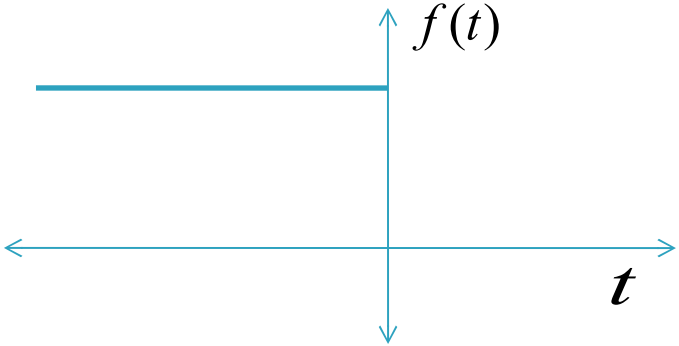
$$f(t)=\begin{cases}1 & t\geq 0\\ 0 & t<0\end{cases}$$




Sol:

$$f(-t)=\begin{cases}1 & -t\geq 0\\ 0 & -t<0\end{cases}$$

$$f(-t)=\begin{cases}1 & t\leq 0\\ 0 & t>0\end{cases}$$



# :Note

- Reflection can be symmetrical or not symmetrical. If symmetrical, then we may have two types:

A. Even Symmetry, when $x(t) = x(-t)$ .
B. Odd Symmetry, when $x(t) = -x(-t)$ .

# :Scaling (3

- Time scaling of signals is the process that expand or compress the signal in time.
- Time scaling of a signal $x(t)$ implies $x(\alpha t)$ leading to:

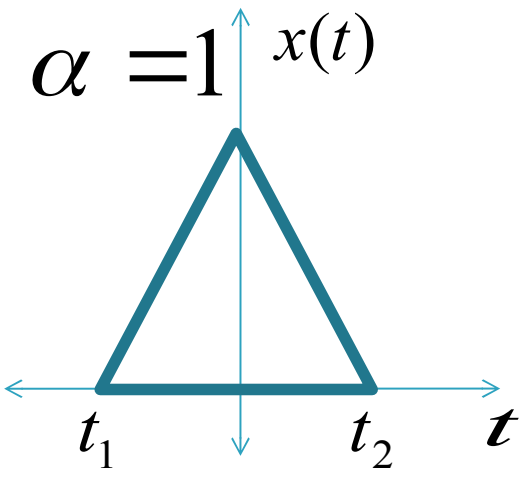
A. If $\alpha > 1$ , then $x(t)$ is compressed by factor $\alpha$ .

B. If $\alpha < 1$ , then $x(t)$ is expanded by factor $\alpha$ .

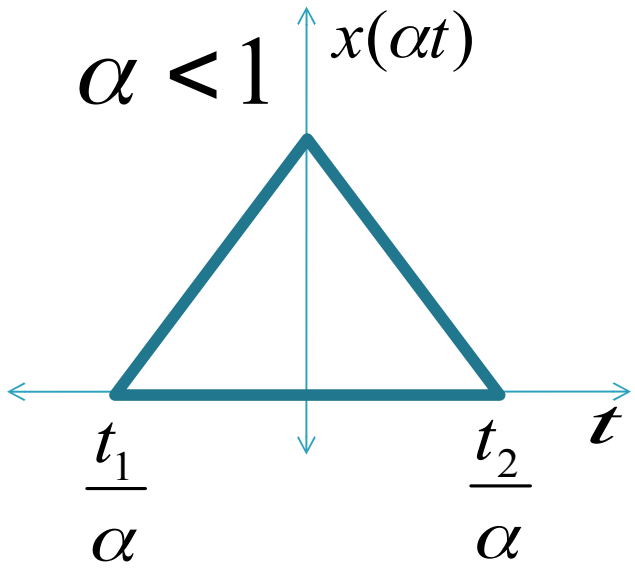
Where $\alpha$ is constant.

# Scaling (3

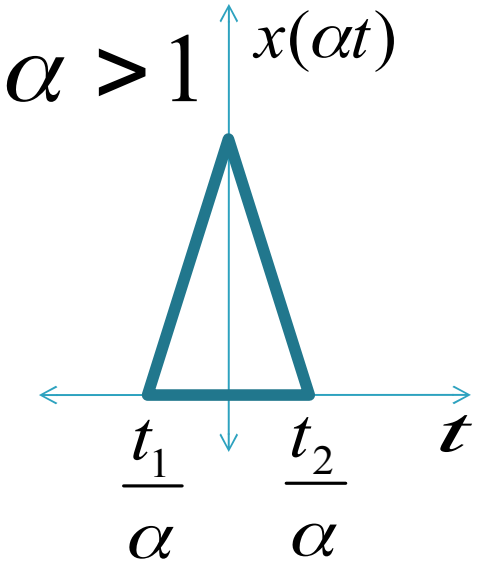
- e.g. the figure (a) below shows the signal $x(t)$, figure (b) shows the signal $x(t)$ after expand it by the factor $\alpha < 1$, figure (c) shows the signal $x(t)$ after compress it by the factor $\alpha > 1$




(a) The signal x(t)




(b) The expanded version of x(t)




(c) The compressed version of x(t)

# :Example 5

- Consider the function:

$$g(t)=\begin{cases}\dfrac{t+4}{4} & -4\le t<0\\[2ex] \dfrac{2-t}{2} & 0\le t<2\end{cases}$$

I. Compress $g(t)$ by 2.
II. Expand $g(t)$ by 4.

# Example 5

Solution:

I. To compress $g(t)$ multiply each $t$ by 2. that means $\alpha = 2$ .

$$g(2t) = \begin{cases} \dfrac{2t+4}{4} & -4 \le 2t < 0 \\ \dfrac{2-t}{2} & 0 \le 2t < 2 \end{cases}$$

$$g(2t) = \begin{cases} \dfrac{t+2}{2} & -2 \le t < 0 \\ 1-t & 0 \le t < 1 \end{cases}$$

# :Example 5

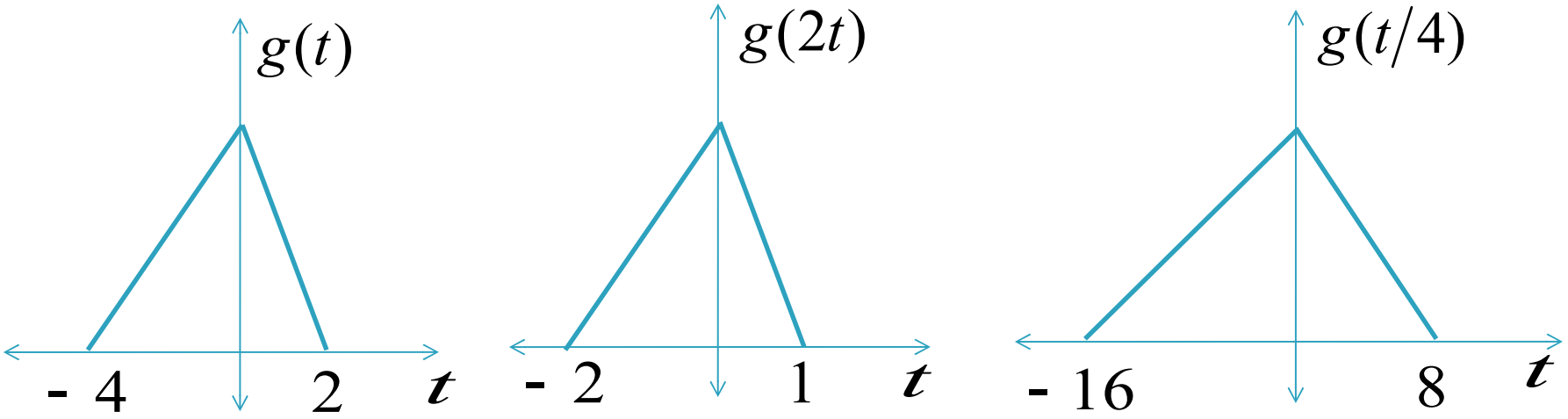
II. To expand $g(t)$ divide each $t$ by 4. that means $\alpha = 1/4$ .

$$g(t/4) = \begin{cases} \dfrac{t/4 + 4}{4} & -4 \le \dfrac{t}{4} < 0 \\ \dfrac{2 - t/4}{2} & 0 \le \dfrac{t}{4} < 2 \end{cases}$$

$$g(t/4) = \begin{cases} \dfrac{t + 16}{16} & -16 \le t < 0 \\ \dfrac{8 - t}{8} & 0 \le t < 8 \end{cases}$$

# :Example 5

- Graphically:



# Combination of three (4 operators

- From the previous operators we conclude that $x(-t)$ is the reflected version of $x(t)$, $x(2t)$ for example is the scaled version of $x(t)$, and $x(t-3)$ is the shifted version.
- But something like $x(2t-6)$ is a scaled and reflected version of $x(t)$ at the same time. That means this function has two operators. And something like $x(-3t+4)$ has three operators; scaling, shifting, and reflection.
- The question is how to do a such operation?

# Combination of three (4 operators

- To do so, use one of the of the following procedure:

1) Do the *shift* first, then *scaling*, then finally *reflection*.

OR

2) Do the *scale* first, then *reflection*, then *shifting*.

# :Example 6

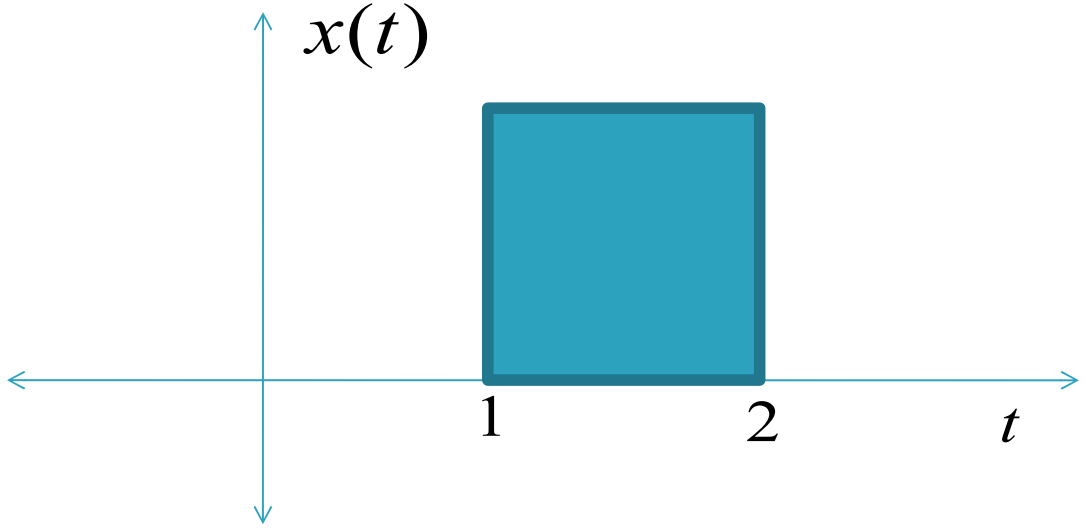
- Consider

$$f(t)=\begin{cases} t+3 & -3 \le t < 0 \\ 3-t & 0 \le t < 3 \end{cases}$$

Find and Sketch $f(3t-6)$ .

# :Example 7

- Draw $x(-\frac{t}{2} - 2)$ , given that $x(t)$ is as shown:



# :Exercise 1

- Let

$$x(t)=\begin{cases} -t+1 & -1\le t<0 \\ t & 0\le t<2 \\ 2 & 2\le t<3 \\ 0 & otherwise \end{cases}$$

A. Sketch $x(t)$ .
B. Sketch $x(-3t-2)$ .
C. Sketch $x(-\frac{2}{3}t-\frac{1}{2})$ .
D. Find the analytical expressions for these functions.

# :Exercise 2

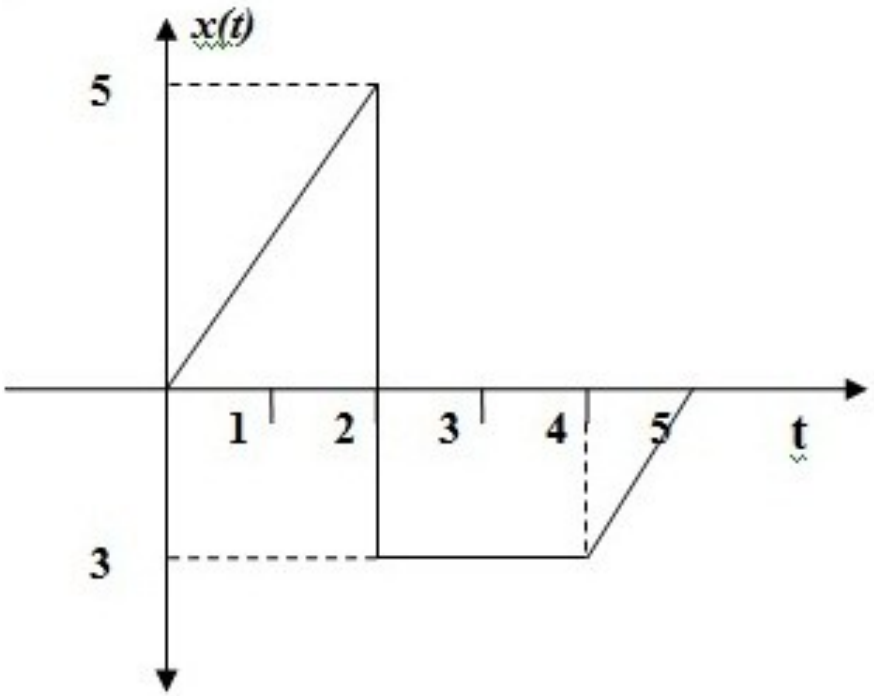
- Sketch the signal

$$x(t) = \sin\left(\frac{\pi}{4}t + 20^{o}\right)$$

# :Exercise 3

- For x(t) shown below, sketch:

  $y(t)=-0.5x(2t-4)+2$



# Representing Signals

1.3: Elementary Signals

# Introduction

- In signal processing we need an appropriate reference frame, so that we can place the signals we try to analyze. Since we deal with signals, any reference frame of interest has to be made of signals with well known properties, which reflect the kind of information we want to extract.

# Introduction

- Several important elementary signals that occur frequently in applications also serve as a basis for representing other signals.
- The following signals are some of the continuous- time elementary signals of interest:
  1. The unit step function.
  2. The Ramp function.
  3. The sampling & sinc signals.
  4. The unit impulse function.

# Objectives

- After completing this session, you will be able to:

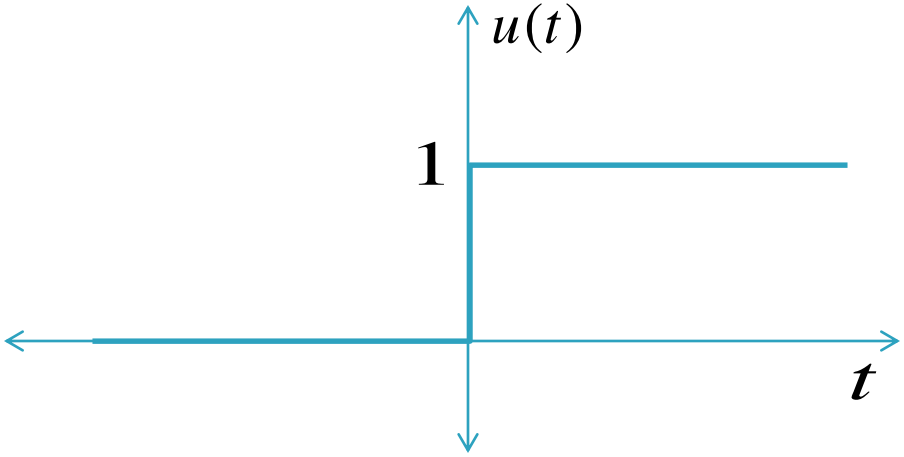
1. To know the elementary signals, and how to define the above continuous time signals mathematically and graphically.
2. To know the properties of the elementary signal.
3. To know how to represent other signals.
4. To simplify the solution of the some engineering problems.

# 1) The Unit Step function

- The unit step $u(t)$ is defined as:

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

- Graphically



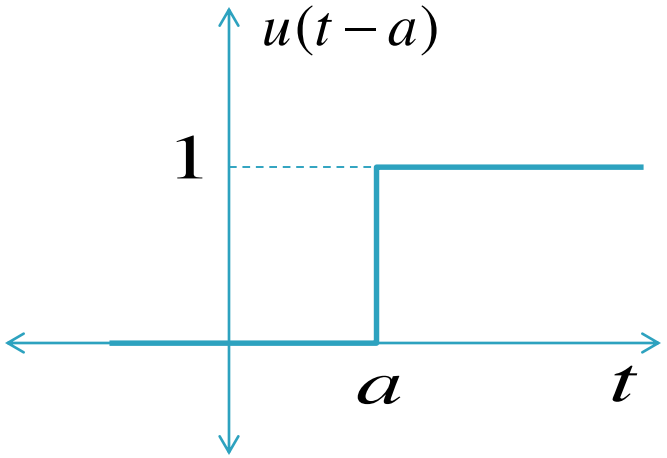
# 1) The Unit Step function

- This signals is an important signal for analytic studies, and it also has many practical applications.
- The function is used in the mathematics of *control theory* and *signal processing* to represent a signal that switches on at a specified time and stays switched on indefinitely.

# 1) The Unit Step function

- It is also used in structural mechanics together with Dirac Delta function (to be studied later on in this section) to describe different types of structural loads.
- The shifted version of the unit step is:

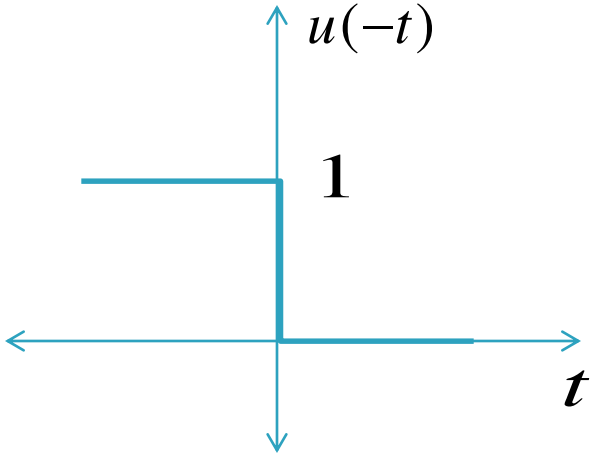
$$u(t-a)=\begin{cases}1 & t>a\\ 0 & t<a\end{cases}$$



# 1) The Unit Step function

- The reflected unit step is:

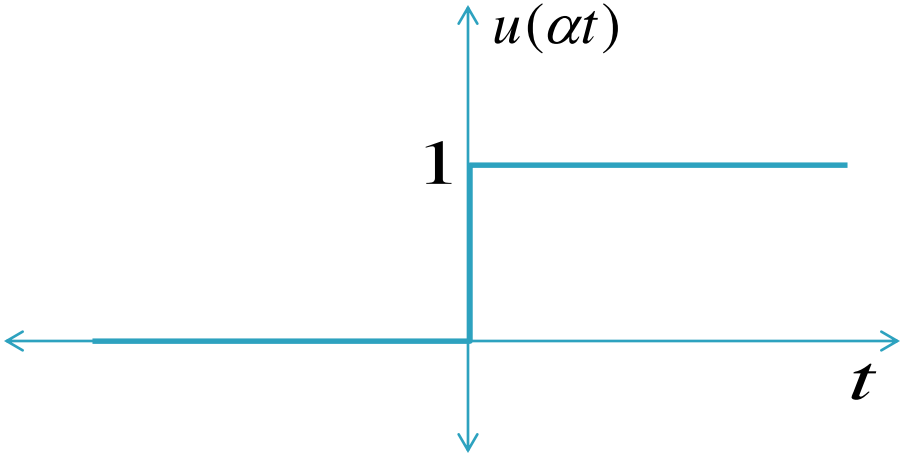
$$u(-t) = \begin{cases} 1 & t < 0 \\ 0 & t > 0 \end{cases}$$



# 1) The Unit Step function

- The scaled unit step remain the same as $u(t)$ , why?

$$u(\alpha t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$



# 1) The Unit Step function

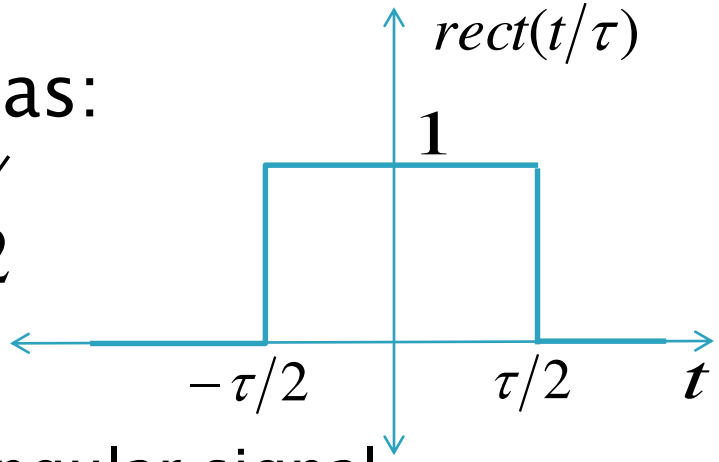
- From the step signal we can represent other types of signals, such as:
  A. Rectangular signal $rect(t/\tau)$.
  B. Signum Signal $Sgn(t)$.

# 1) The Unit Step function

A. Rectangular signal:

- The $rect$ signal is defined as:

$$rect(\frac{t}{\tau}) = \begin{cases} 1 & -\tau/2 \le t \le \tau/2 \\ 0 & otherwise \end{cases}$$




Where $\tau$ is the width of the rectangular signal.

- We can represent $rect(t/\tau)$ using the unit step function:

$$rect(t/\tau) = u(t+\tau/2) - u(t-\tau/2)$$

# 1) The Unit Step function

- The following figures illustrate that subtract figure b from figure a, the result is the rectangular shape as shown in figure c.

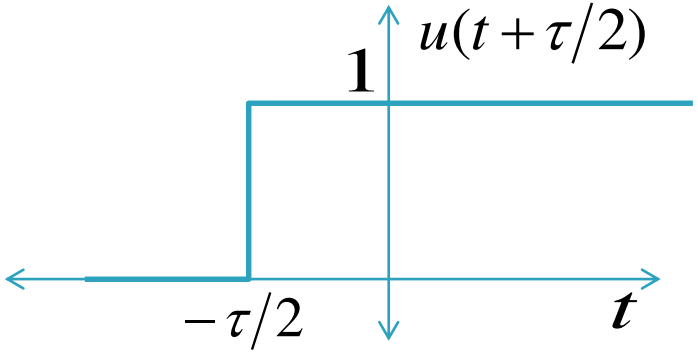



Fig (a)

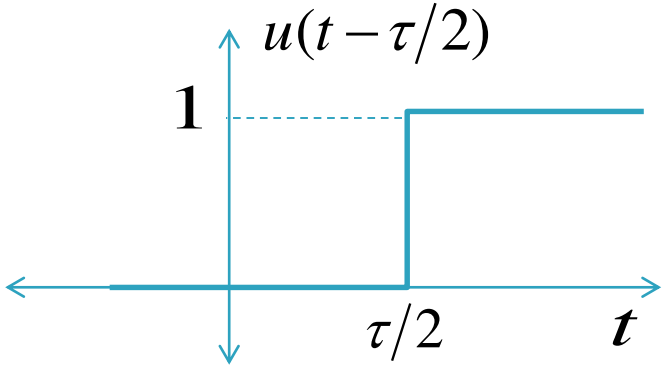



Fig (b)

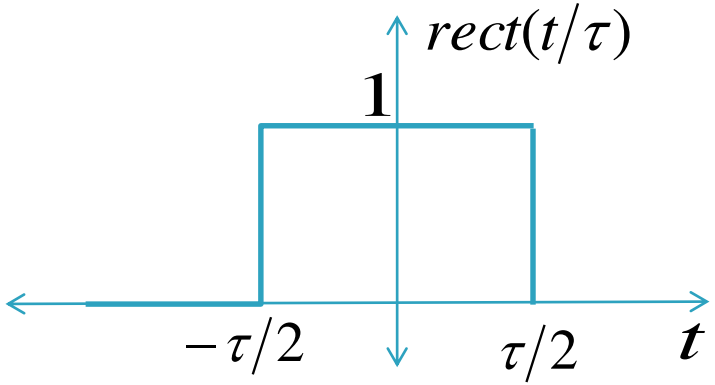


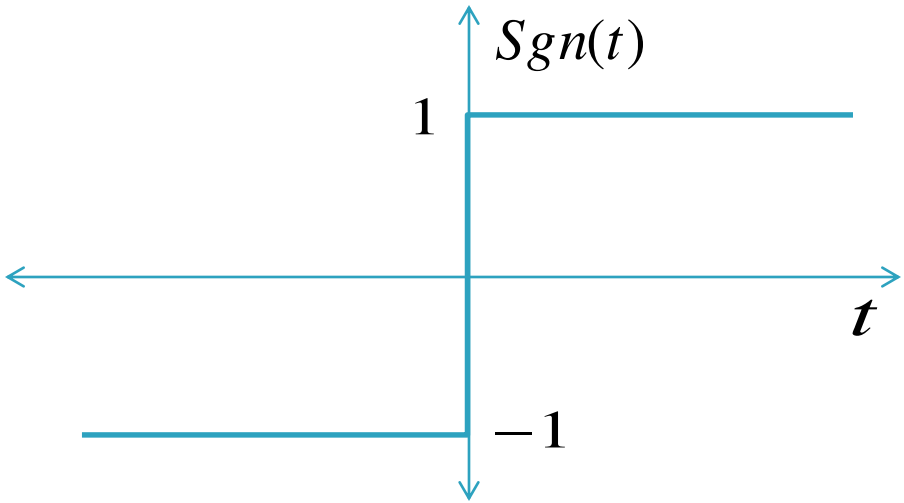

Fig (c)

# 1) The Unit Step function

B. The Signum Signal:

- We define the Signum Signal as:

$$Sgn(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$$

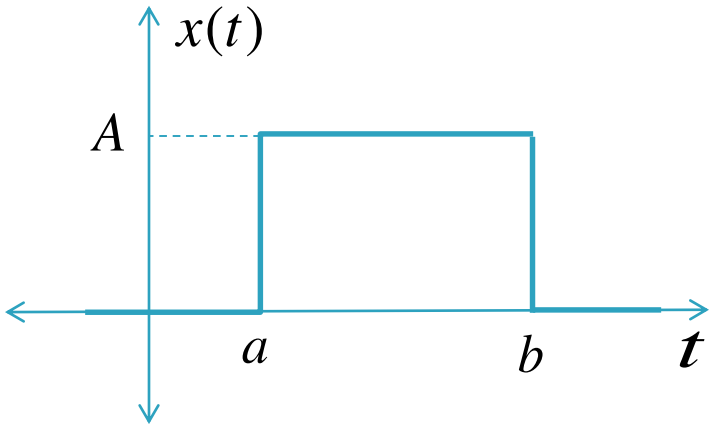



- And we can represent $Sgn(t)$ as:

$$Sgn(t) = u(t) - u(-t)$$

# Example 1:

- Represent the following graph using the:
  i. $rect$ signal.
  ii. The unit step $u(t)$ signal.



# Exercise 1:

- Sketch the following signals:

1) $x_1(t) = u(t) + 5u(t-1) - 2u(t-2)$ .
2) $x_2(t) = u(t).u(t-a), \quad a > 0$ .
3) $x_3(t) = u(t)u(a-t), \quad a > 0.$

# Ramp signal (Function)

- The ramp function is a unary real function, whose graph is shaped like a ramp. It can be expressed by numerous definitions, for example "0 for negative inputs, output equals input for non-negative inputs". The term "ramp" can also be used for other functions obtained by scaling and shifting, the *unit* ramp function (slope 1, starting at 0).
- In mathematics, the ramp function is also known as the positive part.
- Its applications : In machine learning, it is commonly known as a rectifier (neural networks) or ReLU (rectified linear unit) activation function  activation function or a rectifier in analogy to half-wave rectification in electrical engineering.
  In statistics (when used as a likelihood function) it is known as a tobit model.

## 2) The Ramp Signal

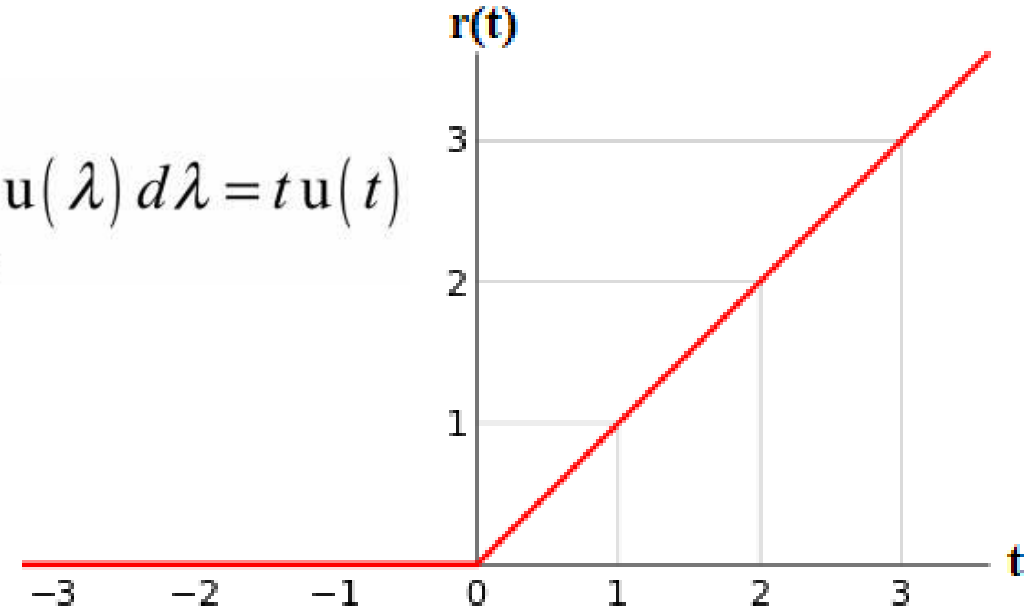
The ramp signal $r(t)$ is defined as a causal signal:

( **causal signal** is **any signal that is zero prior to time zero**. Thus, if $x(t)$ denotes the signal amplitude at time (sample) n, the signal x is said to be causal if $x(t) = 0$ for all $t < 0$)

$$r(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$\text{ramp}(t) = \left\{ \begin{matrix} t & , & t > 0 \\ 0 & , & t \leq 0 \end{matrix} \right\} = \int_{-\infty}^{t} \text{u}(\lambda)\, d\lambda = t\,\text{u}(t)$$

- Graphically:



# Properties of the ramp signal:

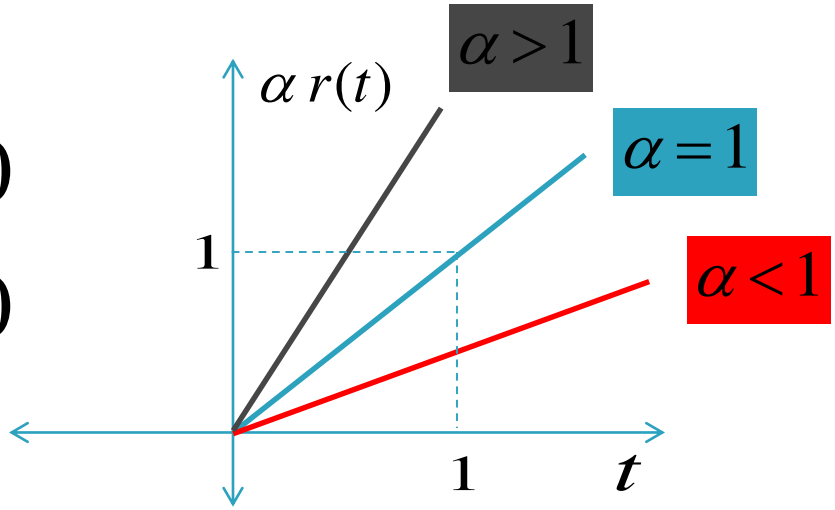
1) $\frac{dr(t)}{dt} = u(t)$ .

2) $r(t) = \int_{-\infty}^{t} u(\tau).d\tau$ .

   - The device that accomplishes this operation is called an integrator.

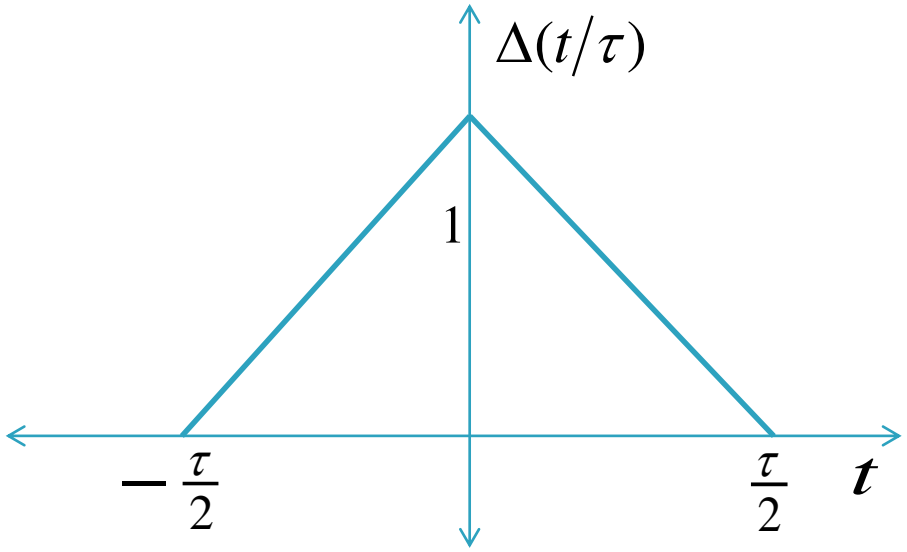
3) $\alpha\, r(t)$ results in changing the slope of $\alpha$ where is the slope.

$$\alpha\, r(t) = \begin{cases} \alpha t & t \geq 0 \\ 0 & t < 0 \end{cases}$$



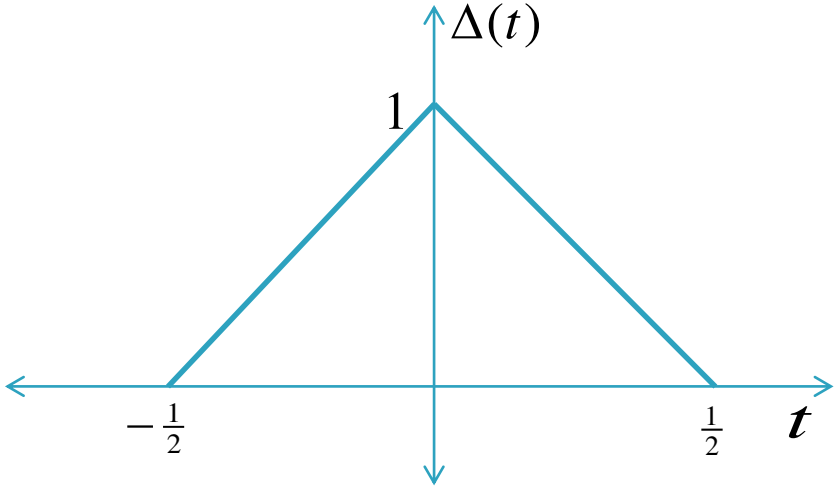
# Properties of the ramp signal:

- The general form of the triangular function is:



# Properties of the ramp signal:

4) From the ramp signal we can represent other types of signals, such as triangular function $\Delta(t)$

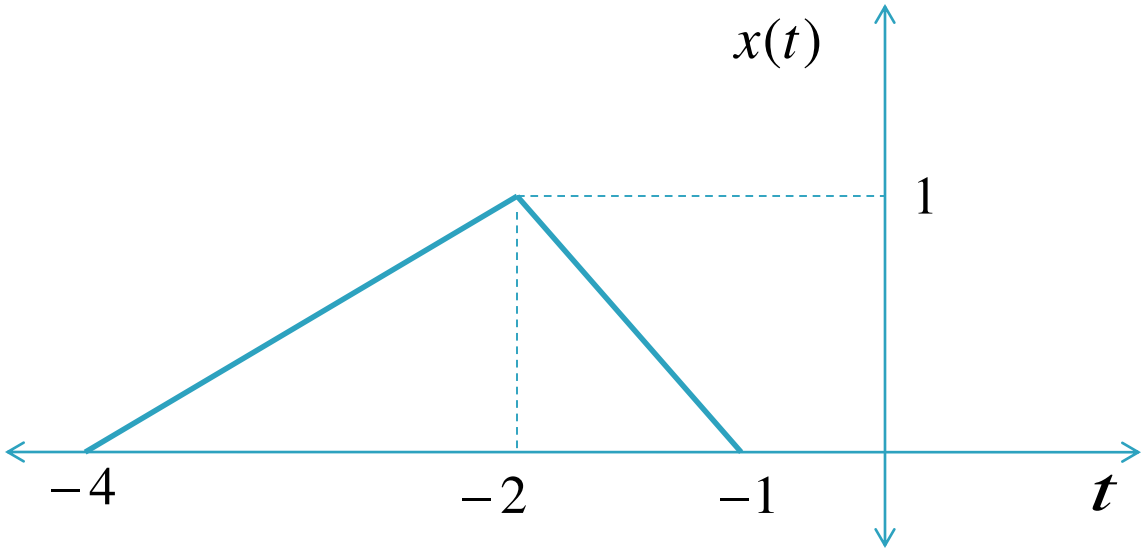



- using the ramp signal:

$$\Delta(t) = 2r(t+0.5) - 4r(t) + 2r(t-0.5)$$

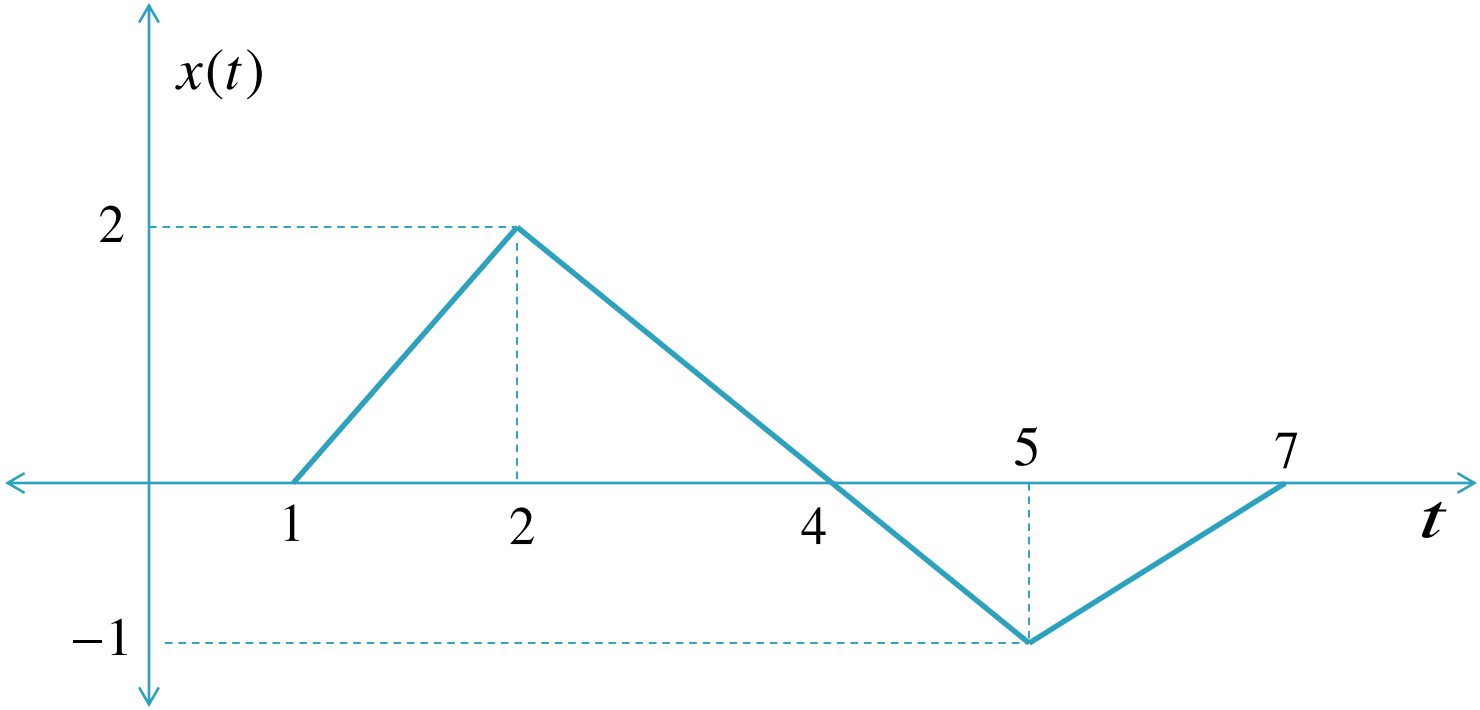
# Exercise 1:

- Express the following function using the Ramp function.



# Exercise 2:HW

- Express the following figure using the ramp signal.

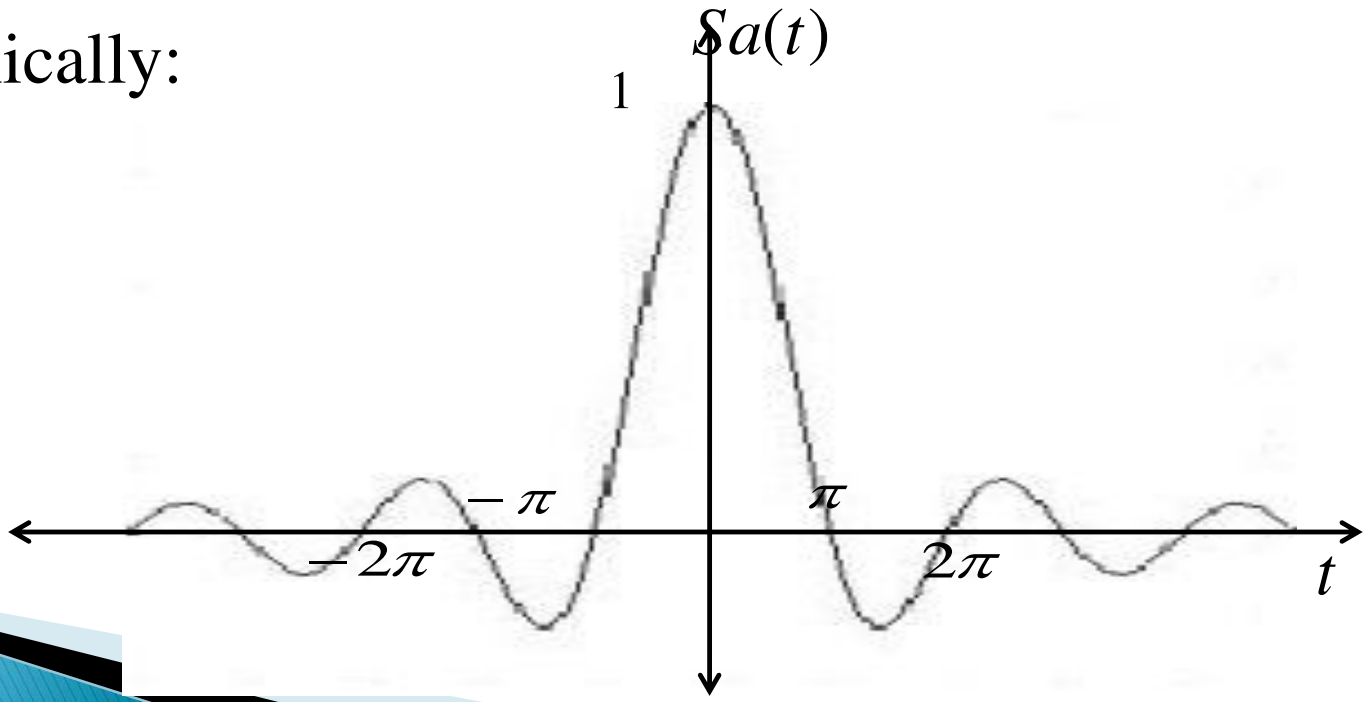


# 3) The Sampling & Sinc Signals

A. The Sampling Signal $Sa(t)$ is defined as:

$$Sa(t) = \begin{cases} 1 & t = 0 \\ \dfrac{\sin(t)}{t} & otherwise \end{cases}$$

- Graphically:



# Properties of the *Sa(t)*:

1. The zero crossing ($Sa(t) = 0$) occurs at:

$$t = \pm k\pi \quad where\ k = 1, 2, 3, \ldots$$

2. The area under the curve of $Sa(t)$ is equal to $\pi$. That means:

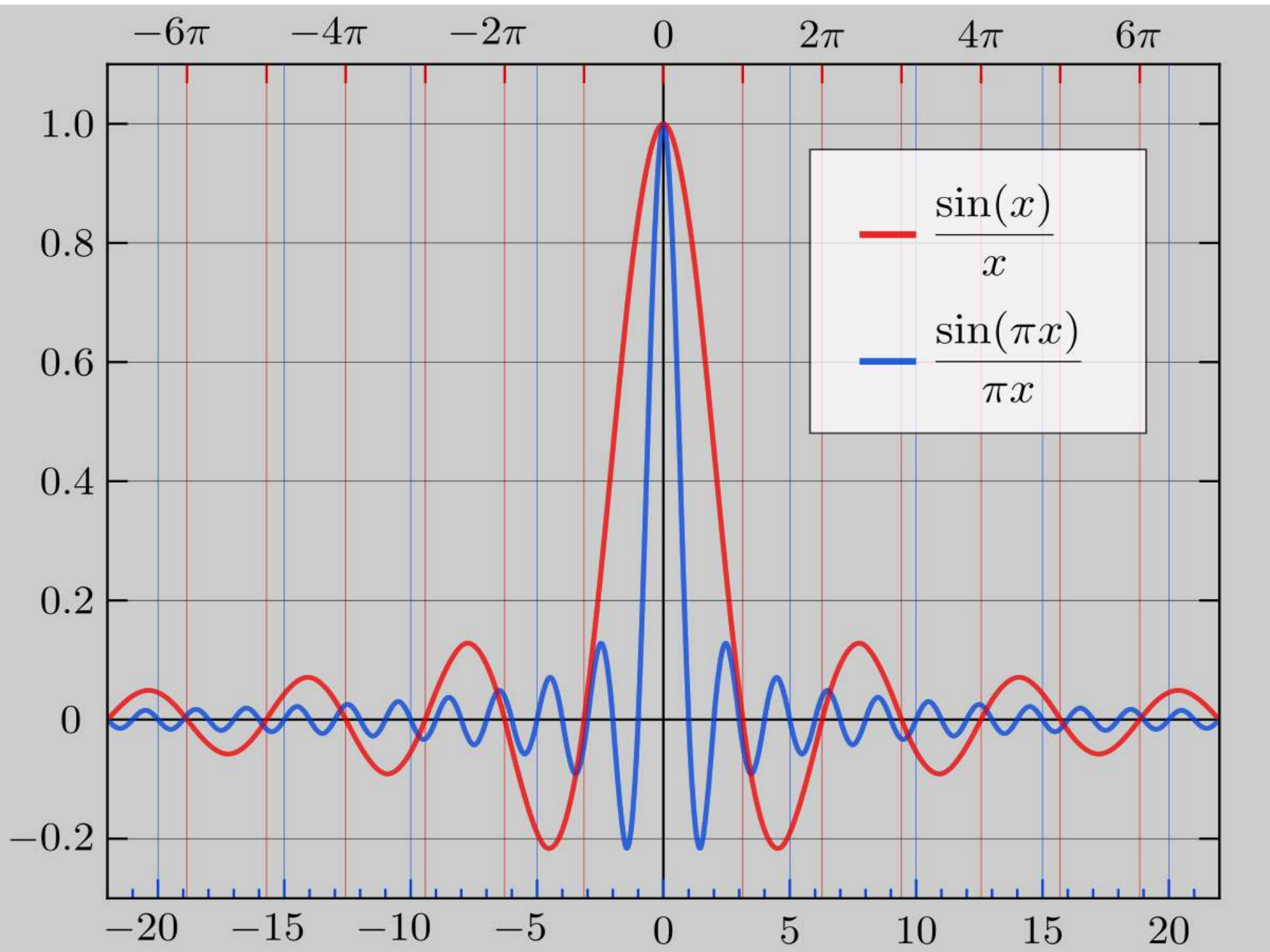
$$\int_{-\infty}^{\infty} Sa(t).dt = \pi$$

# The Sinc Signal:

B. The Sinc signal $Sinc(t)$ is defined:

$$\text{sinc(t)} = \begin{cases} 1 & t = 0 \\ \dfrac{\sin(\pi t)}{\pi t} & otherwise \end{cases}$$

- The only difference between the two definitions ( $Sa(t)$ & $Sinc(t)$ ) is the scaling of the variable t (the x-axis) by a factor .
- That means, $sinc(t)$ is a compressed version of $\pi$ .
- The following figure illustrates the difference between the two mentioned signals.


−6π
−4π
−2π
0
2π
4π
6π
1.0
0.8
0.6
0.4
0.2
0
−0.2
−20
−15
−10
−5
0
5
10
15
20
$\frac{\sin(x)}{x}$
$\frac{\sin(\pi x)}{\pi x}$

# The Sinc Signal:

- $Sinc(t)$ is the normalized $Sa(t)$, the normalization course the definite integral of the function over the real numbers to equal "1" (whereas the same integral of the unnormalized *Sinc* function has a value of $\pi$ ).
- As a further useful property, all of the zeros of the *Sinc* function are integer value *t*.

$$sinc(t) = 0 \ at \ t = \pm 1, \pm 2, \pm 3 ... etc.$$

# The Sinc Signal:

- The Sinc function is the Fourier transform (F.T) of the rectangular function which has many interests in the telecommunication world.
- The Sinc function is fundamental in the concept of reconstructing the original continuous bandlimited signal from uniformly spaced samples of that signal.
- The Sinc signal is commonly used in DSP applications.

# Exercise 3:

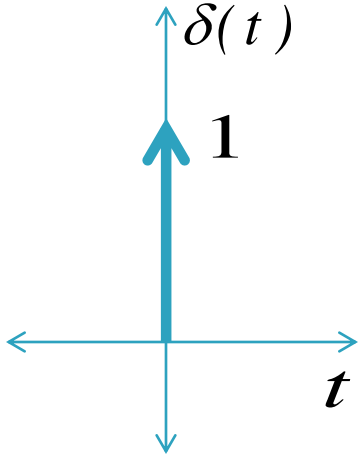
- Draw $Sinc(\frac{t}{4})$ and determine the first five zero crossing points of the negative and the positive side.

# 4) The unit impulse (Dirac Delta or Delta):

- The unit impulse $\delta(t)$ is defined as:

- $$\delta(t) = \begin{cases} 1 & , t = 0 \\ 0 & , otherwise \end{cases}$$

- Graphically:



# 4) The unit impulse (Dirac Delta or Delta):

- The $\delta(t)$ function satisfy:

I. The peak occurs at t=0.
II. Their duration is short, so as $\tau \to 0$. Where $\tau$ is the width of the signal.
III. The total area under each function equal to *one*.
IV. All are even function.

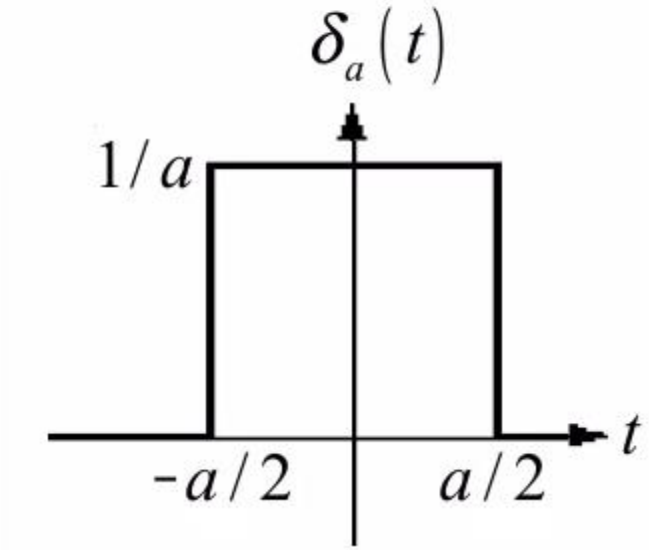
# 4) The unit impulse (Dirac Delta or Delta):

- We can approximate $\delta(t)$ by using the following function:

The total area is equal to one.

$$\int_{-\infty}^{\infty} \delta(t).dt = 1$$

Rectangular pulse, $\delta_a(t) = \begin{cases} 1/a & , \ |t| < a/2 \\ 0 & , \ |t| > a/2 \end{cases}$

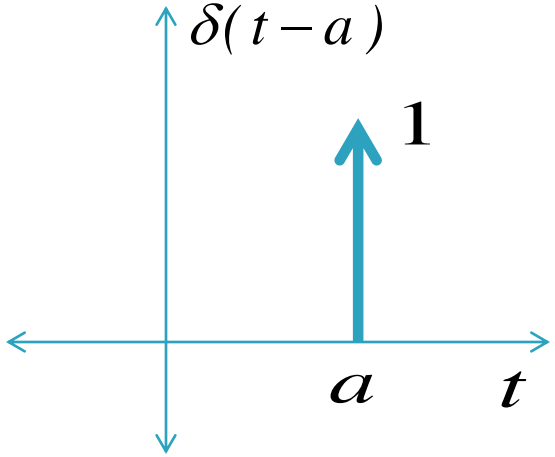


# 4) The unit impulse (Dirac Delta or Delta):

- The shifted delta function is defined:

$$\delta(t-a)=\begin{cases}1 & ,t=a\\ 0\ ,otherwise\end{cases}$$

- Graphically:



# Applications of δ(t):

- The delta function, occupies a central place in signal analysis. Many physical phenomena such as point sources, point charges, concentrated loads on structures, and *voltage or current sources acting for very short times can be modeled as delta functions*.

# Properties of δ(t):

1. $$\int_a^b x(t).\delta(t).dt = x(0) \cdot \quad where \quad a < 0 < b$$

Example: $x(t) = t^2 + 2t + 5$ so:

$$x(0) = 5 \quad and \quad 0 \in [-2,5]$$

$$\therefore \quad \int_{-2}^{5} x(t).\delta(t).dt = x(0) = 5$$

# Properties of δ(t):

2. $$\int_a^b x(t).\delta(t-t_o).dt = x(t_o) \quad for \quad a < t_o < b$$

And zero for $t_o \notin (a,b)$

Which commonly known the *shifting property*.

e.g.: if $x(t) = t^2 + 2t + 5$ , find:

$$\int_{-4}^{4} x(t).\delta(t+3).dt$$

First we find x(-3) then apply the property:

$$x(-3) = 8 \quad and \quad -3 \in [-4, 4]$$

$$\therefore \quad \int_{-4}^{4} x(t).\delta(t+3).dt = x(-3) = 8$$

# Properties of δ(t):

3. $\int_a^b x(t-t_o).\delta(t).dt = x(-t_o) \quad where \quad a \lesssim -t_o < b$

e.g.: if $x(t) = t^2 + 2t + 5$ , find:

$$\int_{-1}^{3} x(t+2).\delta(t).dt$$

First we find x(2) then apply the property:

$$x(2) = 13 \quad and \quad 2 \in [-1, 3]$$

$$\therefore \quad \int_{-1}^{3} x(t+2).\delta(t).dt = x(2) = 13$$

# Properties of δ(t):

4. $$\int_{-\infty}^{\infty} \delta(t).dt = 1 \ .$$

This leads to:

$$u(t) = \int_{-\infty}^{t} \delta(t).dt$$

$$\Rightarrow \frac{du(t)}{dt} = \delta(t)$$

# Properties of δ(t):

5. $x(t).\delta(t-t_o) = x(t_o).\delta(t.-t_o)$

Which commonly known the *sampling theorem*.

e.g.: if $x(t) = t^2 + 2t + 5$ , find:

$$x(t).\delta(t-1)$$

Sol:

$$x(1).\delta(t-1) = 8.\delta(t-1)$$

# Properties of δ(t):

6. $\delta(t) = \delta(-t)$

That means the delta function is even function.

# Properties of δ(t):

7. $$\int_{-\infty}^{\infty} \delta(at - t_o).dt = \frac{1}{|a|} \int_{-\infty}^{\infty} \delta(t - \frac{t_o}{a}).dt$$

e.g.: if $x(t) = t^2 + 2t + 5$ , find:

$$\int_{-5}^{5} x(t).\delta(2t - 8).dt$$

Sol: $$\int_{-5}^{5} x(t).\delta(2t - 8).dt = \frac{1}{2} \int_{-5}^{5} x(t).\delta(t - 4).dt = \frac{1}{2} x(4)$$

# Properties of δ(t):

8. $\int_a^b x(t).\delta'(t-t_o).dt = -x'(t_o)$

e.g.: if $x(t) = t^2 + 2t + 5$, find:

$$\int_{-5}^{5} x(t).\delta'(t-3).dt$$

Sol: $\int_{-5}^{5} x(t).\delta'(t-3).dt = -x'(t_o) \;\; where\; t_o = 3$

$$x'(t) = 2t + 2 \Rightarrow x'(3) = 8$$

$$\int_{-5}^{5} x(t).\delta'(t-3).dt = -x'(3) = -8$$

# Properties of δ(t):

9. $$\int_{-\infty}^{\infty} \delta'(at - t_o).dt = \frac{1}{|a|} \int_{-\infty}^{\infty} \delta'(t - \frac{t_o}{a}).dt$$ .

e.g:

$$\int_{-\infty}^{\infty} \delta'(2t - 3).dt = \frac{1}{2} \int_{-\infty}^{\infty} \delta'(t - \frac{t_o}{2}).dt$$

# Properties of δ(t):

10. $\int_a^b x(t).\delta^{(n)}(t-t_o).dt = (-1)^n x^{(n)}(t_o)$.

e.g: if $x(t) = t^2 + 2t + 5$ , find:

$$\int_{-5}^{5} x(t).\delta''(t-3).dt$$

Sol: $\int_{-5}^{5} x(t).\delta''(t-3).dt = (-1)^2.x''(t_o) \;\; where\; t_o = 3$

$$x'(t) = 2t + 2 \Rightarrow x''(t) = 2 \Rightarrow x''(3) = 2$$

$$\int_{-5}^{5} x(t).\delta''(t-3).dt = 2$$

# Example 3:

Evaluate:

I. $\int_0^3 e^{(t-2)}.\delta(2t-4).dt$

II. $\int_{-4}^{4}(t-2)^2.\delta'(-\frac{1}{3}t+\frac{1}{2}).dt$

III. signals and systems second ed. haykin .

# Example 3:

III. $\int_{0}^{\infty} \sqrt{t+1}.e^{(1-t)}.\delta(t+2).dt$ .

IV. $\int_{-\infty}^{\infty} \tau.u(\tau+1).u(2-\tau).d\tau$ .

# Exercise 4:

Evaluate:

I. $\int_{-\infty}^{\infty} e^{-\alpha}.\delta(3-5\frac{\alpha}{2}).d\alpha$ .

II. $\delta(2+\frac{2\tau}{\pi}).sin(\frac{\tau^2-2}{\pi})$ .

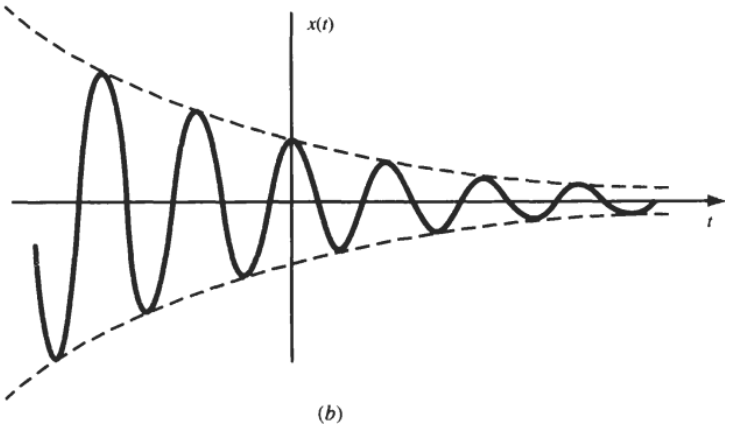
# Exercise 4:

III. $\int_{-\infty}^{\infty} t^3 . \delta^{(2)}(t+2).dt$ .

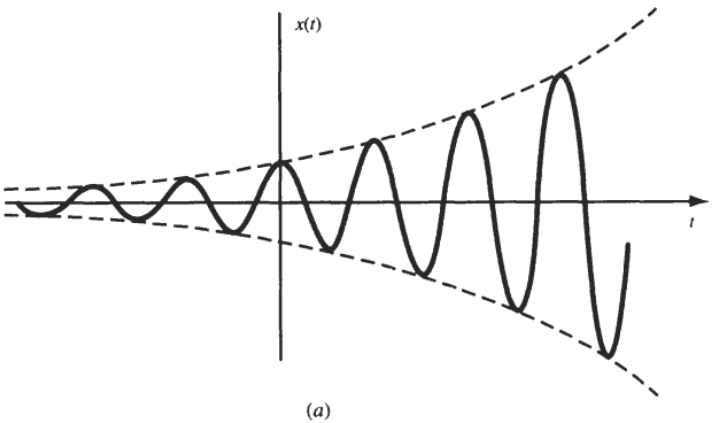
IV. $\int_{-\infty}^{\infty} sin(\frac{\tau}{4}).cos(\frac{\tau^2}{\pi}).\delta(1+\frac{\tau}{\pi}).d\tau$ .

# Complex Exponential Signals:

- The *complex exponential* signal
- $x(t) = e^{j\omega_o t}$




**( b )Exponentially decreasing sinusoidal signal.**




**(a) Exponentially increasing sinusoidal signal;**

- Using Euler's formula, this signal can be defined as

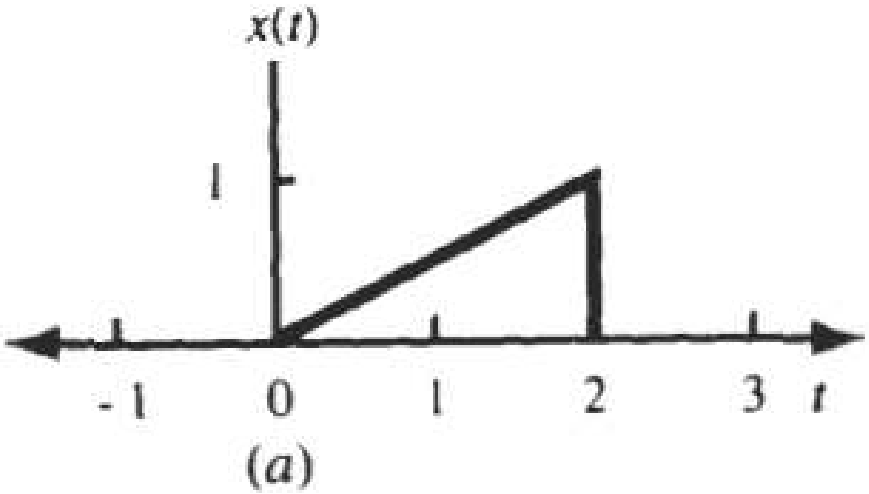
$$x(t) = e^{j\omega_o t} = \cos(\omega_o t) + j\sin(\omega_o t)$$

The fundamental period $T_O$ of $x\ (\ t\ )$ is given by

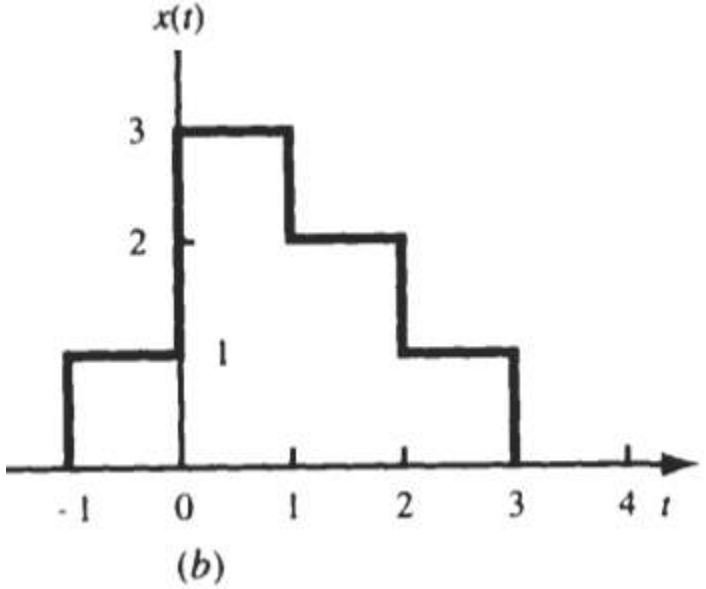
$$T_O = \frac{2\pi}{\omega_0}$$

# Homework

- Express the signals shown in the following figure in terms of unit step functions.



(a)



(b)

# Continuous-Time Systems

2.2 Classification of continuous time systems

# Introduction

- Every physical system is broadly characterized by its ability to accept an input such as Voltage, Current, pressure, force, displacement,..etc, and to produce an output in response to this input.

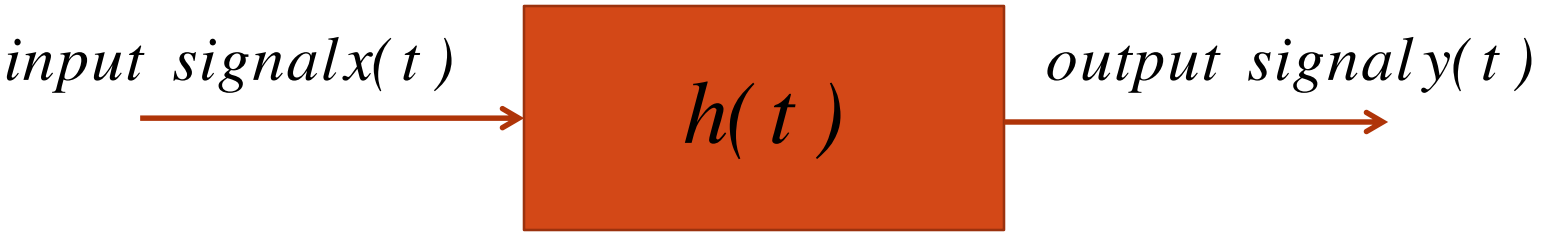
For example:

1. Electrical Filter whose input signal corrupted by noise and interference and whose output signal is the desired signal.
2. A Robot is a system whose input is an electric control signal and whose output is a motion or action.
3. Radar receiver is an electronic system whose input is the reflection of an electromagnetic signal from the target and whose output is a Video signal displayed on the radar screen.

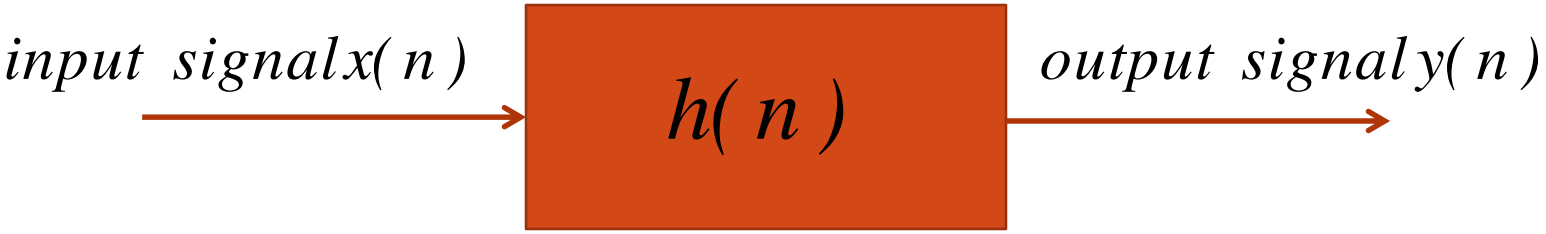
# Introduction

- Systems can be described by one of the following:

1. Continuous-Time Systems.




2. Discrete-Time Systems.

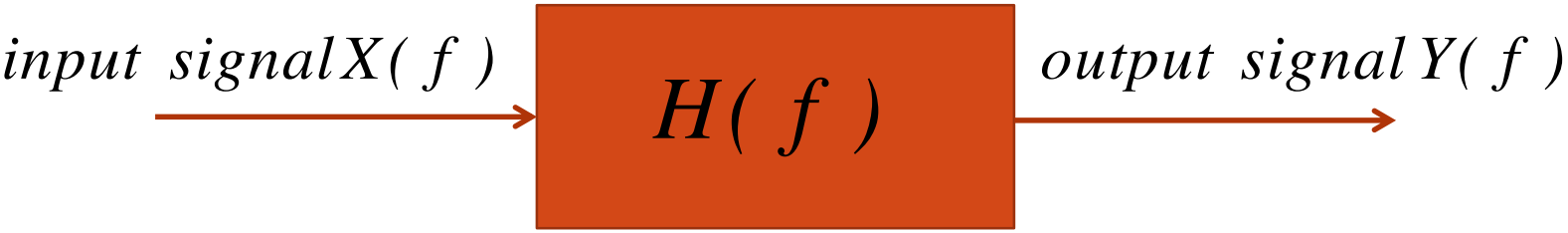


# Introduction

3. S-Domain systems.




4. Frequency Domain Systems.



# Introduction

5. Z- Domain systems.



# Introduction

Note:

A. We are interested in both continuous- time and discrete-time systems.

B. A continuous time system is a system in which continuous-time input signals are transformed into continuous- time output signals.

C. A discrete- time system is a system that transforms discrete- time inputs into discrete- time outputs.

# Introduction

D. S- Domain, frequency Domain, and z- Domain systems are not different systems than the continuous systems or discrete systems. But they described in different domain rather than time to met different fields of interest.

E. For example S- Domain is important for control systems, frequency Domain is used in the telecommunication world.

# Objectives:

- After completing this session, you will be able to classify continuous- time systems with the following classes:
  1. Linear/ non- Linear systems.
  2. Time variant/ Time- invariant systems.
  3. Causal/ Anti-causal systems.
  4. Memory/ memory less systems.
  5. Stable/ unstable systems.

# Classification of Continuous- time systems:

- In this section we are going to illustrate the concept of systems by discussing their classification according to the way the system interacts with the input signal.
- As mentioned, this interaction, which defines the model for the system, can be linear or non-linear, time invariant or time varying, Causal or Non- Causal, memory or memory less, and stable or unstable.

# 1) Linear and Nonlinear systems:

- To make the distinction between linearity and nonlinearity a bit more precise, recall that a mathematical equation can be thought of as a function — something that maps inputs to outputs. The equation $y = x$, for instance, is equivalent to a function that takes as its input a value for x and produces as its output a value for y. The same is true of $y = x^2$.

# 1) Linear and Nonlinear systems:

- While linear functions are easy enough to define, the term "nonlinear" takes in everything else.
- It's impossible to build a theory of nonlinear systems, because arbitrary things can satisfy that definition. Because linear equations are so much easier to solve than nonlinear ones, much research across a range of disciplines is devoted to finding linear approximations of nonlinear phenomena.

# 1) Linear and Nonlinear systems:

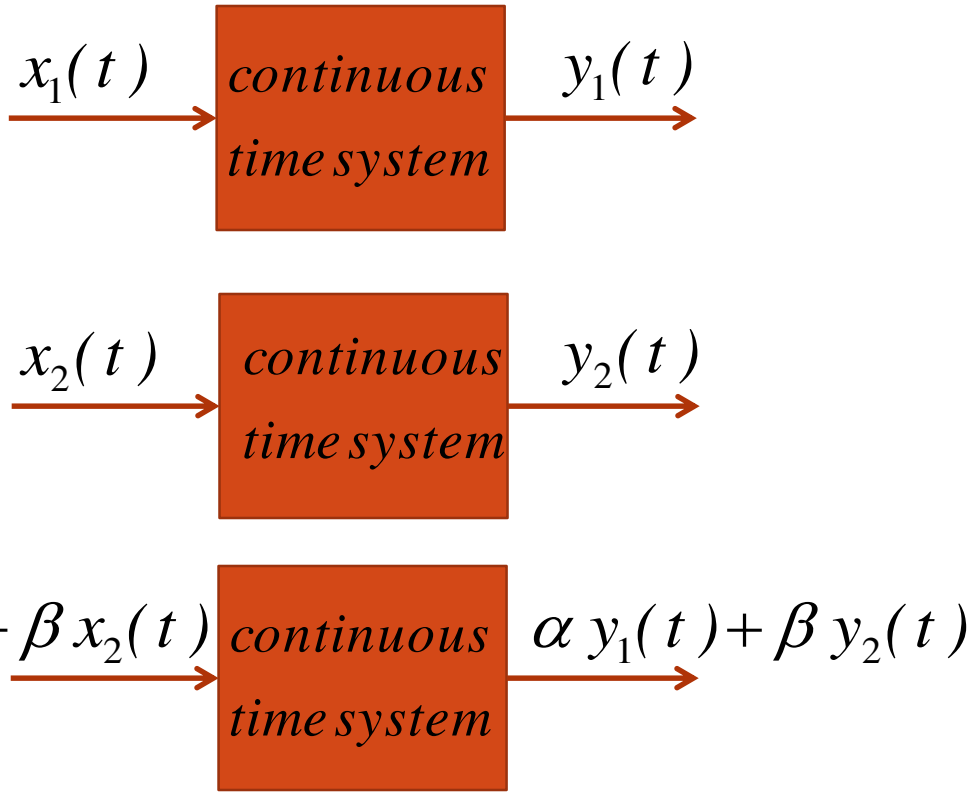
- In other words, In a Linear system the superposition principle is applicable.
- To check the linearity of a system apply the following steps:

1. Apply $x_1(t)$ as an input signal to the system to get $y_1(t)$ as an output signal.
2. Apply $x_2(t)$ as an input signal to get $y_2(t)$ as an output signal.
3. Then apply $\alpha x_1(t) + \beta x_2(t)$ as an input signal to the same system.

# 1) Linear and Nonlinear systems:

4. If the output signal by applying the latest signal is **equal to** the output of applying $\alpha\, x_1(t)$ and $\beta\, x_2(t)$ individually. Then the system is *linear system*.

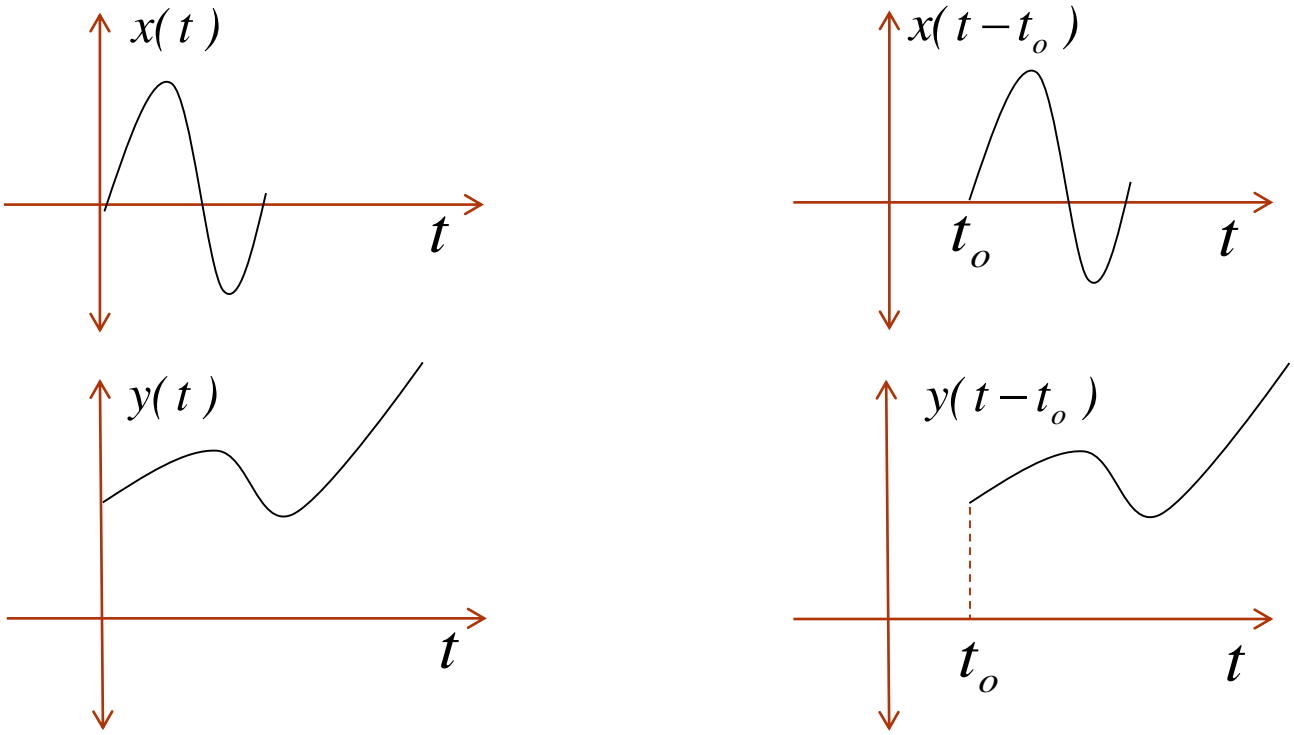
e.g:



# 2) Time Variant/ Invariant systems:

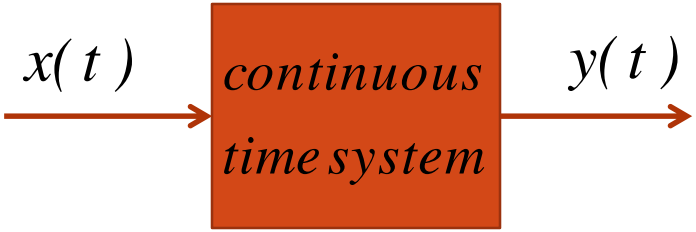
- Time Invariant means that a shift in time of the input signal causes an identical shift in time in the output signal.

e.g.



# 2) Time Variant/ Invariant systems:

- In other words, A time-invariant system is one whose output does not depend explicitly on time.
- To verify time variant/ invariant system as shown in the block diagram below, follow the steps down:




1. Apply the input signal $x_1(t)$ to get the output signal $y_1(t)$.
2. Shift $y_1(t)$ by $t_o$ to get $y_1(t-t_o)$.

# 2) Time Variant/ Invariant systems:

3. Apply the input signal $x_2(t)$ to get the output signal $y_2(t)$.
4. Now replace $x_2(t)$ by $x_1(t-t_o)$ in $y_2(t)$.
5. If $y_2(t)=y_1(t-t_o)$.

# 3) Causal/ Non-Causal system:

- A causal system is a system where the output depends on past and current inputs but not future inputs i.e. the output $y(t_o)$ only depends on the input $x(t)$ for values of $t \leq t_o$
- The causality of systems also plays an important role in digital signal processing, where filters are often constructed so that they are causal.

e.g. Causal Filters

- For a causal system, the impulse response of the system must be 0 for all t<0. That is the sole necessary as well as sufficient condition for causality of a system, linear or non-linear.

# 4) Memory/ Memory less system:

A systems output or response at any instant t generally depends upon the entire past input. However, there are systems for which the output at any instant t depends only on its input at that instant and not on any other input at any other time. Such systems are said to have no memory or is called memoryless.

The only input contributing to the output of the system occurs at the same time as the output. The system has no stored information of any past inputs thus the term memoryless. Such systems are also called static or instantaneous systems. Otherwise, the system is said to be dynamic (or a system with memory). Instantaneous systems are a special case of dynamic systems.

Note that all memoryless systems are causal, but not vice versa.

# 5) Stable/ unstable systems:

A system is stable if a bounded input signal yields a bounded output signal.

A signal is said bounded if its absolute value is less than some finite value for all

time,

$$|x(t)| < \infty, \qquad -\infty < t < \infty.$$

A system for which the output signal is bounded when the input signal is

bounded is called bounded-input-bounded-output (BIBO) stable system.

# LINEAR TIME-INVERIANT SYSTEMS

The Convolution Integral

# Introduction

- Every physical system is broadly characterized by its ability to accept an input such as Voltage, Current, pressure, force, displacement,..etc, and to produce an output in response to this input.

For example:

1. Electrical Filter whose input signal corrupted by noise and interference and whose output signal is the desired signal.
2. A Robot is a system whose input is an electric control signal and whose output is a motion or action.
3. Radar receiver is an electronic system whose input is the reflection of an electromagnetic signal from the target and whose output is a Video signal displayed on the radar screen.

# Introduction

- Linear Time- Invariant systems (LTI), play a fundamental role in signal and system analysis because of the many physical phenomena that can be *modeled* by Linear time- Invariant systems and because a mathematical analysis of the behavior of such systems can be carried out in a fairly straightforward manner.
- The behavior of a linear, continuous- time, time- Invariant system with input signal $x(t)$ and output signal $y(t)$ can be described by the *convolution* integral:



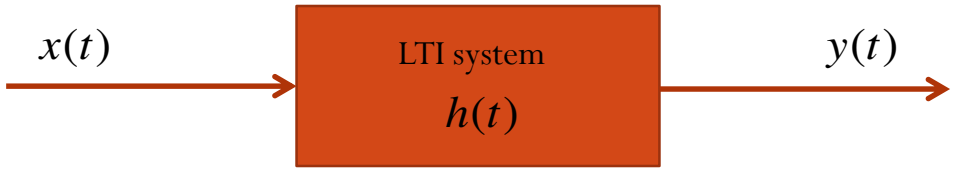
# Objectives

After completing this session, you will be able to:

- To find the response of a system (output signal) if an input signal applied to a continuous time- Invariant system.
- To understand the meaning of convolution integral.
- To know the meaning of impulse response of a system.
- To know how to perform the convolution integral (In time domain) graphically.
- To understand the properties of convolution integration.

# The Convolution Integral (Time domain representation)

- If $x(t)$ an input to LTI system $h(t)$, the output signal $y(t)$ can be evaluated by the convolution process.




- The operation of the convolution is represented symbolically as:

$$y(t) = x(t) * h(t)$$

Convolution symbol

- The convolution integral is:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau).d\tau$$

Reflected & shifted version of x(t)

Where $h(t)$ assumed known, is the response of the system to a unit impulse input $\delta(t)$.

# The Convolution Integral
## (Time domain representation)

- The term $x(t-\tau)$ is the reflected and shifted version of $x(\tau)$.
- The integration with respect to $\tau$ is performed to result $y(t)$.
- So, t is treated as a constant in the integration and it is represent the shifted value of the $x(\tau.)$

# Performing the convolution integral graphically:

- To perform the convolution integral graphically, follow the steps below:
  1. Change $t$ to $\tau$, and draw $x(\tau)$ and $h(\tau)$.
  2. Reflect $x(\tau)$ to produce $x(-\tau)$.
  3. Shift $x(-\tau)$ to the left by $t$ to produce $x(t-\tau)$.
  4. Integrate $h(\tau)x(t-\tau)$ as a function of $\tau$.

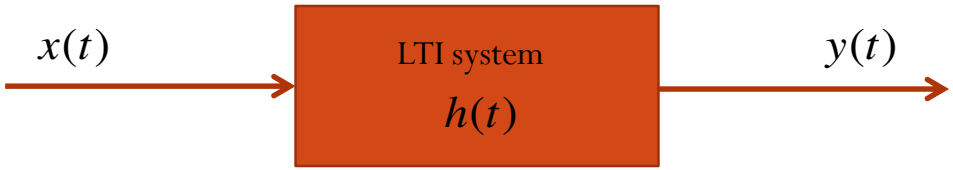
Note:

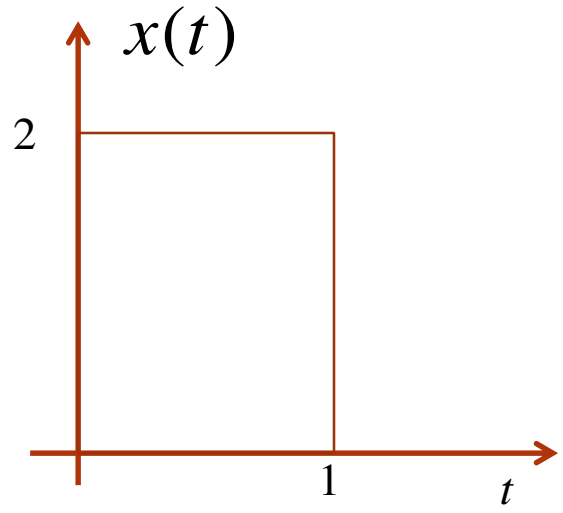
1. After integration with respect to $\tau$ is performed, result will be $y(t)$.
2. The over all result represent the over lap area of $h(t)$ and $x(t)$ when moving the reflected version of $x(t)$ from $-\infty$ to $\infty$.

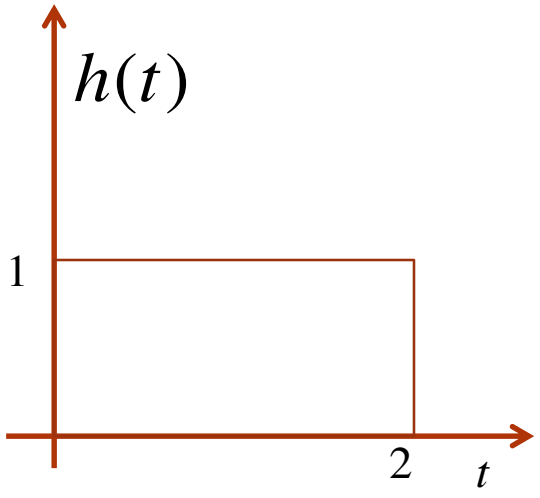
**But: Why we reflect the input signal when performing the convolution graphically?**

# Example1:

The impulse response and the input signal of LTI system is shown below, find the output signal *y(t)*.



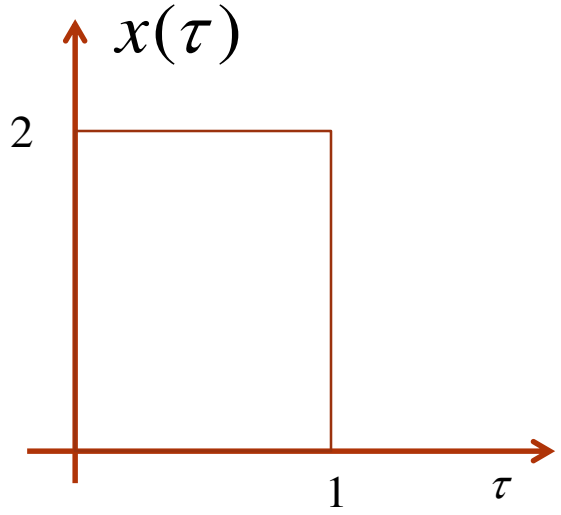
# Example1:
# Solution

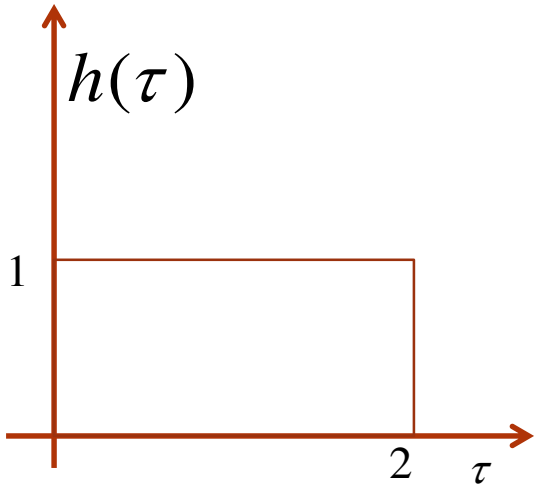
Because the system is LTI system, so y(t) can be obtained using the convolution integral.

$$y(t) = x(t) * h(t)$$

Follow the procedure to get that:

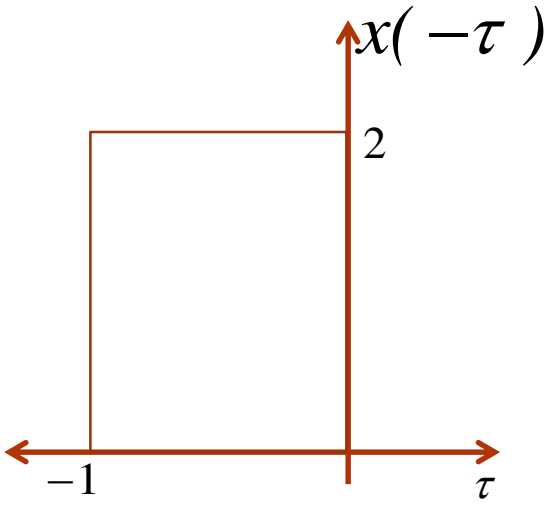
1) Change $t$ *to* $\tau$

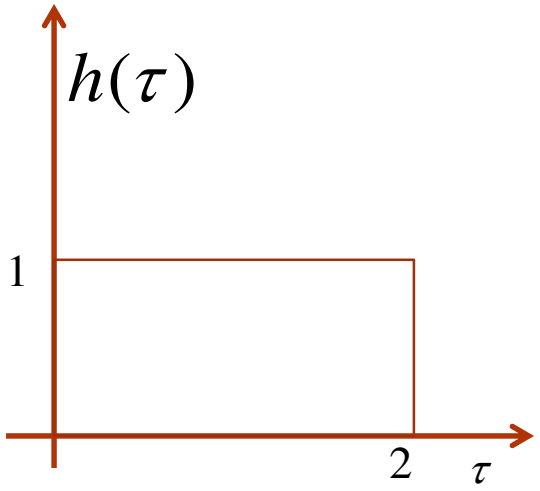


# Example1:
# Solution

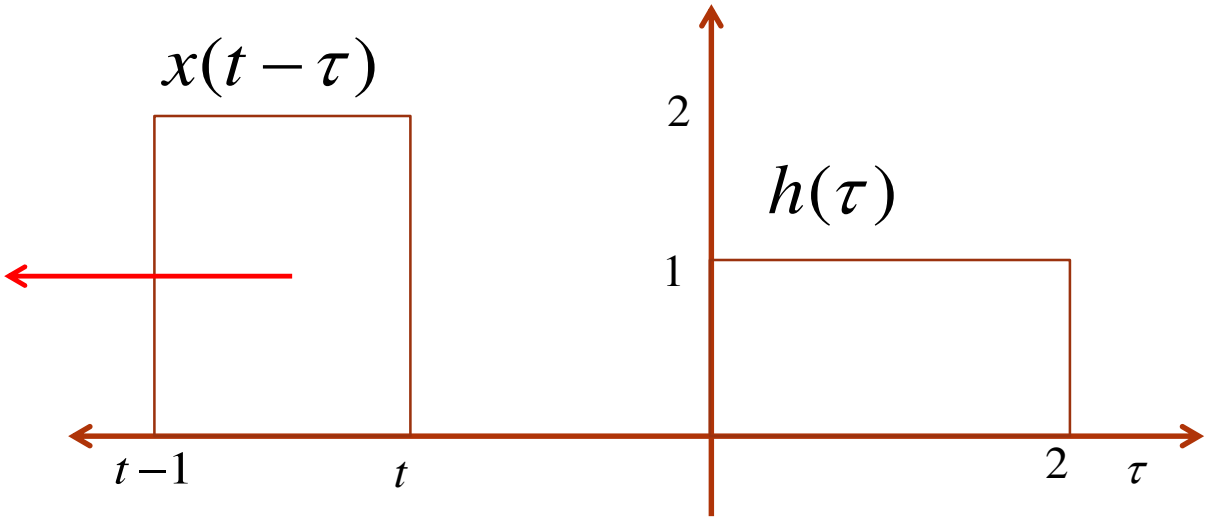
2) Reflect about the y-axis $x(\tau)$ to produce $x(-\tau)$.



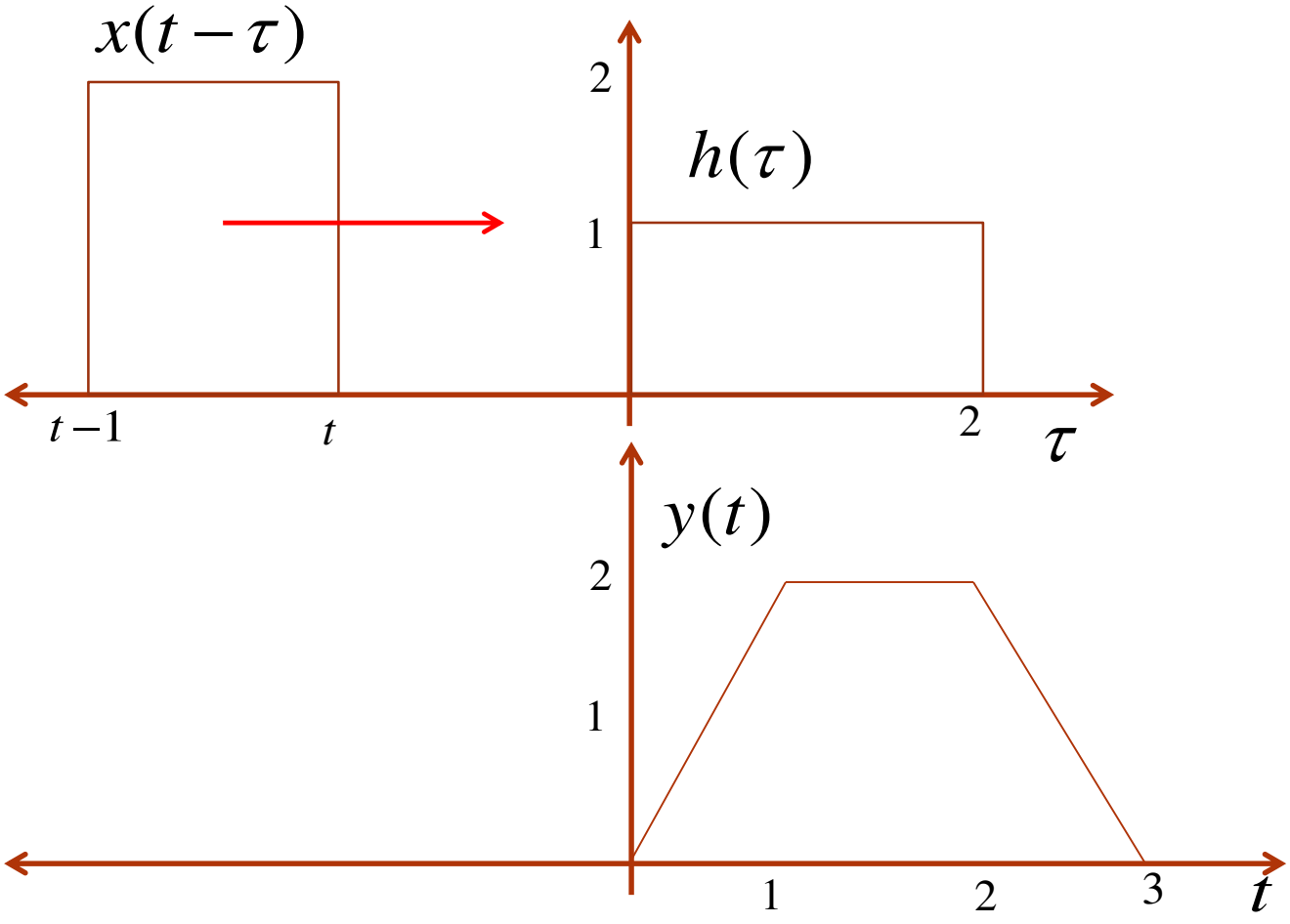
# Example1: Solution

3) Shift $x(-\tau)$ to the left by $t$ to produce $x(t-\tau)$. Draw $x$ & $h$ on the same *t-axis*. Shift $h(\tau)$ to the left until no overlap with $x(-\tau)$.



# Example1:

4) Integrate $h(\tau)x(t-\tau)$ as a function of $\tau$ . This can be done by moving $x(t-\tau)$ to the right gradually and do the integration for each case individually. The integration results represent the overlap area of $x(t-\tau)$ & $h(\tau)$ .



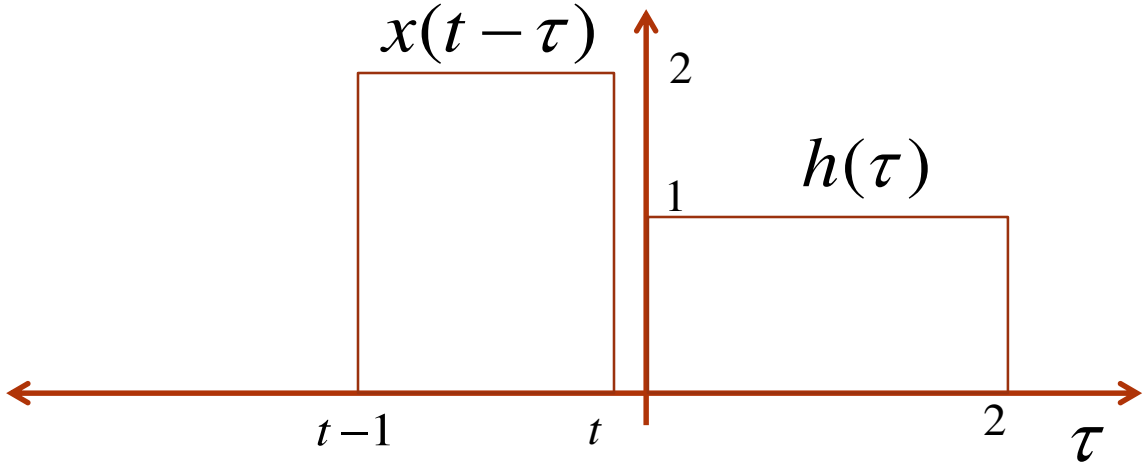
# Example1:

Let's now do the integration step by step to find *y(t)*:

Step 1: when No overlap, that's occur when *t* is less that 0.

$$t < 0$$

$$y(t) = 0$$

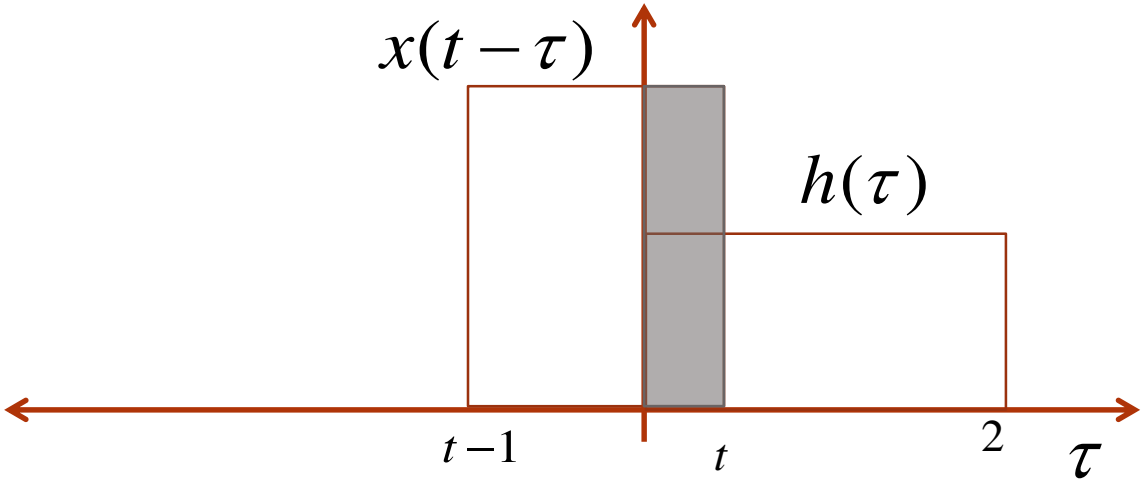


# Example1:

Step 2: when overlap start, that's occur after t>0. the area in this case increase until $t$ become larger than 1. To find the overlap area integrate from $0 \rightarrow t$.

$$y(t) = \int_0^t h(\tau).x(t-\tau).d\tau = \int_0^t (1).(2).d\tau$$

$$y(t) = 2\tau\Big|_0^t = 2t$$

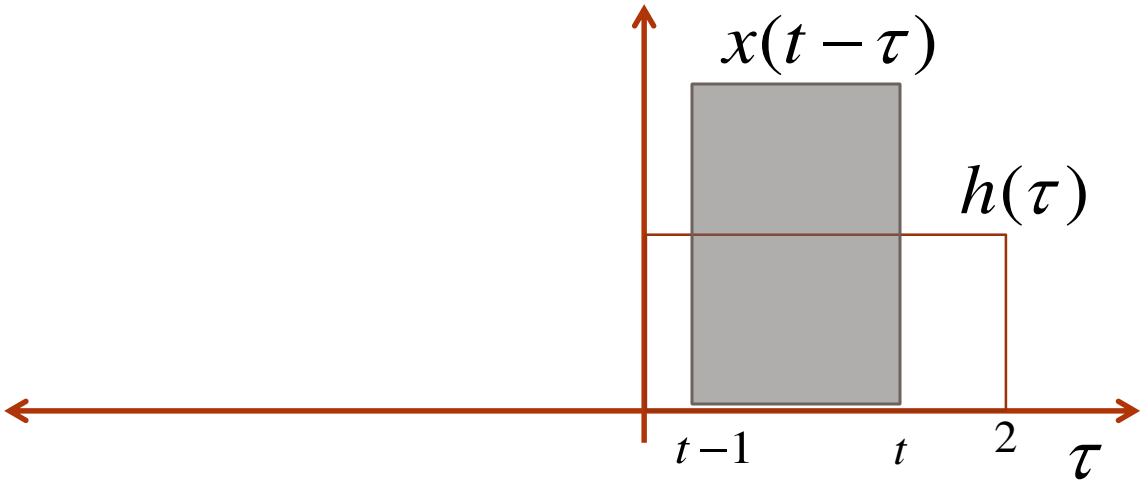


# Example1:

Step 3: when $1 \le t < 2$, the overlap area become fixed, To find the overlap area integrate from $(t-1) \to t$.

$$y(t) = \int_{t-1}^{t} h(\tau).x(t-\tau).d\tau = \int_{t-1}^{t} (1).(2).d\tau$$

$$y(t) = 2\tau\Big|_{t-1}^{t} = 2(t-(t-1)) = 2$$

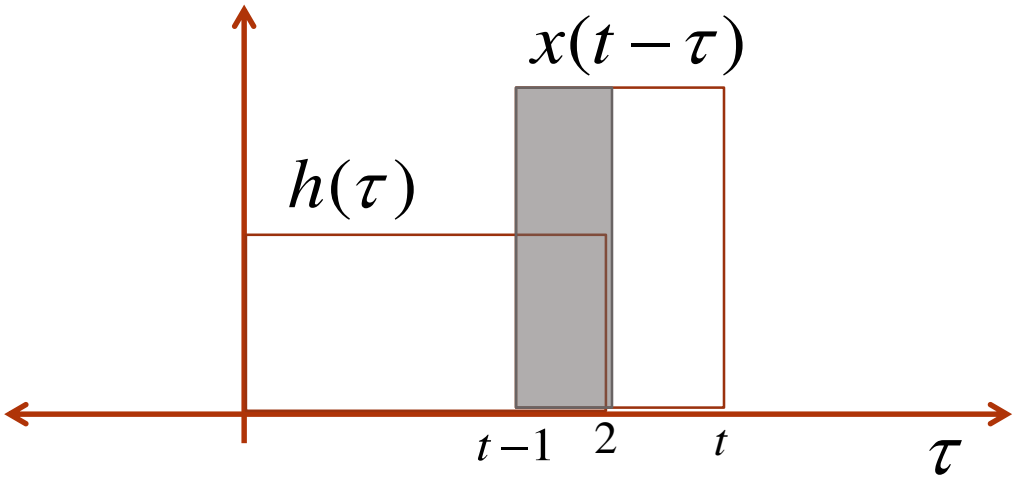


# Example1:

Step 4: when $2 \le t \le 3$, the overlap area decrease. In this case to find the overlap area integrate from $(t-1) \to 2$

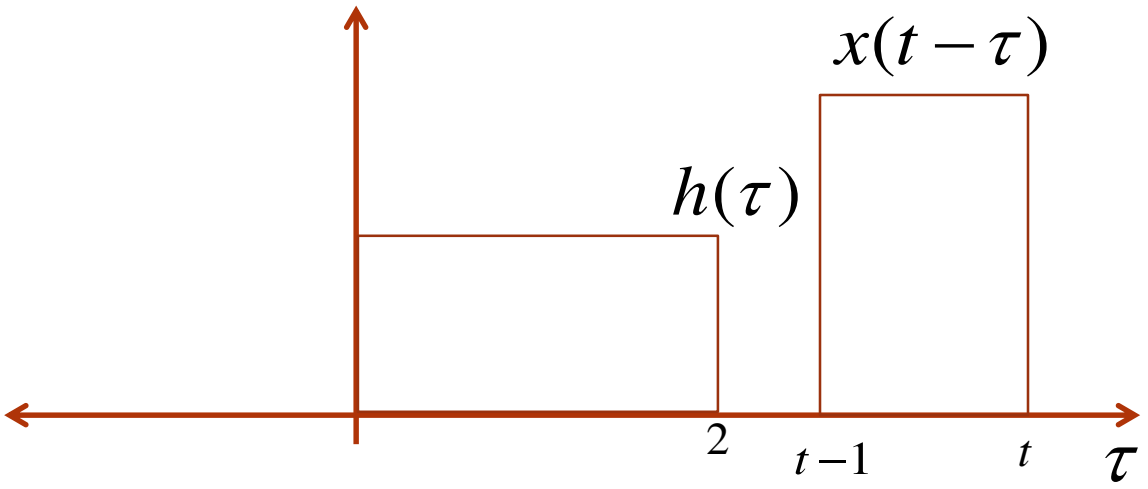
$$y(t) = \int_{t-1}^{2} h(\tau).x(t-\tau).d\tau = \int_{t-1}^{2} (1).(2).d\tau$$

$$y(t) = 2\tau\Big|_{t-1}^{2} = 2(2-(t-1)) = 6-2t$$



# Example1:

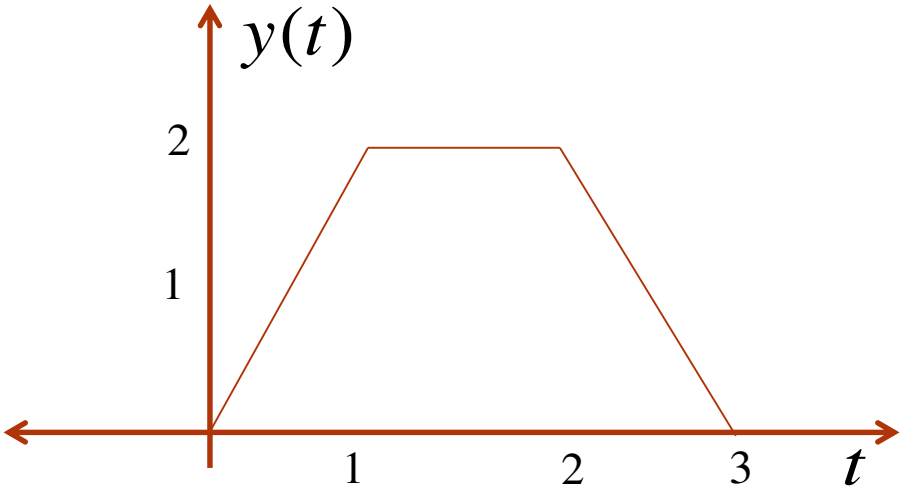
Step 5: when $t > 3$ , in this case $x(t-\tau)$ completely outside $h(\tau)$, resulting $y(t) = 0$.



# Example1:

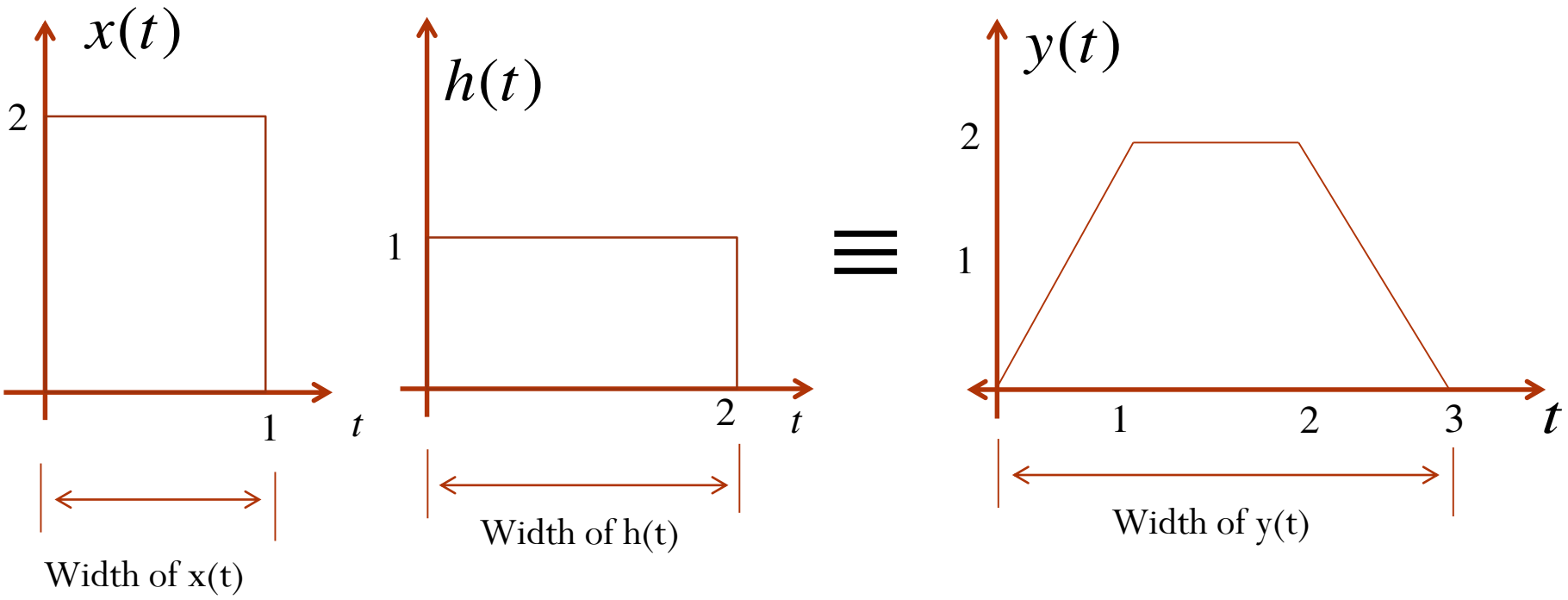
- Now combine all the results together:

$$y(t)=\begin{cases}0 & , \quad t<0\\ 2t & , \quad 0\le t<1\\ 2 & , \quad 1\le t<2\\ 6-2t & , \quad 2\le t\le 3\\ 0 & , \quad t>3\end{cases}$$



# Example1:

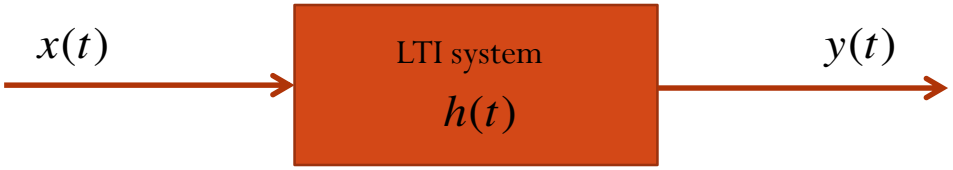
- We can conclude that the result y(t) width ( the time that y(t) is not zero) is the summation of x(t) width with h(t) width, where in the case x(t) has one unit of width and h(t) has two units. The total width is three units.

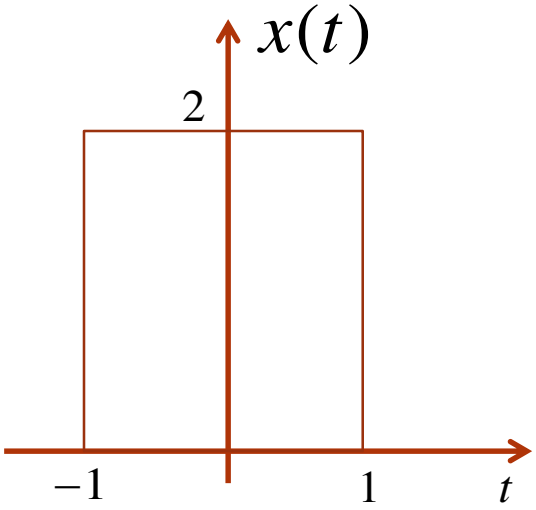


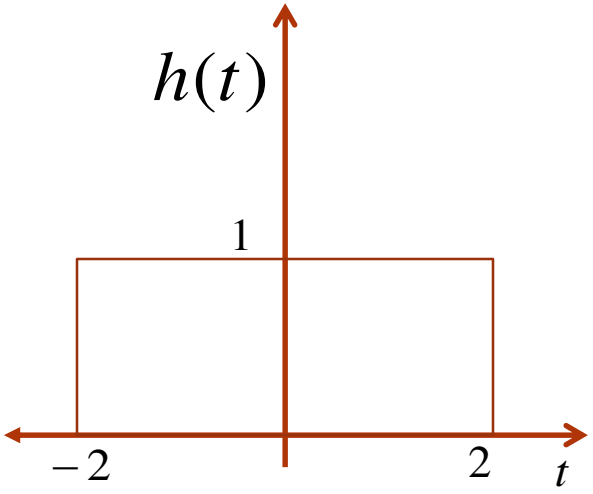
# Example2:

For the signal *x(t)* and *h(t)* shown below, find the output signal

. $y(t) = x(t) * h(t)$

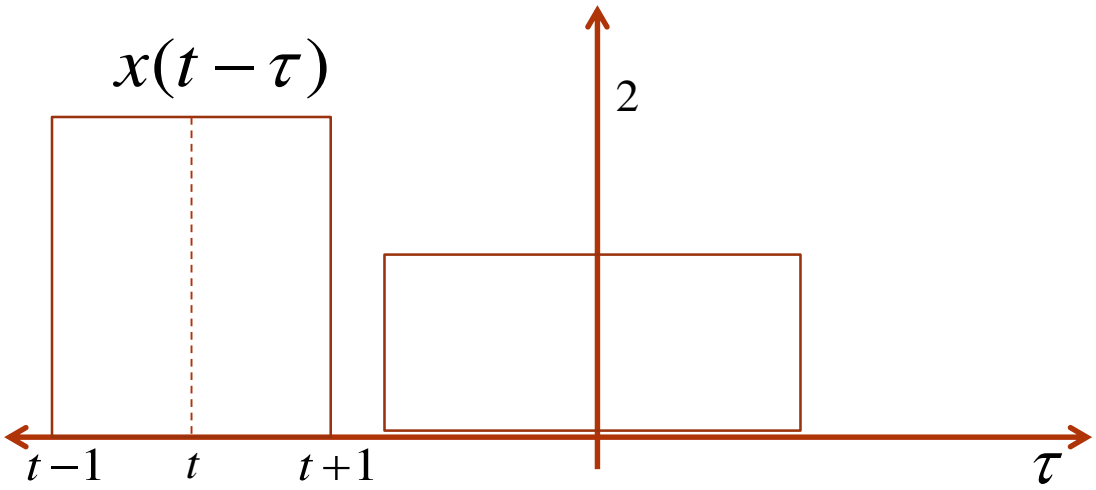


# Example2:

- Change $t$ *to* $\tau$, then Reflect and Shift to the left $x(\tau)$.



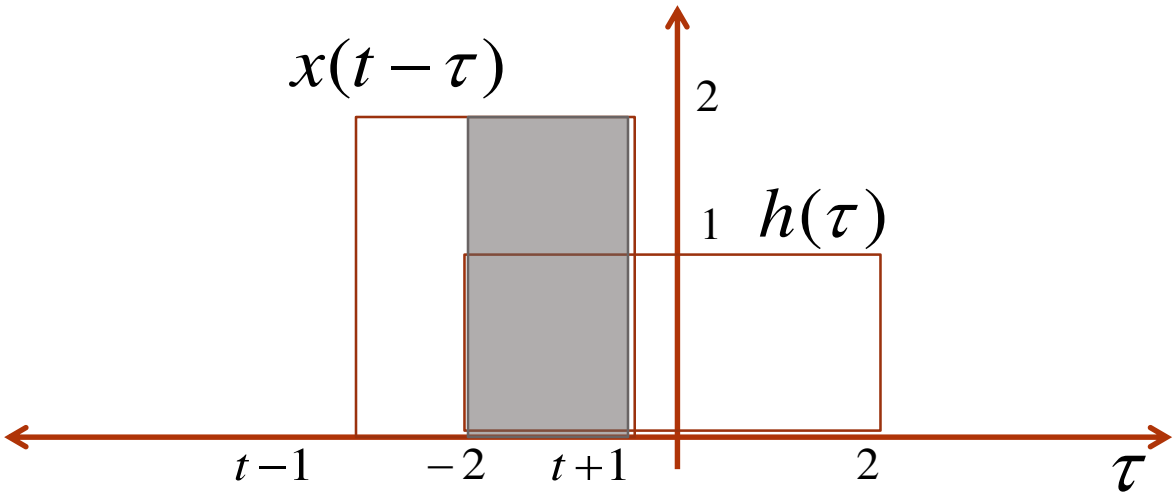

1) for $t+1<-2$. And that is equivalent to $t<-3$.

$$y(t)=0 \qquad (No\ overlap)$$

# Example2:

2) For $t+1 \geq -2 \;\&\; t-1 < -2$

$$\Rightarrow -3 \leq t \leq -1$$




$$y(t) = \int_{-2}^{t+1} h(\tau).x(t-\tau).d\tau = \int_{-2}^{t+1} (1).(2).d\tau$$

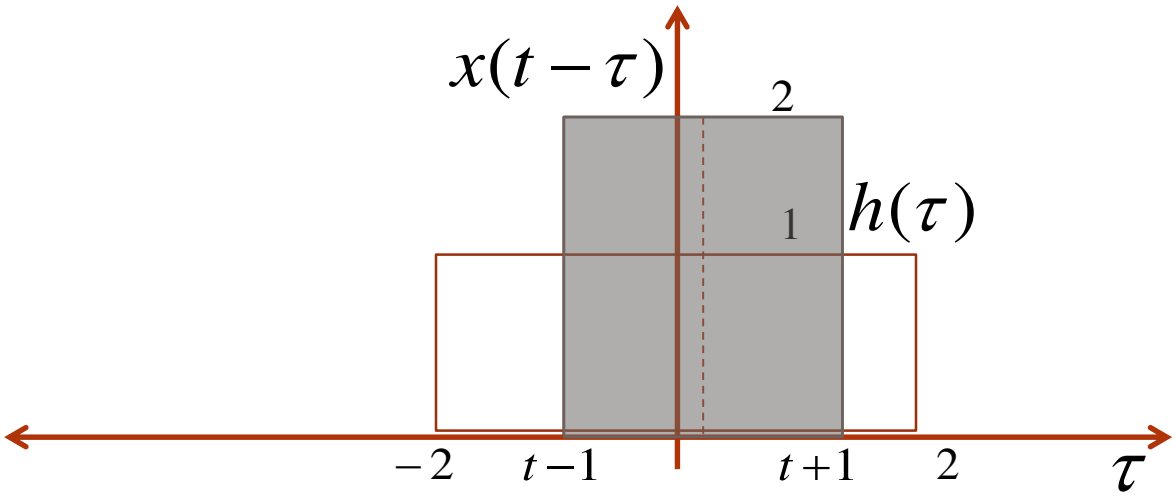
$$y(t) = 2\tau\Big|_{-2}^{t+1} = 2(t+1-(-2))$$

$$\boxed{y(t) = 2t+6}$$

# Example2:

3) For $t-1 \geq -2 \;\&\; t+1 < 2$

$$\Rightarrow -1 \leq t < 1$$




$$y(t) = \int_{t-1}^{t+1} h(\tau).x(t-\tau).d\tau = \int_{t-1}^{t+1} (1).(2).d\tau$$

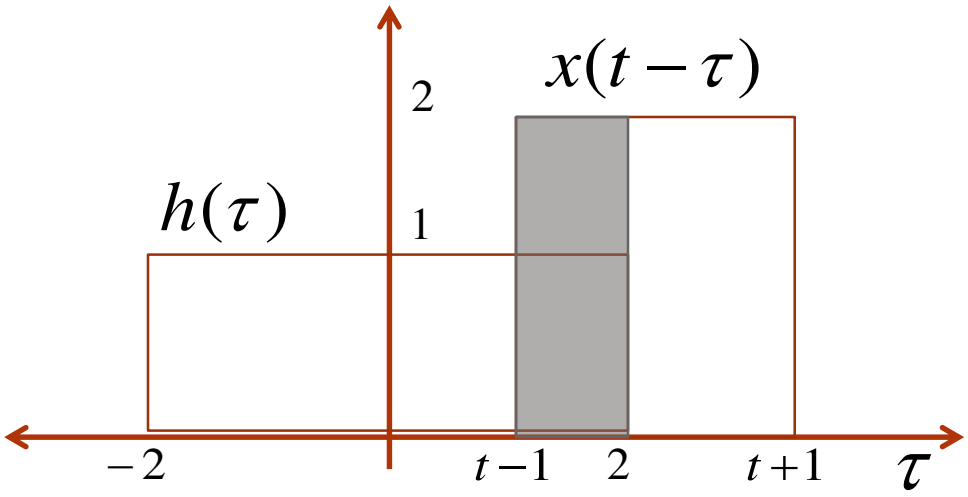
$$y(t) = 2\tau\Big|_{t-1}^{t+1} = 2(t+1-(t-1))$$

$$\boxed{y(t) = 4}$$

# Example2:

4) For $t-1 \le 0 \;\&\; t+1 \ge 2$

$$\Rightarrow 1 \le t \le 3$$




$$y(t) = \int_{t-1}^{2} h(\tau).x(t-\tau).d\tau = \int_{t-1}^{2} (1).(2).d\tau$$

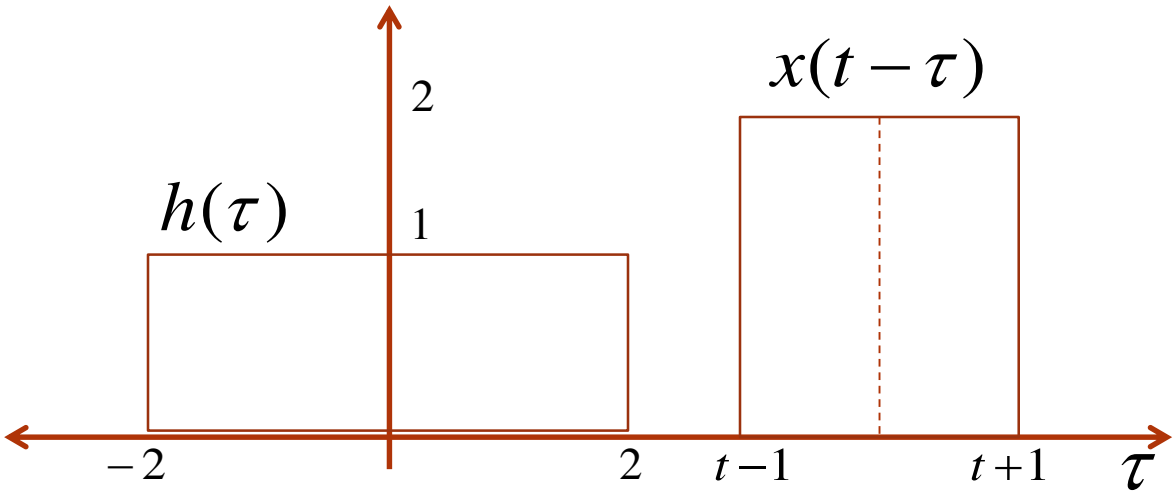
$$y(t) = 2\tau\Big|_{t-1}^{2} = 2(2-(t-1))$$

$$\boxed{y(t) = 6-2t}$$

# Example2:

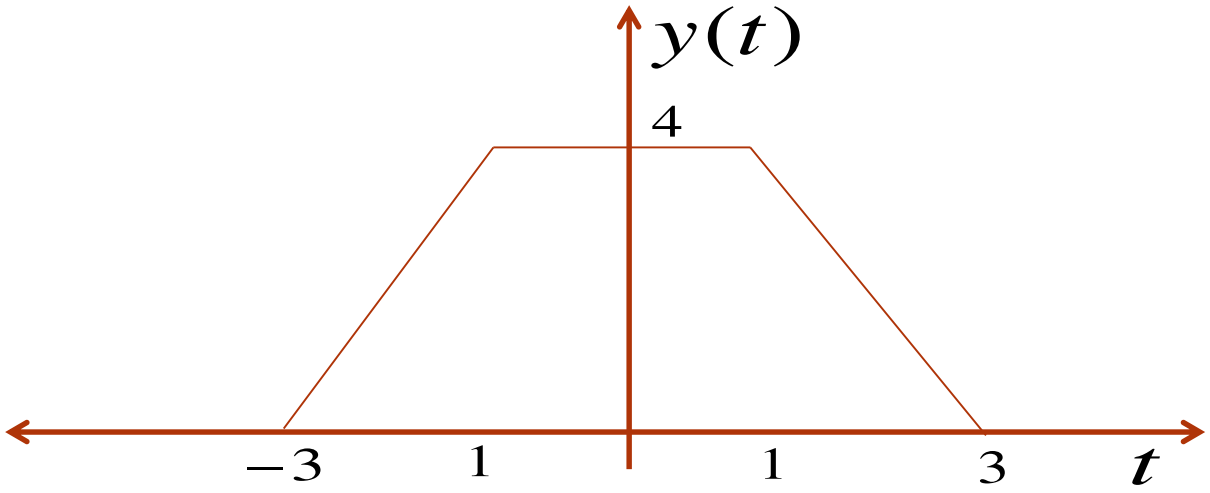
5) For $t-1>2$

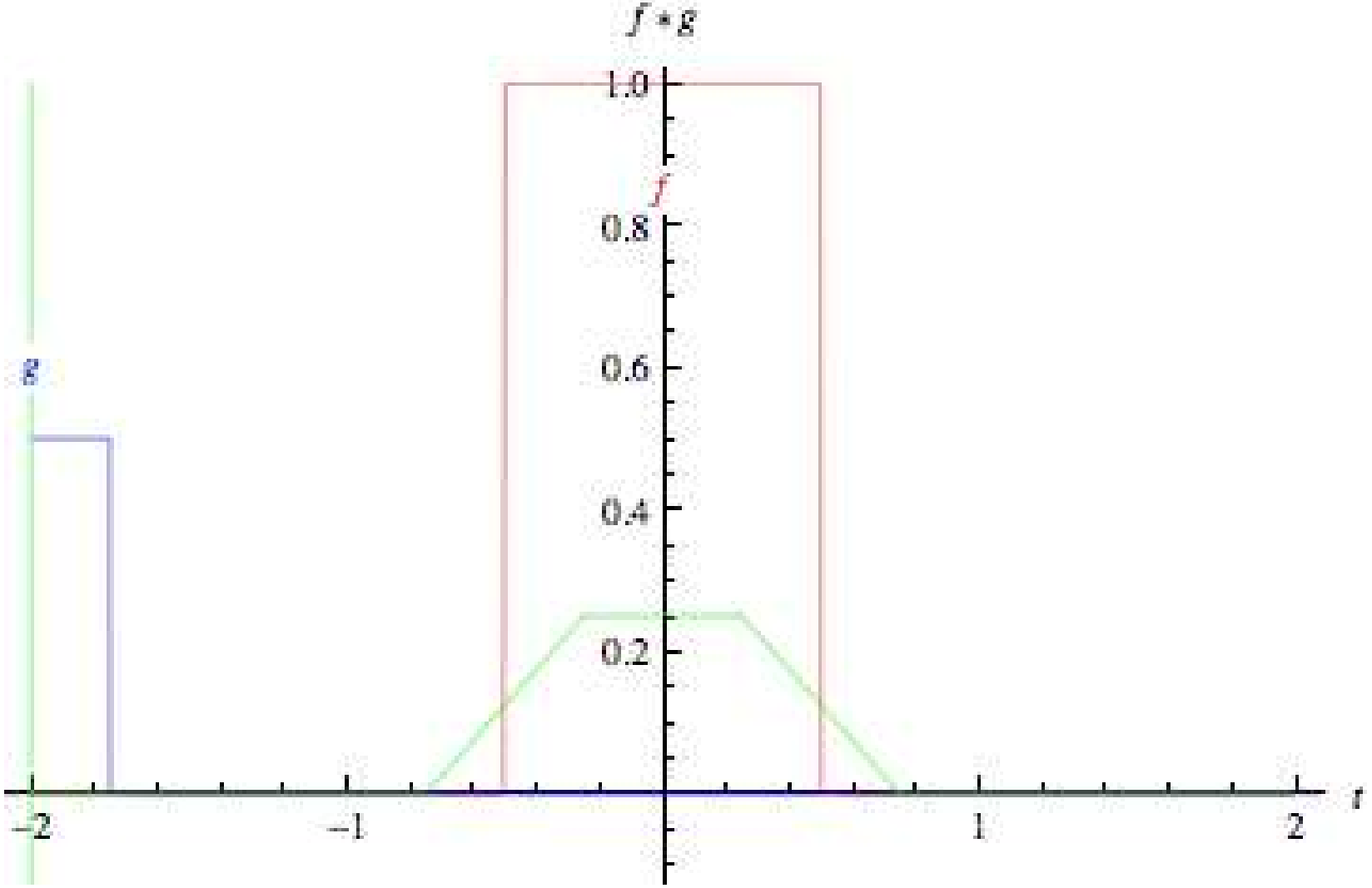
$$y(t)=0 \qquad (No\ overlap)$$

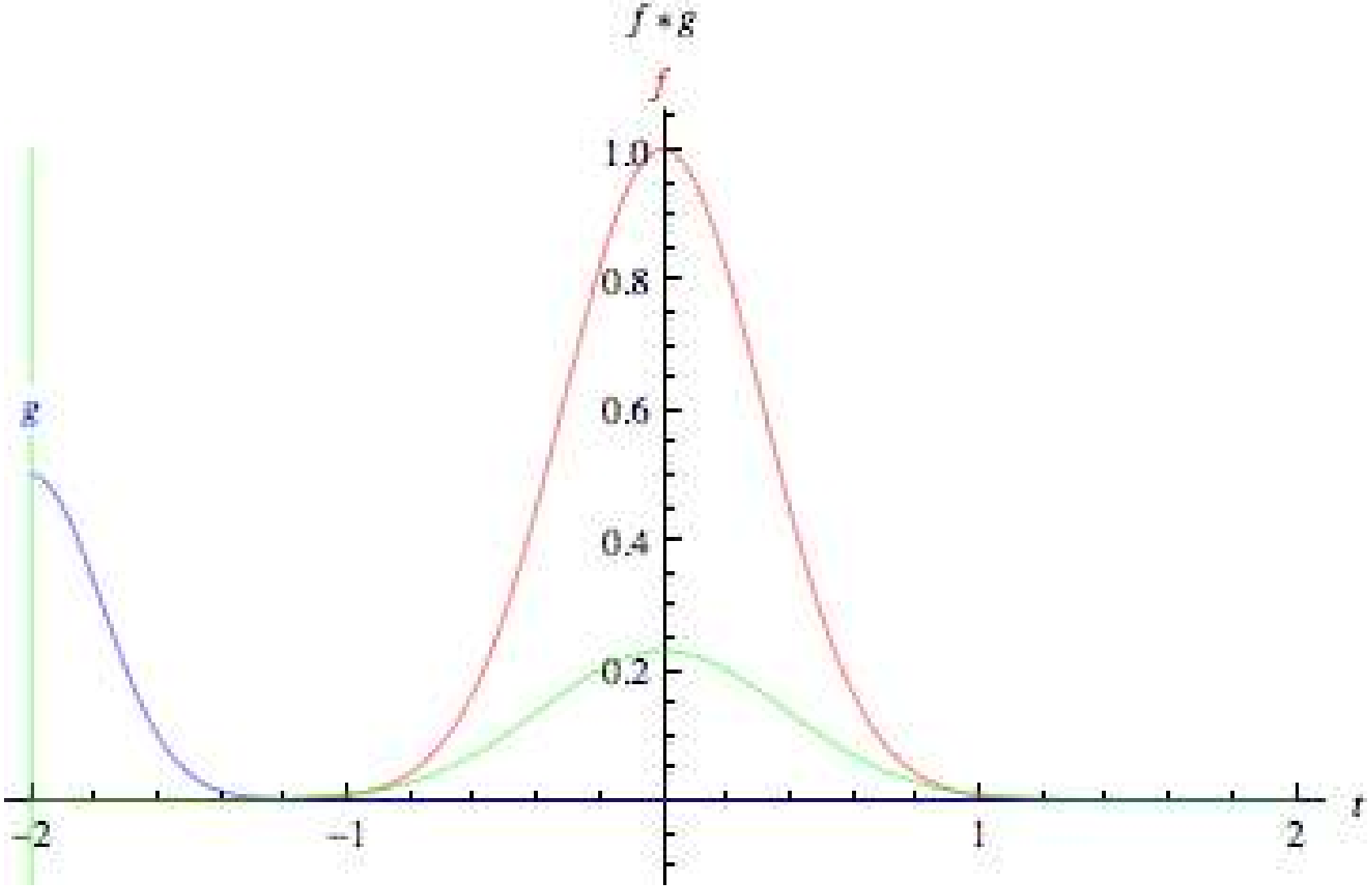


# Example2:

$$y(t)=\begin{cases}0 & , \quad t<-3\\ 2t+6 & , \quad -3\le t<-1\\ 4 & , \quad -1\le t<1\\ 6-2t & , \quad 1\le t\le 3\\ 0 & , \quad t>3\end{cases}$$




f * g
1.0
f
0.8
0.6
g
0.4
0.2
−2
−1
1
2
t


f * g
f
g
1.0
0.8
0.6
0.4
0.2
−2
−1
1
2
t

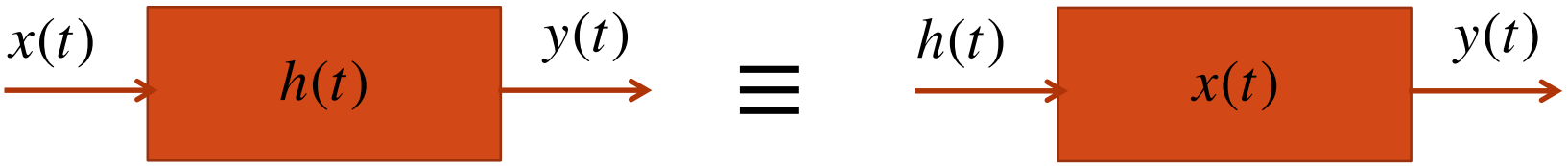
# Properties of convolution integral

- Continuous- time convolution satisfies the following important properties:

1. **Commutative:**

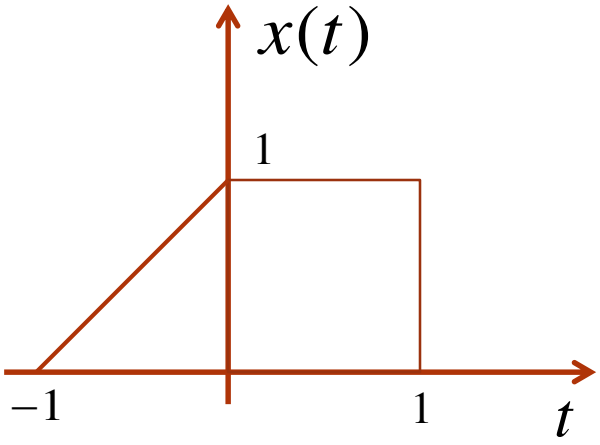
$$x(t) * h(t) = h(t) * x(t)$$

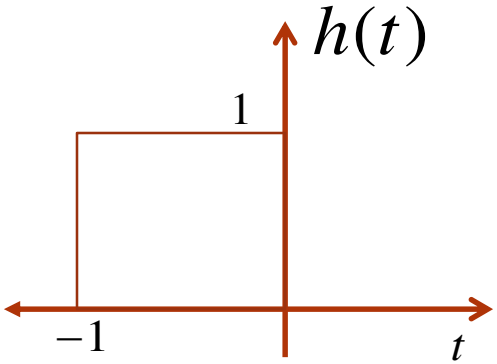
- This property is proved by substitution of variables. The property implies that the roles of the input signal and the impulse response are interchangeable. This property is illustrated in the following block diagram



# Properties of convolution integral

Example 3: Evaluate the convolution of the following two signals shown below.



# Properties of convolution integral

Example 3: In this example, it is easier to reflect and shift x(t) instead of h(t). Because the commutative property allow us to do so. We can use:

$$y(t) = h(t) * x(t)$$

Instead of

$$y(t) = x(t) * h(t)$$

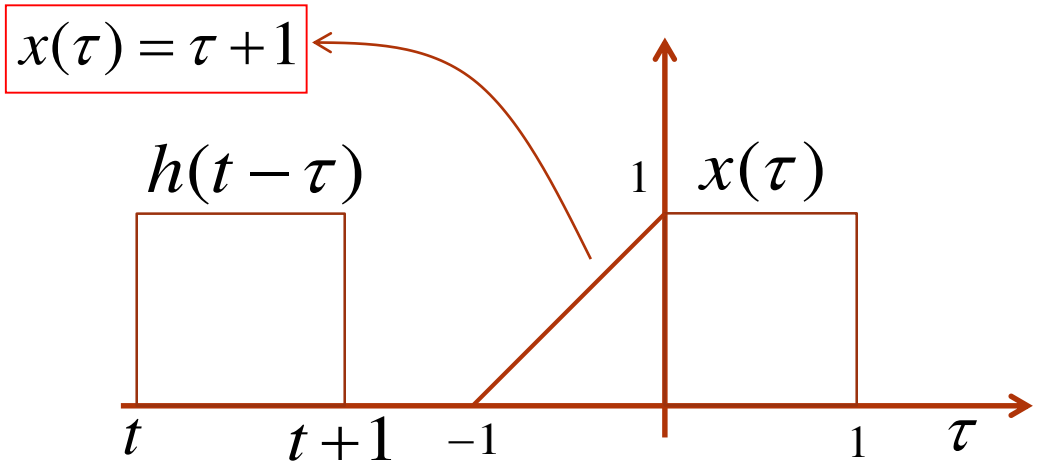
And simply we can follow the convolution procedure we used in Example 1 & 2

# Properties of convolution integral

## Example3:

- Change $t$ *to* $\tau$, then Reflect and Shift to the left $h(\tau)$.

Find the line equation for the line joining (-1,0) & (0,1).



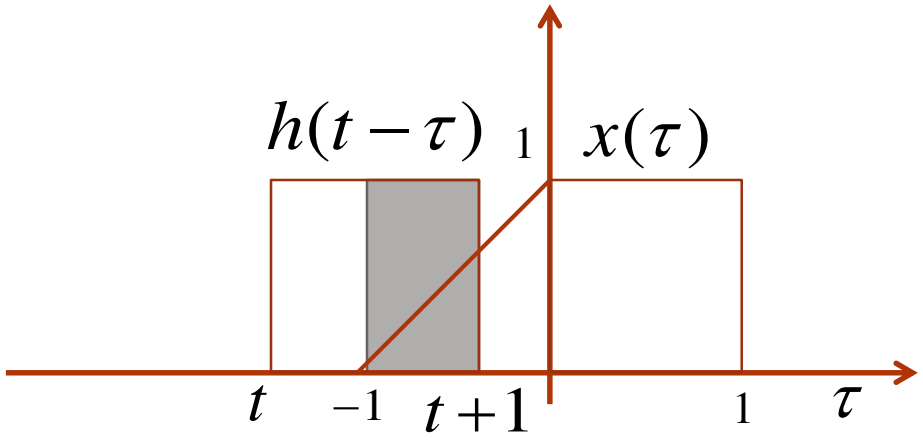

1) For $t+1<-1$ . And this is equivalent to $t<-2$.

$$y(t) = 0 \qquad (No\ overlap)$$

# Example3:

2) For $t+1 \geq -1 \;\&\; t < -1$

$$\Rightarrow -2 \leq t \leq -1$$




$$y(t) = \int_{-1}^{t+1} x(\tau).h(t-\tau).d\tau = \int_{-1}^{t+1} (\tau+1).(1).d\tau$$

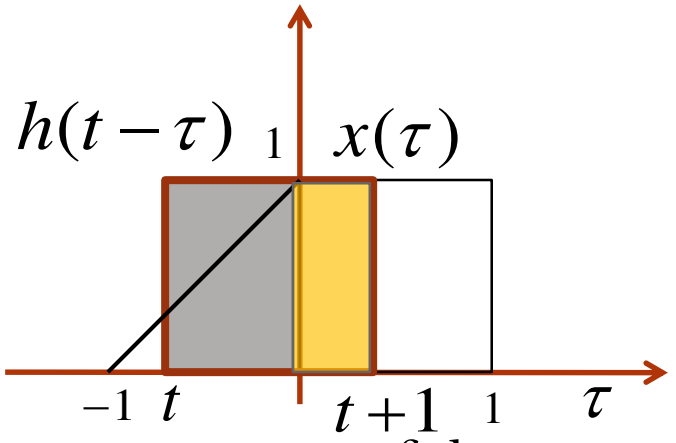
$$y(t) = \left(\frac{\tau^2}{2} + \tau\right)\Big|_{-1}^{t+1} = \frac{t^2}{2} + 2t + 2$$

$$\boxed{y(t) = \frac{t^2}{2} + 2t + 2}$$

# Example3:

3) For $t+1 \geq 0 \;\&\; t < 0$

$$\Rightarrow -1 \leq t < 0$$




- In this case, to obtain *y(t)* we need to use two terms of the integration. Because the multiplication of $x(\tau).h(t-\tau)$ to be integrated from $t \rightarrow 0$(the gray area) is different than $0 \rightarrow t+1$ (the yellow area). To compute the overlap area in gray color you need to integrate $(\tau+1).(1)$, to compute the overlap area in yellow color you need to integrate $(1).(1)$ as follow:

# Example3:

$$y(t)=\int_{t}^{t+1} x(\tau).h(t-\tau).d\tau=\int_{t}^{0}(\tau+1).(1).d\tau+\int_{0}^{t+1}(1).(1).d\tau$$

$$y(t)=\int_{t}^{0}(\tau+1).d\tau+\int_{0}^{t+1}(1).d\tau$$

Integration part to compute the gray area

Integration part to compute the yellow area

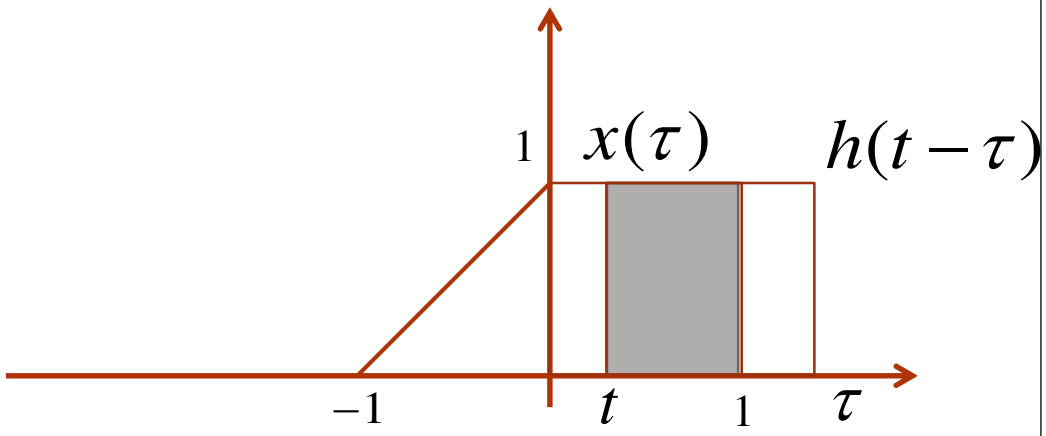
$$y(t)=(\frac{\tau^2}{2}+\tau)\Big|_{t}^{0}+\tau\Big|_{0}^{t+1}=-\frac{t^2}{2}+1$$

$$y(t)=-\frac{t^2}{2}+1$$

# Example3:

4) For $t+1 \geq 1 \;\&\; t < 1$

$$\Rightarrow 0 \leq t \leq 1$$




$$y(t) = \int_{t}^{1} x(\tau).h(t-\tau).d\tau = \int_{t}^{1} (1).(1).d\tau$$

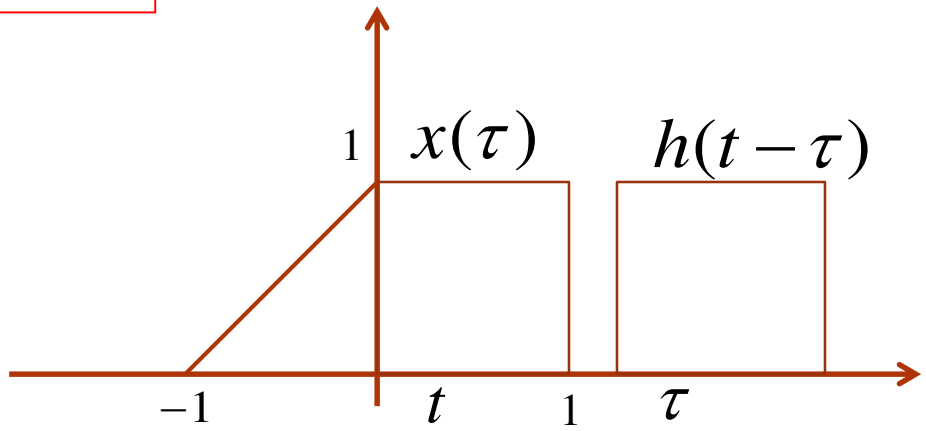
$$y(t) = (\tau)\Big|_{t}^{1} = 1-t$$

$$\boxed{y(t) = 1-t}$$

# Example3:

5) For $t > 1$

$$y(t) = 0 \qquad (No\ overlap)$$



# Properties of convolution integral

2. **Associative:**

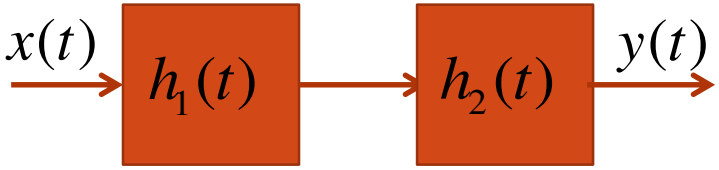
$$x(t) * h_1(t) * h_2(t) = [x(t) * h_1(t)] * h_2(t)$$
$$= x(t) * [h_1(t) * h_2(t)]$$

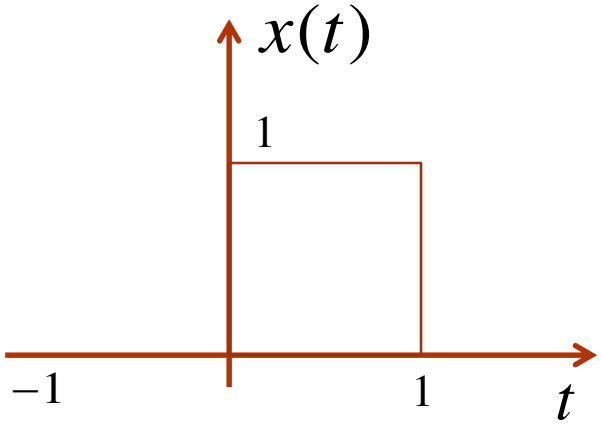
- This property is proved by changing the orders of integration. Associatively, implies that a cascade combination of LTI system can be replaced by a single system whose impulse response is the convolution of the individual impulse response. This property is illustrated in the following block diagram

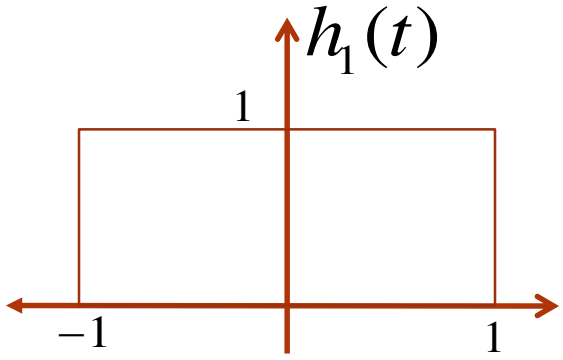


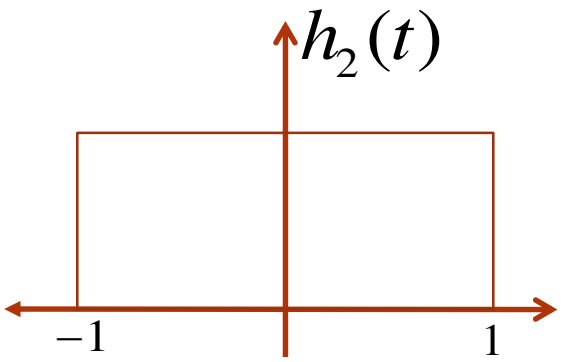
# Properties of convolution integral

Example 4: Evaluate y(t) for the signals shown if the system construction like:



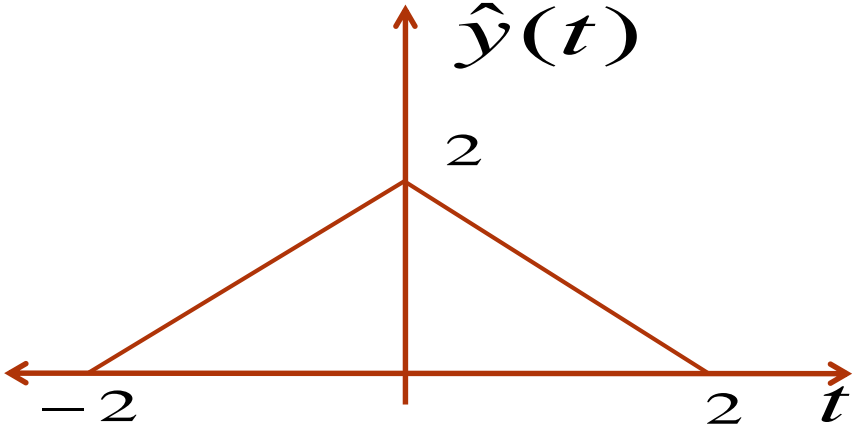



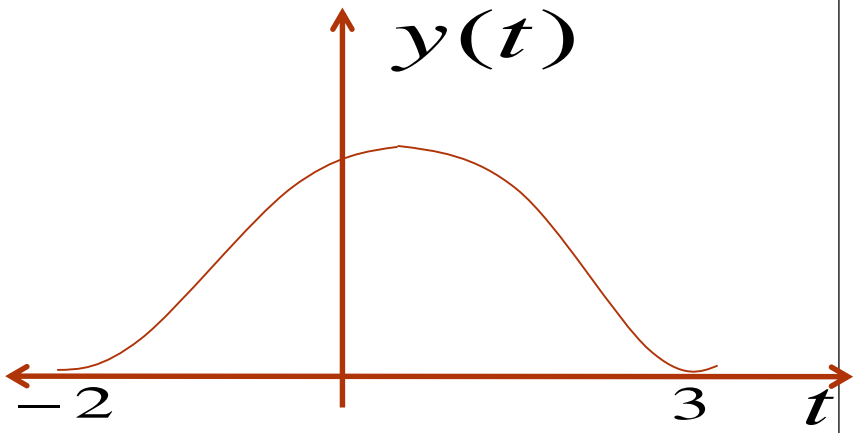
# Example 4:

- First, we can evaluate the convolution for $h_1(t) * h_2(t)$ then the result will be convolved with x(t):

1. $\hat{y}(t) = h_1(t) * h_2(t)$




2. $y(t) = \hat{y}(t) * x(t)$

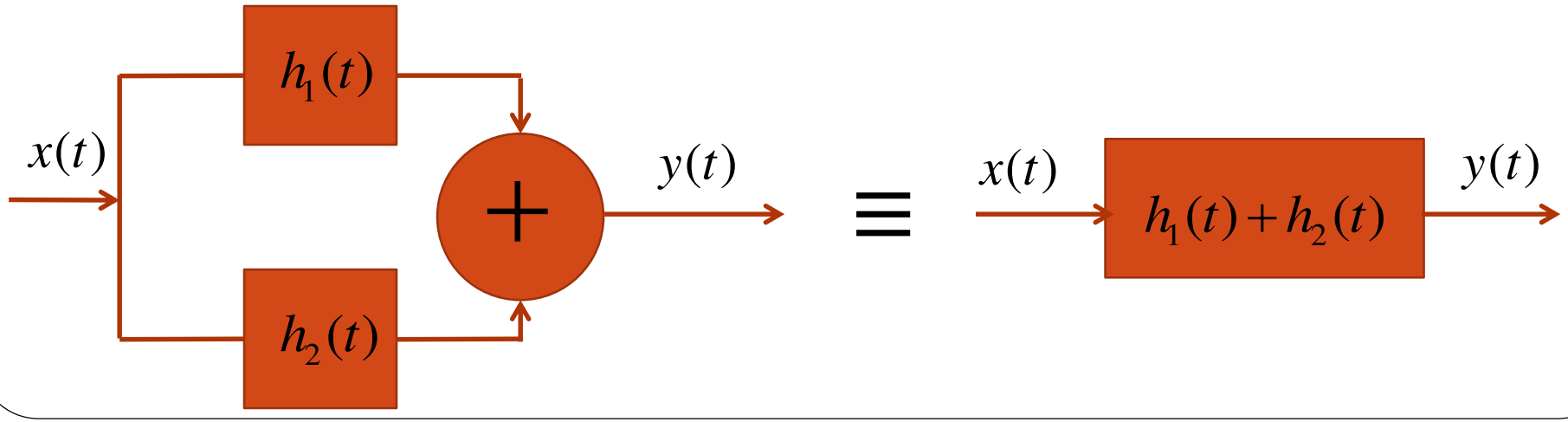


# Properties of convolution integral

3. **Distributivity:**

$$x(t)*[h_1(t)+h_2(t)]=[x(t)*h_1(t)]+[x(t)*h_2(t)]$$

- This property follows directly as a result of the linear property of integration. Distributivity states that a parallel combination of LTI systems is equivalent to a single system whose impulse response is the sum of the individual impulse responses in the parallel configuration. This property is illustrated in the following block diagram.



## Properties of convolution
## Example 5

a) Consider the LTI system with impulse response:

$$h(t) = \exp[-at].\ u(t)$$

If the input to the system is:

$$x(t) = \exp[-bt].u(t), \qquad b \neq a$$

then the output y(t) is:

$$y(t) = \int e^{-b\tau}.u(\tau).\ e^{-a(t-\tau)}.u(t-\tau).d\tau$$

Note that:

$$u(\tau).u(t-\tau) = 1, \qquad 0 < \tau < t$$
$$= 0, \qquad otherwise$$

## Properties of convolution
## Example 5

Therefore:

$$y(t) = \int_0^t e^{-at} . e^{(a-b)\tau} . d\tau$$

$$= \frac{1}{a-b}[e^{-bt} - e^{-at}]. \, u(t)$$

## Properties of convolution

## Example 5

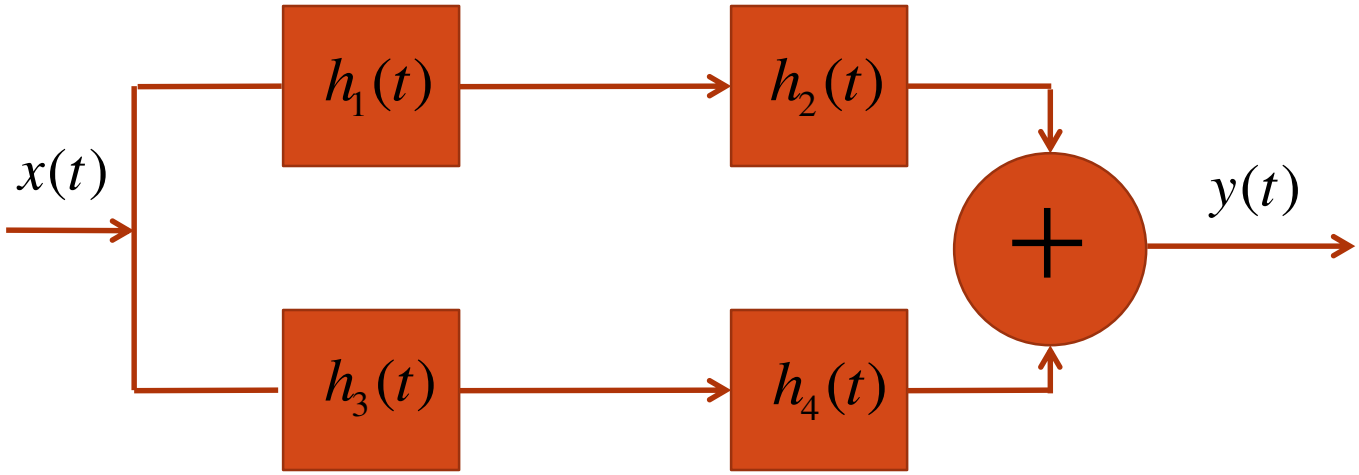
b) Find the impulse response of the system shown in the figure below:

$$h_1(t) = e^{-2t}.u(t)$$

$$h_2(t) = 2e^{-t}.u(t)$$

$$h_3(t) = e^{-3t}.u(t)$$

$$h_4(t) = 4\delta(t)$$



# Properties of convolution
## Example 5

By using the associative and distributive properties of the impulse response it follows that $h(t)$ for the system of the figure shown below is:

$$h(t) = h_1(t) * h_2(t) + h_3(t) * h_4(t)$$

$$h(t) = [\, e^{-t} - e^{-2t} \,].u(t) + 4e^{-3t}.u(t)$$

Where

$$h_3(t) * h_4(t) = 4h_3(t) = 4e^{-3t}.u(t)$$

Because

$$x(t) * \delta(t) = x(t) \ldots\ldots \text{Impulse Response of the system}$$

# Properties of convolution integral

**4. Convolution integral with unit step signal : $u(t)$**

- If the impulse response $h(t) = u(t)$ the convolution integral can be obtained directly using:

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau).u(t-\tau).d\tau = \int_{-\infty}^{t} x(\tau).d\tau$$

- Consequently, an LTI system with response $h(t) = u(t)$ is a perfect integration.

# Properties of convolution integral

Example 6: convolve the unit step function with *u(t)* with LTI system *h(t)* where:

$$h(t) = \begin{cases} e^{-at} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Because h(t) has a value just when t>0, so we are going to integrate from 0 to *t*:

$$u(t) * h(t) = \int_0^t e^{-a\tau} . d\tau$$

$$= \frac{-1}{a} e^{-a\tau} \Big|_0^t = \frac{1}{a}(1 - e^{-at})$$

# Properties of convolution integral

**5. Convolution integral with unit impulse signal : $\delta(t)$**

- If the impulse response $h(t) = \delta(t)$ the convolution integral result is the same input signal:

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau).\delta(t-\tau).d\tau = x(t)$$

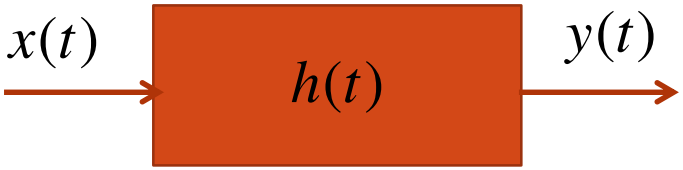
- Therefore, an LTI system with impulse response $h(t) = \delta(t)$ is the identity system. As well as, the previous equation leads to:

$$x(t) * \delta(t-a) = x(t-a)$$

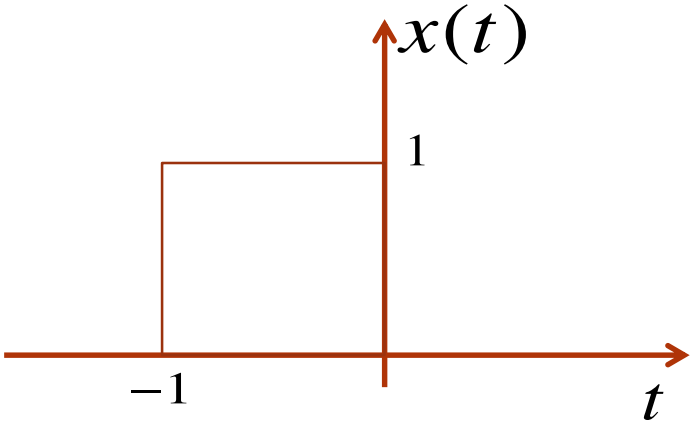
# Properties of convolution integral

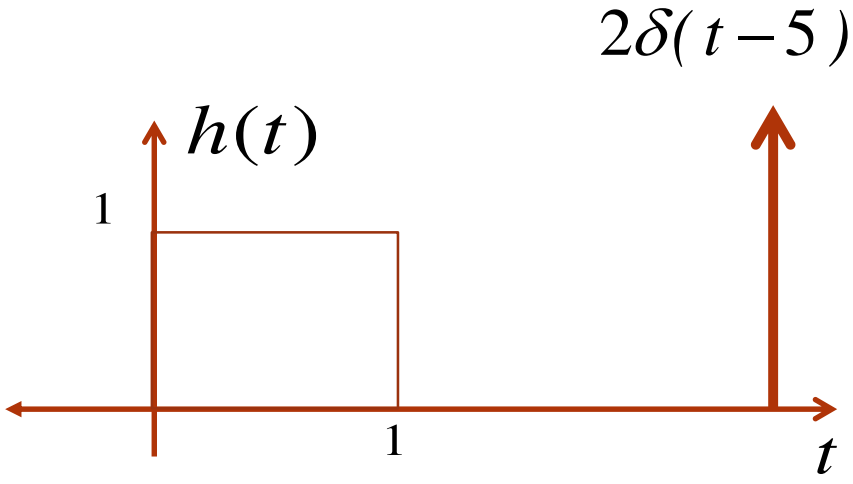
Example 7:

An LTI system with the block diagram shown:




Given that $x(t)$ & $h(t)$ as shown in the figure below. Find and sketch $y(t)$.

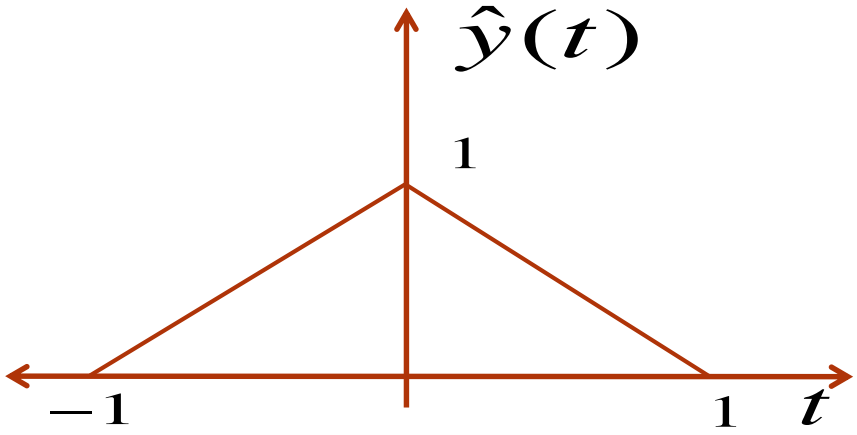


# Properties of convolution integral

## Example 7:

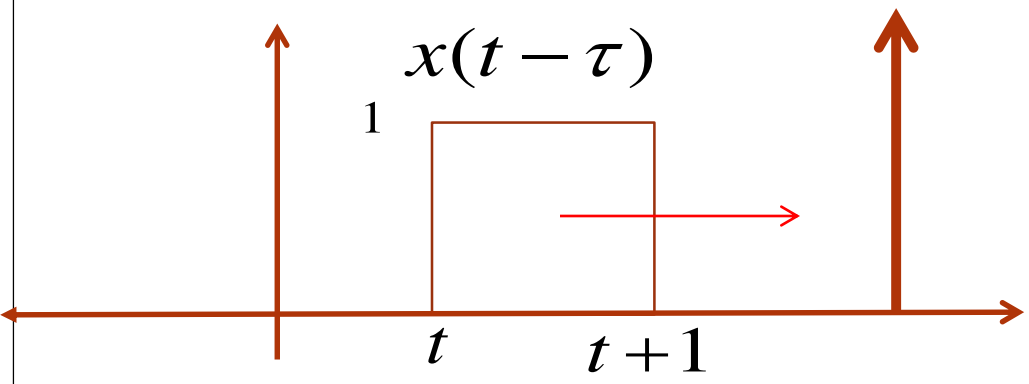
- In this example, we can do the convolution integral by convolving *x(t)* with first part of *h(t)* individually, which is the *rect* function. Then convolve *x(t)* with the second part of *h(t)*, which is the $\delta(t)$ function. because the system is linear we can sum the two result together. So:
- Convolving *x(t)* with the first part results:



# Properties of convolution integral

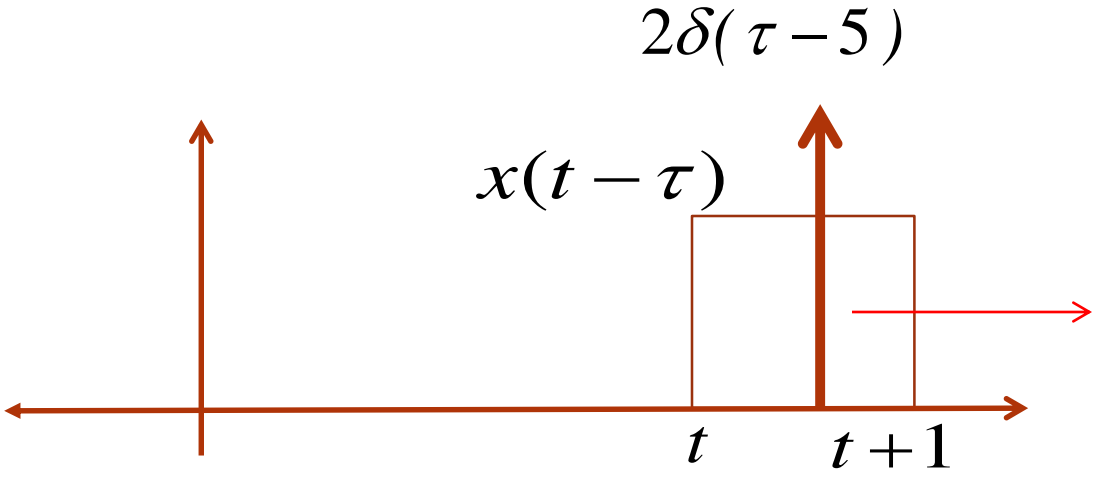
## Example 7:

- Convolving *x(t)* with the second part of *h(t)* ($2\delta(t+5)$) results the same function x(t) shifted to the right by five units with amplitude multiplied by 2.

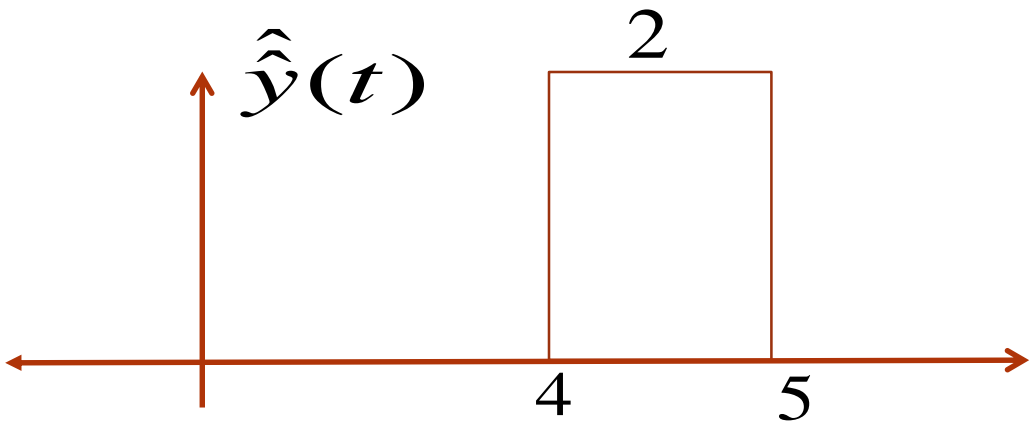


# Properties of convolution integral

## Example 7:




- The final result of the second part is:

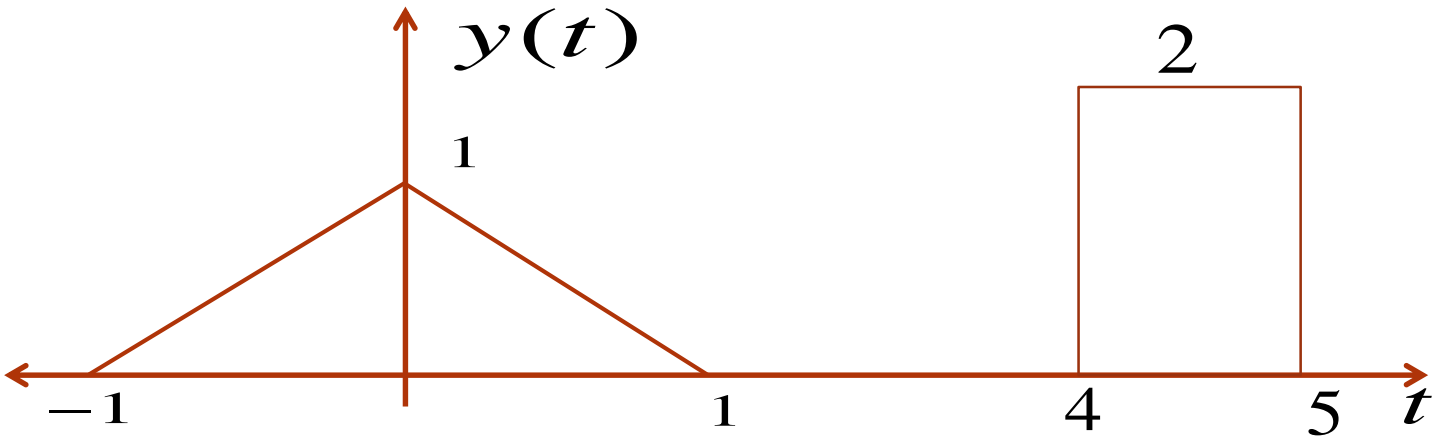


# Properties of convolution integral

## Example 7:

- Now sum the result of the two parts together:

$$y(t) = \hat{y}(t) + \hat{\hat{y}}(t)$$



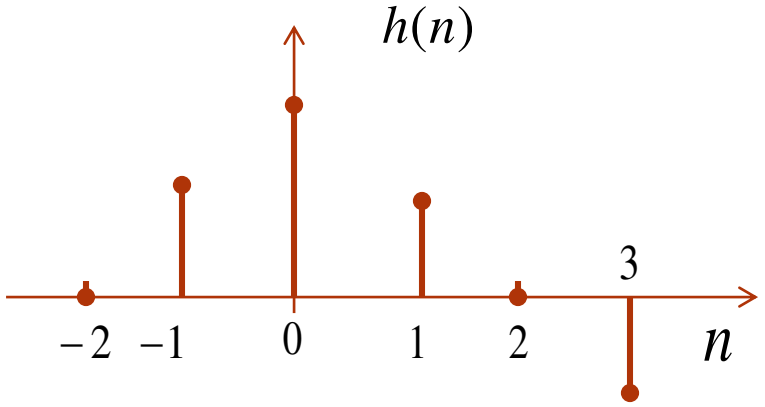
# Discrete-Time System

Discrete Convolution

# Introduction

- Discrete time Signals will take values at selected time values.

$$x(n) = \frac{1}{2}\cos 3n$$




- A discrete-time is a system in which all the signals are discrete-time Signals.
- A discrete-time system transforms discrete-time inputs into discrete-time output.

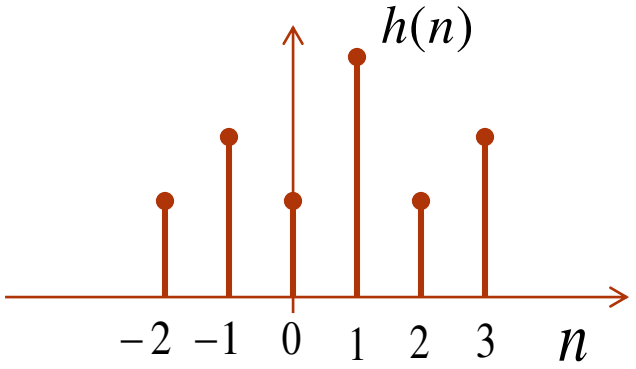
# Objectives:

- To Know how to represent and draw Discrete Signals.
- To know how to perform the convolution of discrete Signals.

# Representing Discrete time Signals

- Discrete Signals can be represented as follow:

$$h(n) = \{1, 2, \underset{\uparrow}{1}, 3, 1, 2\}$$




- The arrow below the third element represent the current value of the signal (at n=0).
- The Signal at any time between 0 and 1 is equal to zero as well as between 1 and 2 and so on. that mean, the signal will take values at selected time values. e.g.: *h(1)*=3, *h(2)*=1….etc.

# Discrete Convolution

- Consider a Linear, Shift- invariant, discrete- time system with input $x(n)$. Any arbitrary Signal $x(n)$ can be written as the weighted sum of shifted unit- sample functions:

$$x(n) = \sum_{k=-\infty}^{\infty} x(k).\delta(n-k)$$

- To convolve $x(n)$ with the system $h(n)$, first replace $n$ by $k$, then shift and reflect $h$ by $n$ to obtain $h(n-k)$ .
- Then fix $n$, let $k$ vary from $-\infty to \infty$, and take the sum.

# Discrete Convolution

- Then, the output of the system at time $n$ is given in term of the input as:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k).h(n-k)$$

- The right-hand side of the above equation is referred to as the *convolution sum* of the two sequences $x(n)$ and $h(n)$ is represented symbolically as $x(n) * h(n)$.

Note: the properties of discrete convolution are exactly like the properties of Continues LTI system.

# Example1:

- For the discrete signals:

$$x(n) = \{2, \underset{\uparrow}{1}, 0, 1, 2\}$$

$$h(n) = \{1, 1, \underset{\uparrow}{-1}, -1\}$$

- Evaluate the Convolution:

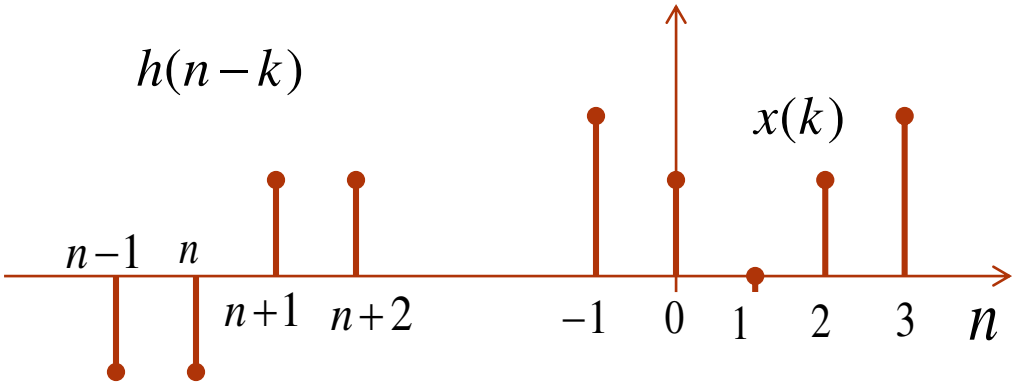
$$y(n) = x(n) * h(n)$$

- First we replace *n* by *k* then reflect *h(k)* :

$$h(-k) = \{-1, \underset{\uparrow}{-1}, 1, 1\}$$

# Example1:

- Then we shift it by $n$. The result looks as shown:




- Now start moving *h(n-k)* to the right to overlap *x(k)*.
- Note: 1) The overlap period starts for n+2=-1 or n=-3. and ends for n-1=3 or n=4.

2) No need shift the signal h(n-k) to the right gradually like the continuous time signal. Actually you need to shift it one unit every time. Why? The result is:

$$y(k)=\{2,3,-1,\underset{\uparrow}{-2},2,1,-3,-2\}$$

# Example1:

- Also you can use MATLAB to perform this using the following code in the command window:

```
>> x=[2 1 0 1 2];
>> h=[1 1 -1 -1];
>> y=conv(x,h)
```

# CHAPTER 3: FOURIER SERIES

## 3.2: Orthogonal Representation

# Introduction:

- A complex signal, in general, is a combination, in relative proportion, of a large number of independent functions.
- So a function can be represented as a summation of these independent orthogonal functions, weighted by their relative presence in the composite function representing the signal.

# Introduction:

- There are several reasons for doing this:
  1. It is mathematically convenient to represent an arbitrary signal as a weighted sum of orthogonal waveforms, since many of the calculations involving signals are simplified by using such representation.
  2. It is possible to visualize the signal as a vector in an orthogonal coordinate system, with the orthogonal waveforms being coordinate.
  3. Representation in terms of orthogonal basis functions provides a convenient means of solving for the response of linear systems to arbitrary signal over a finite interval in terms of some set of orthogonal basis functions

# Objectives:

- After completing this session, you will be able to:

1. To check the orthogonality of a set of signals.
2. To represent an arbitrary signal as weighted sum of orthogonal set of signals.
3. To be familiar with some applications of the orthogonality in the digital systems.

# Orthogonal set:

- Consider the signals set $\theta_i$ . Where $i = \pm 1, \pm 2, \ldots$ are the bases set of the signals $\theta_i$ . And they are orthogonal over [a,b] if:

$$\int_a^b \theta_i(t).\theta_k^*(t).dt = \begin{cases} E_k & i = k \\ 0 & i \neq k \end{cases}$$

$$where \quad k = \pm 1, \pm 2, \ldots$$

- If $Ek = 1$, then $\theta_i$ is a set of orthonormal signals.

# Example 1:

- Check the orthogonality over [-π,π], Considering the set:

$$\theta_m(t) = \sin(mt) \quad where\ m = 1,2,3,\ldots$$

# Example 2:

- Check the orthogonality over [0,T], Considering the set:

$$\theta_k(t) = e^{\frac{j2\pi kt}{T}} \qquad where\ k = 0, \pm 1, \pm 2, \ldots$$

# Applications:

- Orthogonal sets leads to series representation of signals in relatively simple way.
- A signal such as x(t) can be represented in terms of $\theta_i$ by a convergent series as:

$$x(t) = \sum_{-\infty}^{\infty} C_i \theta_i(t) \Rightarrow \textit{Generalized Fourier series}$$

$$\theta_k(t) = e^{\frac{j2\pi kt}{T}} \qquad \textit{where } k = 0, \pm 1, \pm 2, \ldots$$

$$\textit{where } C_i = \int_a^b x(t).\theta_i^*(t).dt$$

# Applications

- To represent signals such as x(t) in series expansion require a summation of infinite sum.
- In practice, however, we can use only a finite number of terms. So, we will get an approximation $x(t)$ to the original signal $\hat{x}(t)$ .
- When we use only M terms, the representation error is:

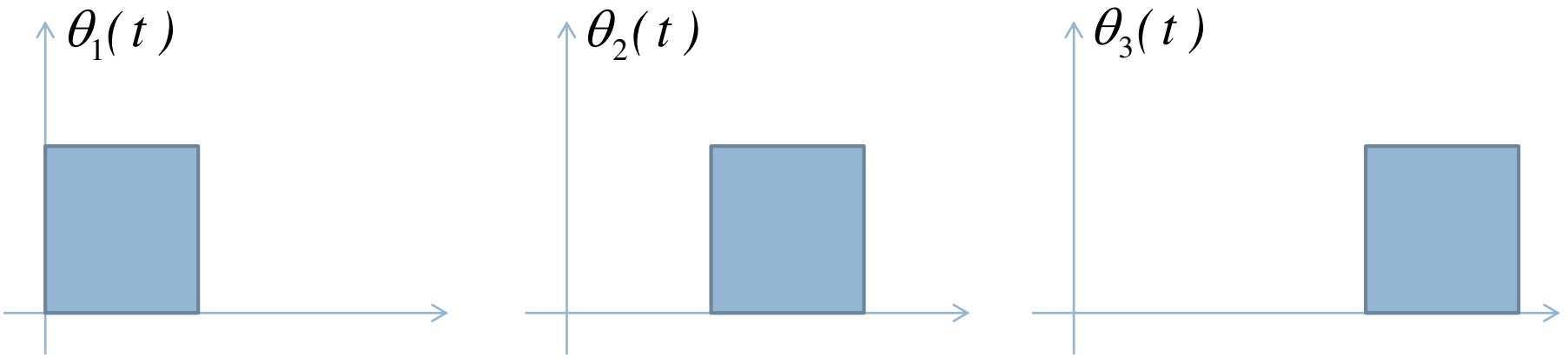
$$e_m(t) = x(t) - \sum_{i=1}^{M} C_i . \theta_i(t)$$

# Applications:

- Therefore, the energy of the error can be computed to know if this finite number of terms is enough to represent the signal in the real world.
- In some applications, e.g. digital communications, x(t) can be expressed in terms of finite number of orthogonal signal (expanded).
- Pseudo Noise signal (PN) is one of most important applications in the digital communications world depending on the orthogonal concepts.

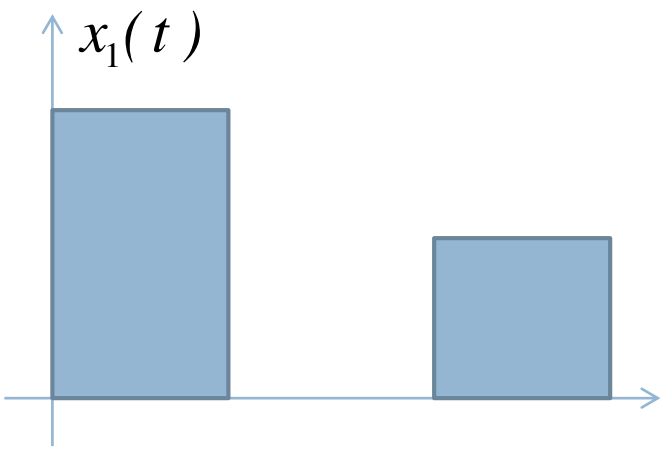
# Example 3:

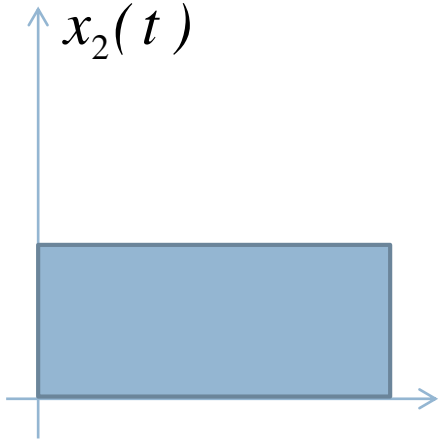
- Consider the orthogonal set below:

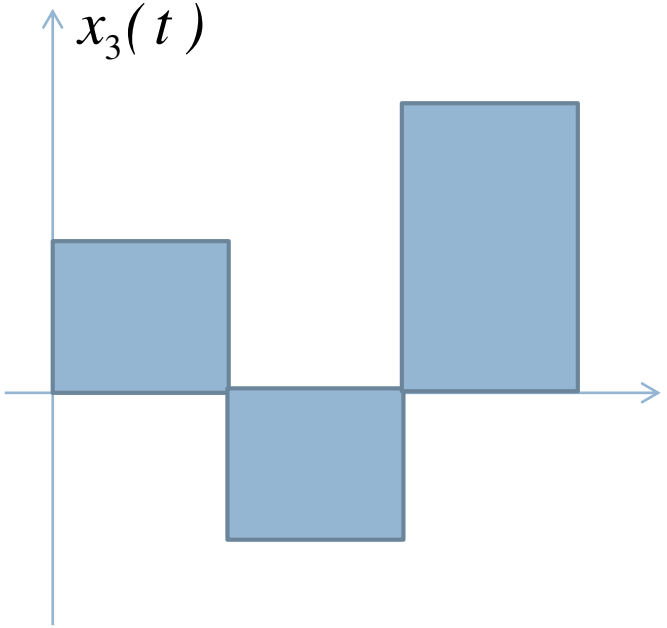


# Example 3:

- Represent the signals in weighted sum of the above orthogonal set.



We call a set of signals $\{\Psi_n(t)\}$ *orthogonal* on an interval $(a, b)$ if any two signals $\Psi_m(t)$ and $\Psi_k(t)$ in the set satisfy the condition

$$\int_a^b \Psi_m(t)\Psi_k^*(t)\,dt = \begin{cases} 0 & m \neq k \\ \alpha & m = k \end{cases}$$

where $*$ denotes the complex conjugate and $\alpha \neq 0$. Show that the set of complex exponentials $\{e^{jk\omega_0 t}: k = 0, \pm 1, \pm 2, \ldots\}$ is orthogonal on any interval over a period $T_0$, where $T_0 = 2\pi/\omega_0$.

For any $t_0$ we have

$$\int_{t_0}^{t_0+T_0} e^{jm\omega_0 t}\,dt = \frac{1}{jm\omega_0} e^{jm\omega_0 t}\Big|_{t_0}^{t_0+T_0} = \frac{1}{jm\omega_0}\left(e^{jm\omega_0(t_0+T_0)} - e^{jm\omega_0 t_0}\right)$$

$$= \frac{1}{jm\omega_0} e^{jm\omega_0 t_0}\left(e^{jm2\pi} - 1\right) = 0 \qquad m \neq 0$$

# Example

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T_0}$$

Multiplying both sides of this equation by $e^{-jm\omega_0 t}$ and integrating the result from $t_0$ to $(t_0 + T_0)$, we obtain

$$\int_{t_0}^{t_0+T_0} x(t) e^{-jm\omega_0 t}\, dt = \int_{t_0}^{t_0+T_0} \left( \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \right) e^{-jm\omega_0 t}\, dt$$

$$= \sum_{k=-\infty}^{\infty} c_k \int_{t_0}^{t_0+T_0} e^{j(k-m)\omega_0 t}\, dt$$

Then reduces to

$$\int_{t_0}^{t_0+T_0} x(t) e^{-jm\omega_0 t}\, dt = c_m T_0$$

Changing index $m$ to $k$, we obtain

$$c_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)e^{-jk\omega_0 t}\,dt$$

We shall mostly use the following two special cases for $t_0 = 0$ and $t_0 = -T_0/2$, respectively. That is,

$$c_k = \frac{1}{T_0}\int_{0}^{T_0} x(t)e^{-jk\omega_0 t}\,dt$$

$$c_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)e^{-jk\omega_0 t}\,dt$$

# Example: Derive the trigonometric Fourier series from the following complex exponential Fourier series

Derive the trigonometric Fourier series from the following complex exponential Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} = c_0 + \sum_{k=1}^{\infty} \left( c_k e^{jk\omega_0 t} + c_{-k} e^{-jk\omega_0 t} \right)$$

and using Euler's formulas

$$e^{\pm jk\omega_0 t} = \cos k\omega_0 t \pm j \sin k\omega_0 t$$

we have

$$x(t) = c_0 + \sum_{k=1}^{\infty} \left[ (c_k + c_{-k}) \cos k\omega_0 t + j(c_k - c_{-k}) \sin k\omega_0 t \right]$$

Setting

$$c_0 = \frac{a_0}{2} \qquad c_k + c_{-k} = a_k \qquad j(c_k - c_{-k}) = b_k$$

becomes

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t)$$

$$= \frac{1}{jm\omega_0} e^{jm\omega_0 t_0}\left(e^{jm2\pi} - 1\right) = 0 \qquad m \neq 0$$

since $e^{jm2\pi} = 1$. When $m = 0$, we have $e^{jm\omega_0 t}|_{m=0} = 1$ and

$$\int_{t_0}^{t_0+T_0} e^{jm\omega_0 t}\, dt = \int_{t_0}^{t_0+T_0} dt = T_0$$

we conclude that

$$\int_{t_0}^{t_0+T_0} e^{jm\omega_0 t}\left(e^{jk\omega_0 t}\right)^* dt = \int_{t_0}^{t_0+T_0} e^{j(m-k)\omega_0 t}\, dt = \begin{cases} 0 & m \neq k \\ T_0 & m = k \end{cases}$$

which shows that the set $\{e^{jk\omega_0 t}: k = 0, \pm 1, \pm 2, \ldots\}$ is orthogonal on any interval over a period $T_0$.

Determine the complex exponential Fourier series representation for each of the following signals:

(a) $x(t) = \cos \omega_0 t$

(b) $x(t) = \sin \omega_0 t$

(c) $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$

(d) $x(t) = \cos 4t + \sin 6t$

(e) $x(t) = \sin^2 t$

$$\cos \omega_0 t = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right) = \frac{1}{2}e^{-j\omega_0 t} + \frac{1}{2}e^{j\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Thus, the complex Fourier coefficients for $\cos \omega_0 t$ are

$$c_1 = \tfrac{1}{2} \qquad c_{-1} = \tfrac{1}{2} \qquad c_k = 0, \; |k| \neq 1$$

(*b*) In a similar fashion we have

$$\sin \omega_0 t = \frac{1}{2j}\left(e^{j\omega_0 t} - e^{-j\omega_0 t}\right) = -\frac{1}{2j}e^{-j\omega_0 t} + \frac{1}{2j}e^{j\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Thus, the complex Fourier coefficients for $\sin \omega_0 t$ are

$$c_1 = \frac{1}{2j} \qquad c_{-1} = -\frac{1}{2j} \qquad c_k = 0, \; |k| \neq 1$$

(c) The fundamental angular frequency $\omega_0$ of $x(t)$ is 2. Thus,

$$x(t) = \cos\left(2t + \frac{\pi}{4}\right) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt}$$

Now
$$x(t) = \cos\left(2t + \frac{\pi}{4}\right) = \frac{1}{2}\left(e^{j(2t+\pi/4)} + e^{-j(2t+\pi/4)}\right)$$

$$= \frac{1}{2}e^{-j\pi/4}e^{-j2t} + \frac{1}{2}e^{j\pi/4}e^{j2t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt}$$

Thus, the complex Fourier coefficients for $\cos(2t + \pi/4)$ are

$$c_1 = \frac{1}{2}e^{j\pi/4} = \frac{1}{2}\frac{1+j}{\sqrt{2}} = \frac{\sqrt{2}}{4}(1+j)$$

$$c_{-1} = \frac{1}{2}e^{-j\pi/4} = \frac{1}{2}\frac{1-j}{\sqrt{2}} = \frac{\sqrt{2}}{4}(1-j)$$

$$c_k = 0 \qquad |k| \neq 1$$

(*d*) the fundamental period $T_0$ of $x(t)$ is $\pi$ and $\omega_0 = 2\pi/T_0 = 2$. Thus,

$$x(t) = \cos 4t + \sin 6t = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt}$$

Again using Euler's formula, we have

$$x(t) = \cos 4t + \sin 6t = \frac{1}{2}(e^{j4t} + e^{-j4t}) + \frac{1}{2j}(e^{j6t} - e^{-j6t})$$

$$= -\frac{1}{2j}e^{-j6t} + \frac{1}{2}e^{-j4t} + \frac{1}{2}e^{j4t} + \frac{1}{2j}e^{j6t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt}$$

Thus, the complex Fourier coefficients for $\cos 4t + \sin 6t$ are

$$c_{-3} = -\frac{1}{2j} \qquad c_{-2} = \frac{1}{2} \qquad c_2 = \frac{1}{2} \qquad c_3 = \frac{1}{2j}$$

(e) the fundamental period $T_0$ of $x(t)$ is $\pi$ and $\omega_0 = 2\pi/T_0 = 2$. Thus,

$$x(t) = \sin^2 t = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt}$$

Again using Euler's formula, we get

$$x(t) = \sin^2 t = \left(\frac{e^{jt} - e^{-jt}}{2j}\right)^2 = -\frac{1}{4}(e^{j2t} - 2 + e^{-j2t})$$

$$= -\frac{1}{4}e^{-j2t} + \frac{1}{2} - \frac{1}{4}e^{j2t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt}$$

Thus, the complex Fourier coefficients for $\sin^2 t$ are

$$c_{-1} = -\tfrac{1}{4} \qquad c_0 = \tfrac{1}{2} \qquad c_1 = -\tfrac{1}{4}$$

# CHAPTER 3
# FOURIER SERIES

## 3.3: The Fourier Series:

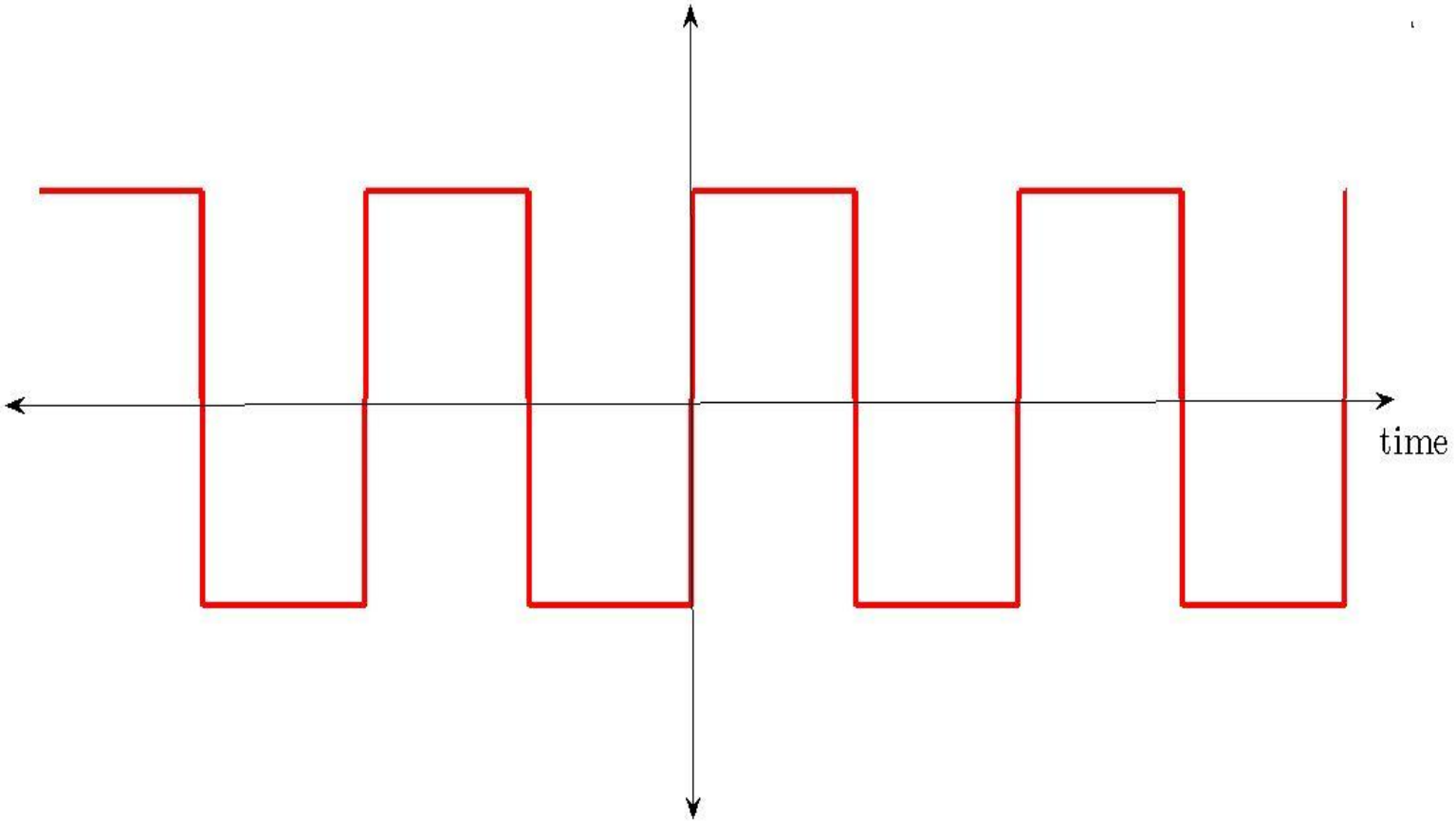
# Introduction:

- Any periodic signal x(t) with period $T_o$ will have a Fourier Series (FS) representation. Aperiodic signals will have a Fourier Transform.
- The sum of basis functions (Orthogonal Functions); sine, cosine, and exponential will be used to represent the signals.
- When we say series, that means we are going to expand the signal to many terms.

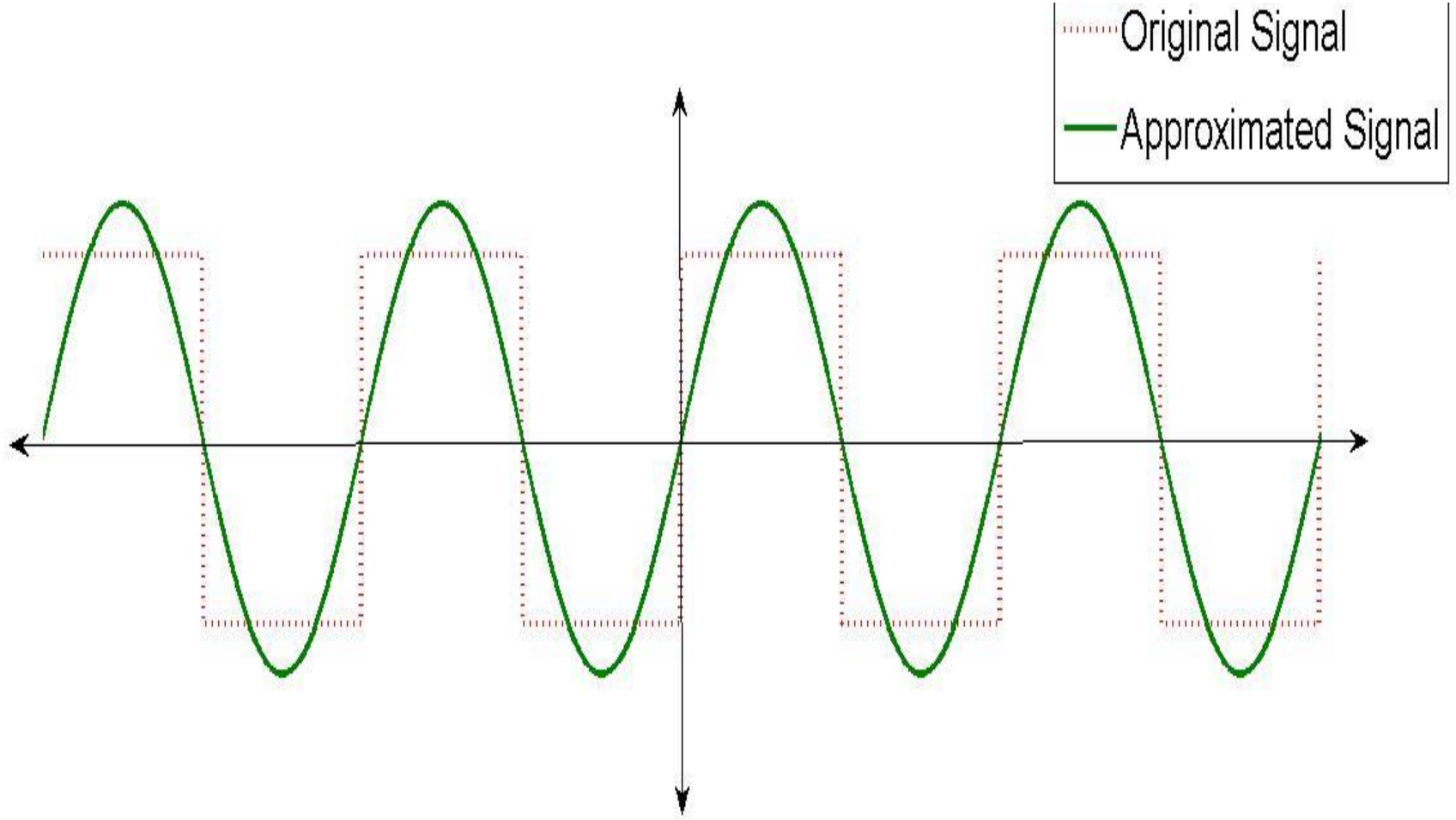
# Introduction:

- The expansion of the signal using FS can be exactly as x(t) if we take infinity number of terms of the series.
- Finite number of terms means that we have an approximation of x(t).

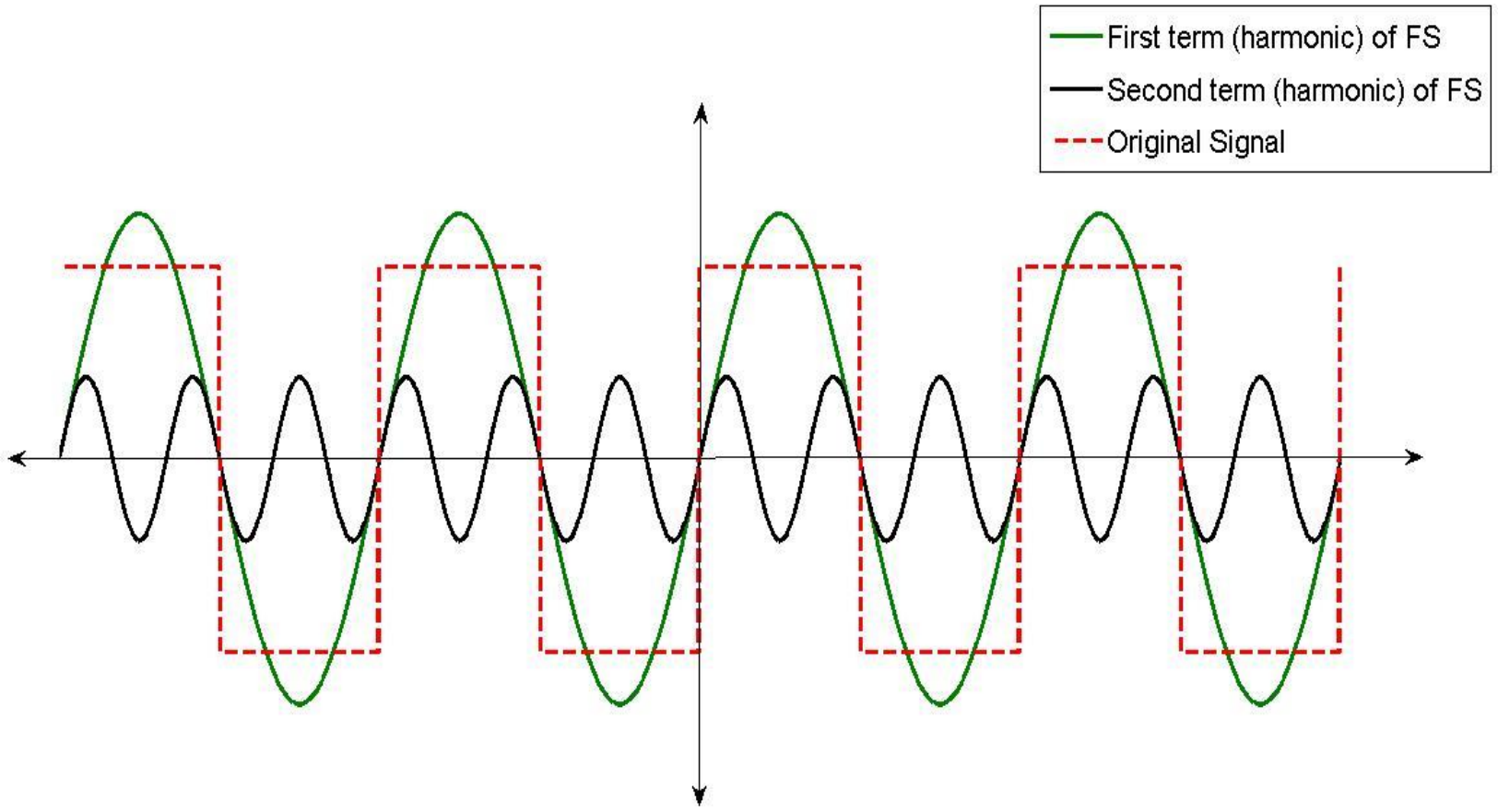
# Approximating Periodic signal:



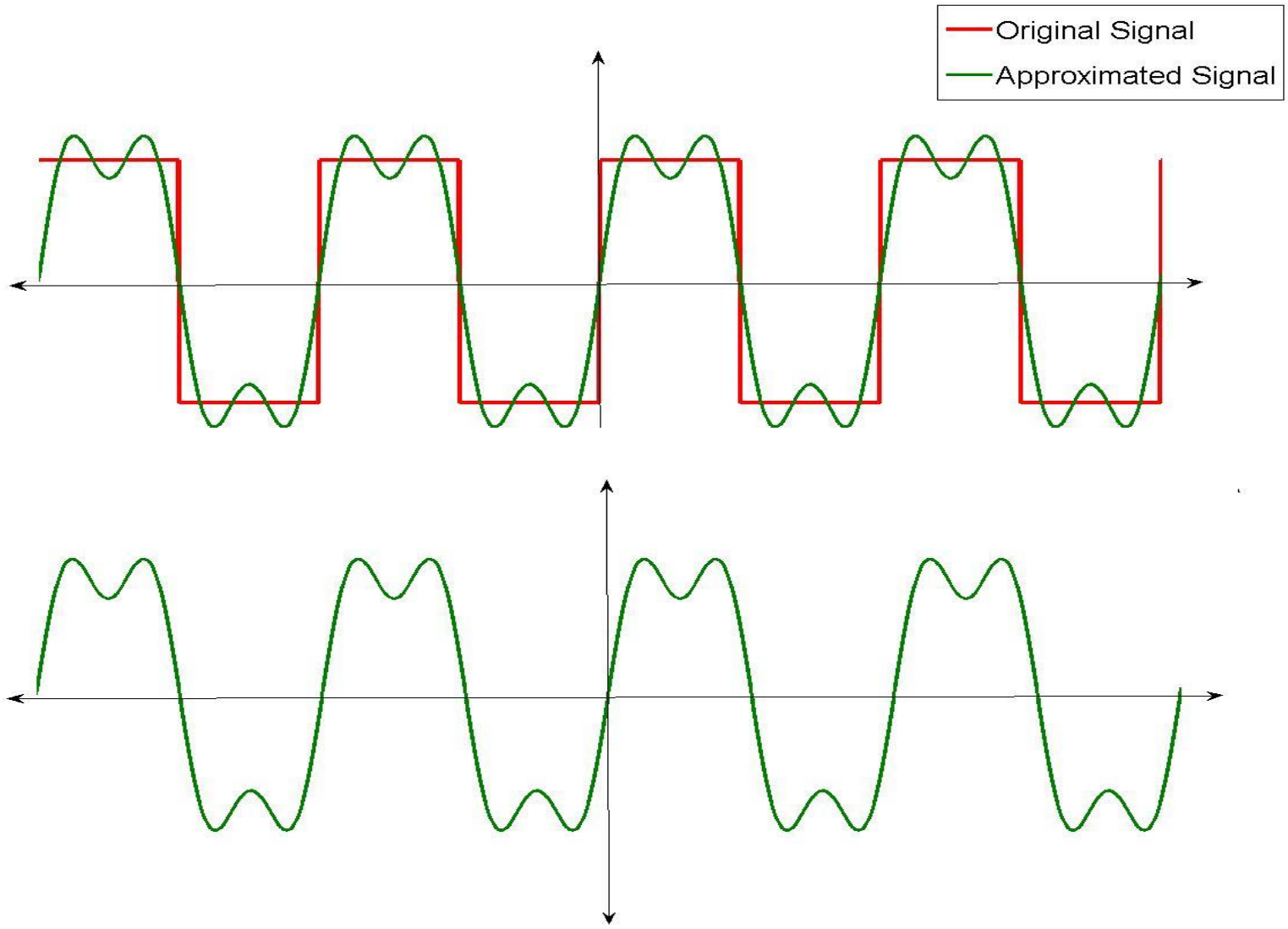
# Approximating signal with one orthogonal signals:



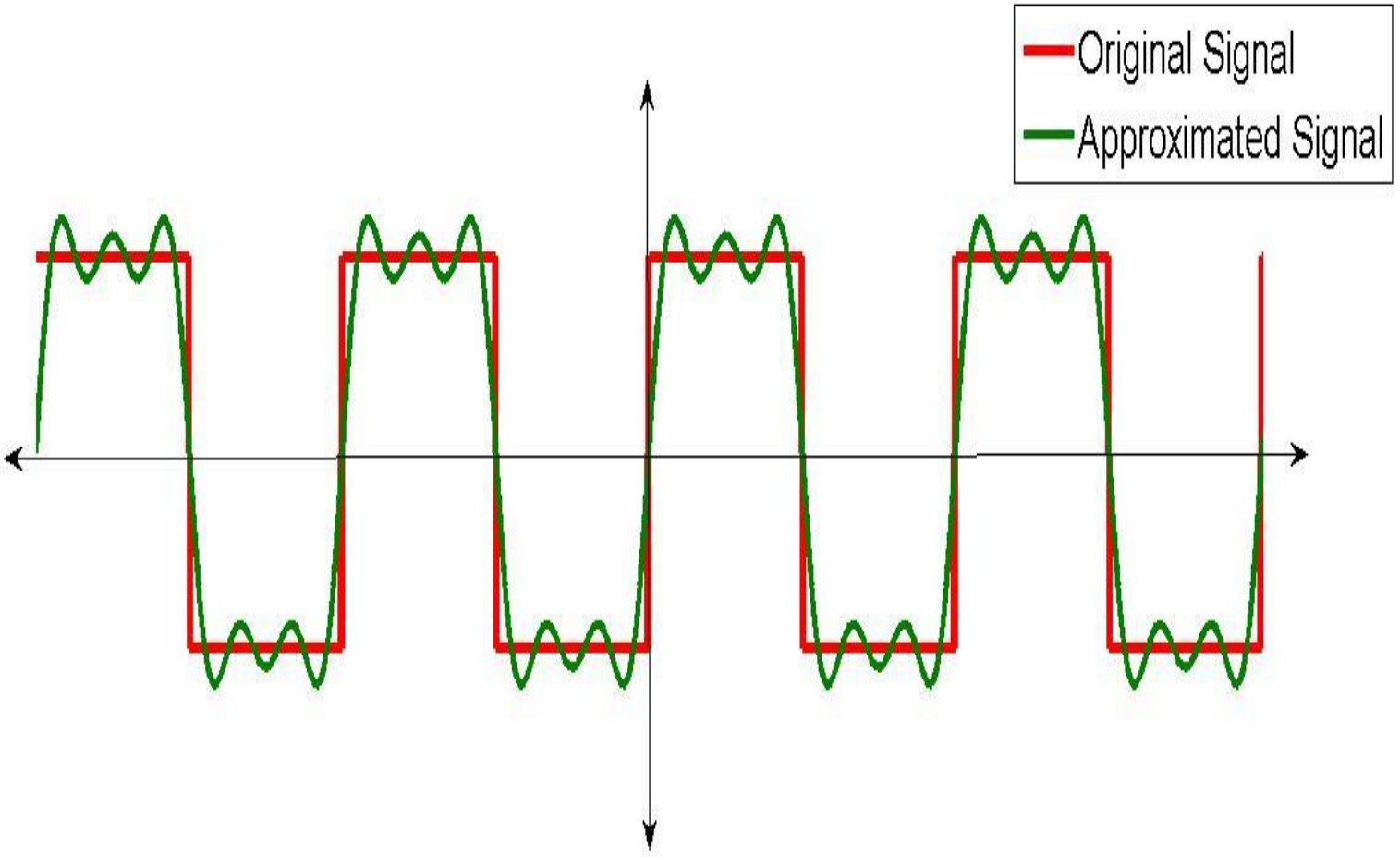
# Approximating signal with two orthogonal signals:



# Resulting:



# Approximating signal with three orthogonal signals:



# Objective:

- After completing this session you will be able to:
  1. Represent the periodic time domain signals in Fourier Series Representation.
  2. Understand the meaning of “the harmonics of the signal”.
  3. Plot the spectrum (frequency domain) of arbitrary signal.
  4. Know the three forms of Fourier Series and the meaning of them.

# Fourier Series Forms:

1. The Exponential Fourier Series (EFS).
2. The Trigonometric Fourier Series (TFS).
3. The Compact Fourier Series (CFS).

# 1)The Exponential Fourier Series:

- The basis set (Orthogonal) is:

$$e^{jk\omega_o t}$$

where:

$$\omega_o = \frac{2\pi}{T_o} = 2\pi f_o \quad represents the\ fundamentd\ frequency$$

$$\&\ k = 0, \pm 1, \pm 2, \pm 3, .. \ represents the\ k^{th}\ harmonic$$

# 1)The Exponential Fourier Series:

- So, the periodic signal x(t) can be represented as follow:

$$x(t) = \sum_{-\infty}^{\infty} C_k . e^{jk\omega_o t}$$

- Where $C_k$ is the $k^{th}$ EFS Coefficients.
- $C_k$ and the fundamental frequency ($\omega_o$) will be different from signal to another.

# 1)The Exponential Fourier Series:

- $\omega_o$ can be determined from the time domain signal.
- So, All what we need to find the EFS of an arbitrary signal is to calculate Ck as follow:

$$C_k = \frac{1}{T_o} \int_{(T)} x(t).e^{-jk\omega_o t}.dt$$

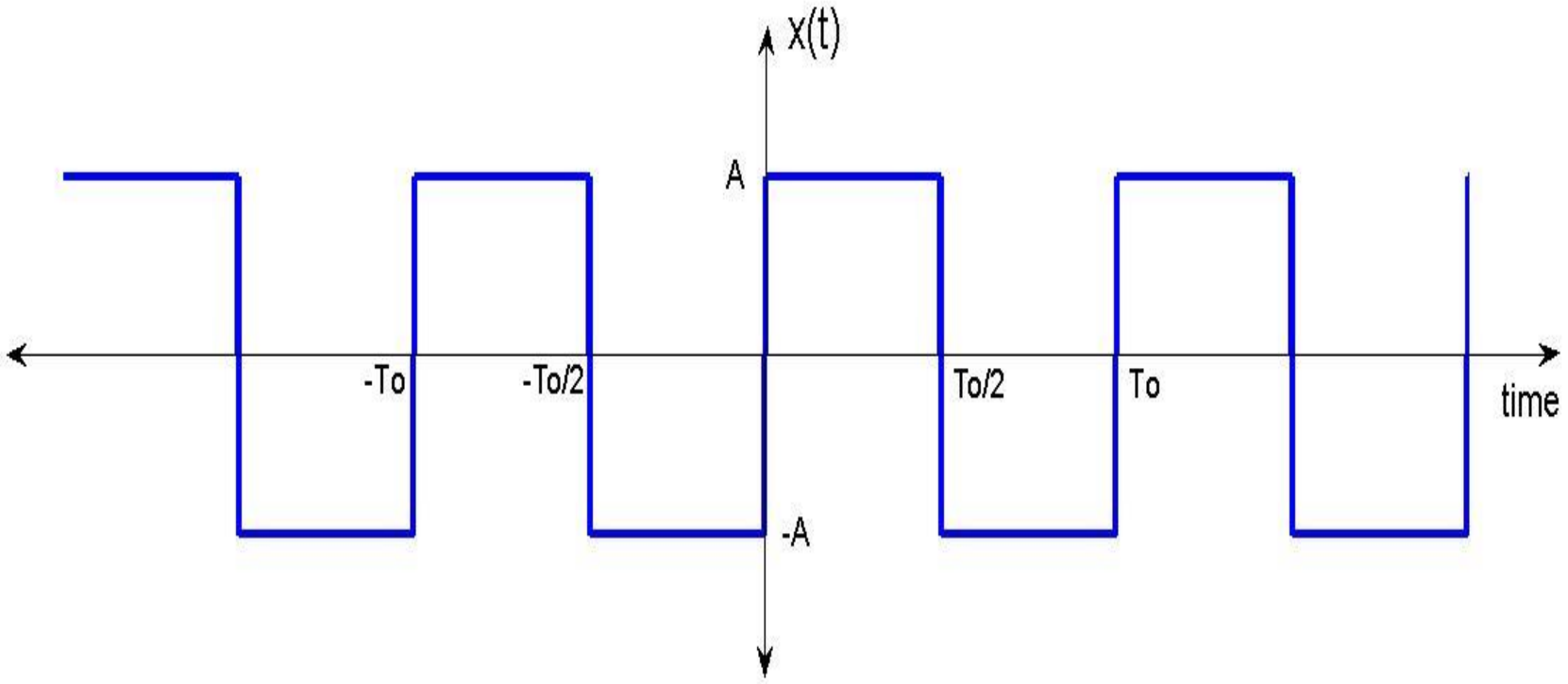
# 1)The Exponential Fourier Series:

- So, $C_o$ can be found as follow:

$$C_o = \frac{1}{T_o} \int_{(T)} x(t).dt$$

# Example 1:

- Express x(t) shown below using EFS:



# Example 1:

- We will find that $C_k$ is:

$$C_k = \begin{cases} \dfrac{-j4A}{2\pi k} & k \text{ is odd} \\ 0 & k \text{ is even} \end{cases}$$



# Example 1:

- So that the expansion of x(t) is:

$$x(t) = \sum_{\substack{k=-\infty \\ k\,is\,odd \\ k\neq 0}}^{\infty} \frac{-j2A}{k\pi}.e^{jk\omega_o t}$$

# Drawing of the Spectrum

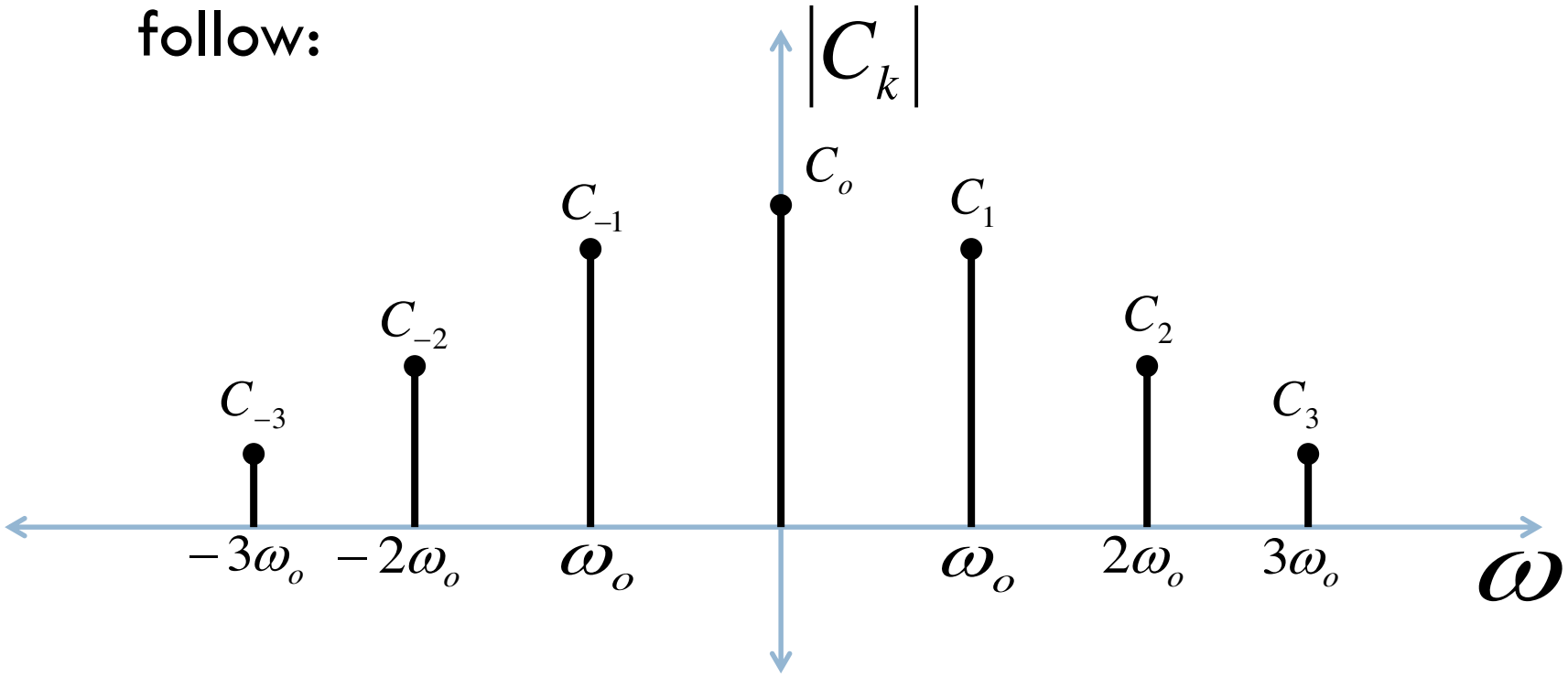
- Spectrum means drawing the frequency domain of the signal (magnitude Vs frequency).
- To do so, you need to expand the signal using the Fourier series as we did in the previous example.
- The following notes are helpfully to find the spectrum of a signal:

# Drawing of the Spectrum

A. $C_k$ is found for harmonic numbers $k=0,\pm1,\pm2,\ldots$ such as:
   1) $k=0$, $C_o$ represent the zero harmonic of the signal (DC content).
   2) $k=1$, $C_1$ represent the first harmonic of the signal (fundamental frequency).
   3) $k=2$, $C_2$ represent the second harmonic (which is double of the fundamental frequency).
   4) $k=n$, $C_n$ represent the $n^{th}$ harmonic of the signal.

# Drawing of the Spectrum

B. Now draw the magnitude of $C_k$ harmonics as discrete values verses frequency (x-axis) at $k\omega_o$ as follow:



# Drawing of the Spectrum

C. The lines at $k\omega_o$ are called the spectra lines.

D. The $-k\omega_o$ represent the negative frequency components. In reality there is no negative frequency. However; this theoretical representation due to the fact that for each $C_k$ there is $C_{-k}$ associated with the basis function ( $e^{jk\omega_o t}$ & $e^{-jk\omega_o t}$ )

# Drawing of the Spectrum

E. $C_k$ is complex and can be expressed as:

$$C_k = C_{kR} + j\,C_{kI}$$

and can be converted to polar or complex form as:

$$C_k = |C_k| \angle \theta_k = |C_k| e^{j\theta_k}$$

# Drawing of the Spectrum

F. EFS Spectrum are two graph plots; the magnitude of $C_k$, and the phase $\Theta_k$ at $k\omega_o$.
G. Usually; drawing of $C_k$ as an even function and $\Theta_k$ as odd function.
H. The harmonics of the signal called sometimes as the “frequency components of the signal”

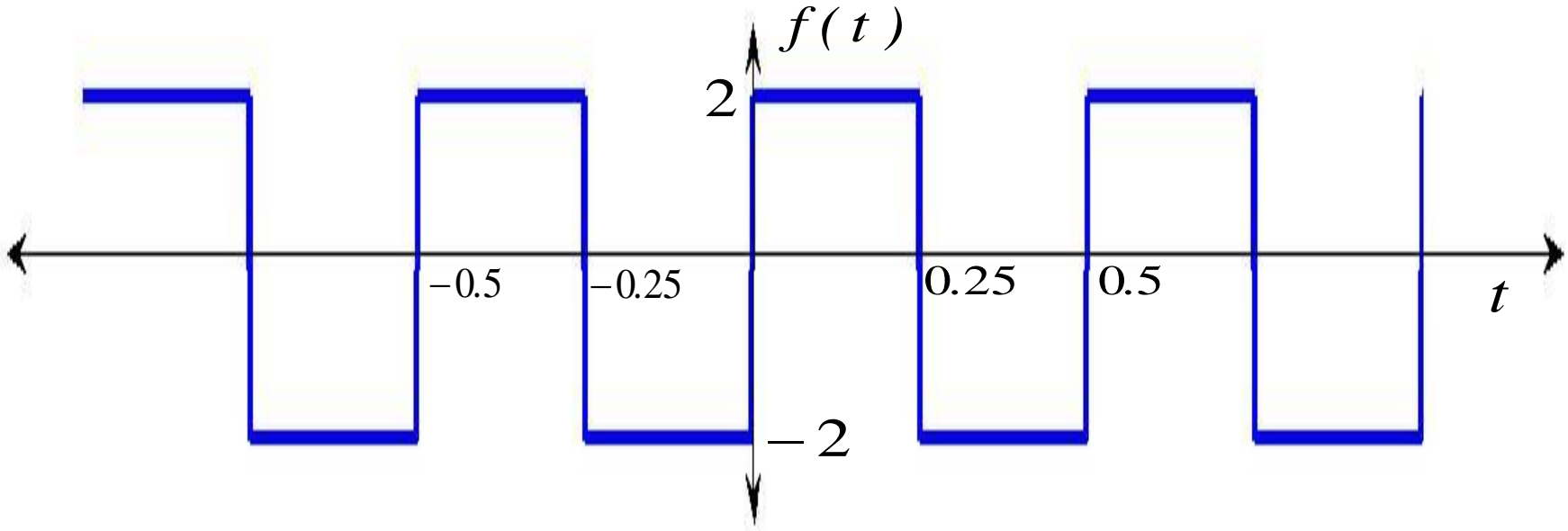
# Example 2:

- Draw the spectrum for the following expanded signal:

$$f(t) = 3.e^{j\frac{\pi}{3}}.e^{-j7\omega_o t} - 5.e^{-j3\omega_o t} + 2 + 5.e^{j3\omega_o t} - 3.e^{j\frac{\pi}{3}}e^{j7\omega_o t}$$

# Exercise 1:

- Find the EFS for the following signal.



# 2) The Trigonometric FS (TFS):

- If we consider the case where x(t) is real valued signal then:

$$x(t) = \sum_{-\infty}^{\infty} C_k . e^{jk\omega_o t}$$

and

$$C_k = \frac{1}{T_o} \int_{(T)} x(t) . e^{-jk\omega_o t} . dt$$

$$C_k^* = \frac{1}{T_o} \int_{(T)} x(t) . e^{jk\omega_o t} . dt = -C_k$$

# 2) The Trigonometric FS (TFS):

- That means:

$$|C_{-k}| = |C_k| \quad \ldots. \quad even\ symmetry$$

$$\& \ \angle C_k = -\angle C_{-k} \quad \ldots. \quad odd\ symmetry$$

- Now let's do some mathematical manipulation of EFS form, we can express x(t) as sum of three terms; negative, positive, & zero terms as follow:

# 2) The Trigonometric FS (TFS):

$$x(t) = \sum_{-\infty}^{-1} C_k . e^{jk\omega_o t} + C_o + \sum_{k=1}^{\infty} C_k . e^{jk\omega_o t}$$

$$x(t) = \sum_{k=1}^{\infty} C_{-k} . e^{-jk\omega_o t} + C_o + \sum_{k=1}^{\infty} C_k . e^{jk\omega_o t}$$

then

$$x(t) = C_o + \sum_{k=1}^{\infty} \left( C_{-k} . e^{-jk\omega_o t} + C_k . e^{jk\omega_o t} \right)$$

# 2) The Trigonometric FS (TFS):

- Now split both of the coefficients $C_k$ and $C_{-k}$ to two parts; real & imaginary, as well as the exponential:

$$C_k = C_{kR} + jC_{kI}$$

and

$$C_{-k} = C_{kR} - jC_{kI}$$

$$e^{jk\omega_o t} = cos(k\omega_o t) + j\,sin(k\omega_o t)$$

$$e^{-jk\omega_o t} = cos(k\omega_o t) - j\,sin(k\omega_o t)$$

# 2) The Trigonometric FS (TFS):

$$x(t) = C_o + \sum_{k=1}^{\infty} \left[ \begin{array}{l} (C_{kR} - jC_{kI})(\cos(k\omega_o t) - j\sin(k\omega_o t)) \\ \quad + (C_{kR} + jC_{kI})(\cos(k\omega_o t) + j\sin(k\omega_o t)) \end{array} \right]$$

$$= C_o + \sum_{k=1}^{\infty} \left[ \begin{array}{l} (C_{kR}\cos(k\omega_o t) - jC_{kR}\sin(k\omega_o t)) \\ \quad * (jC_{kI}\cos(k\omega_o t) + j^2 C_{kI}\sin(k\omega_o t)) \\ + (C_{kR}\cos(k\omega_o t) + jC_{kR}\sin(k\omega_o t)) \\ \quad * (jC_{kI}\cos(k\omega_o t) - j^2 C_{kI}\sin(k\omega_o t)) \end{array} \right]$$

# 2) The Trigonometric FS (TFS):

- finally;

$$x(t) = C_o + \sum_{k=1}^{\infty} \left[ \left( 2C_{kR} \cos(k\omega_o t) - 2C_{kI} \sin(k\omega_o t) \right) \right]$$

- Now rename the coefficients as follow:

$$a_o = C_o$$

$$a_k = 2C_{kR}$$

$$b_k = -2C_{kI}$$

# 2) The Trigonometric FS (TFS):

- We get the **TFS** formula; which is meaningful expression:

$$x(t) = a_o + \sum_{k=1}^{\infty} \left( a_k . cos(k\omega_o t) \right) + \sum_{k=1}^{\infty} \left( b_k \, sin(k\omega_o t) \right)$$

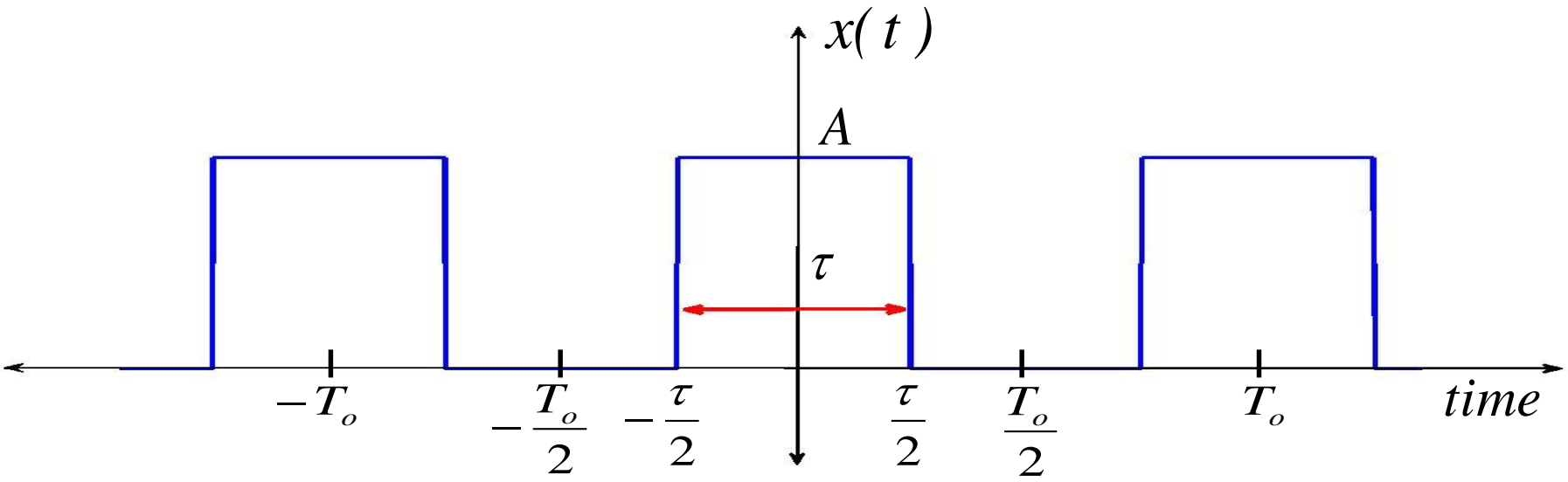
$$\therefore \quad a_o = C_o = \frac{1}{T_o} \int_{(T)} x(t).dt \qquad (DC\ content)$$

$$a_k = 2C_{kR} = \frac{2}{T_o} \int_{(T)} x(t).cos(k\omega_o t).dt$$

$$b_k = -2C_{kI} = \frac{2}{T_o} \int_{(T)} x(t).sin(k\omega_o t).dt$$

# Example 3:

- Find the TFS for the signal shown below:



# Example 3:

- The TFS coefficient are:

$$a_o = \frac{A\tau}{T_o}$$

$$b_o = 0 \qquad ??$$

$$a_k = \frac{2A}{kT_o\omega_o} sin\left(\frac{k\omega_o\tau}{2}\right) = \frac{2A\tau}{T_o}\,\mathrm{sinc}\left(\frac{k\tau}{T_o}\right) \qquad ??$$

# Example 3:

- The TFS will be:

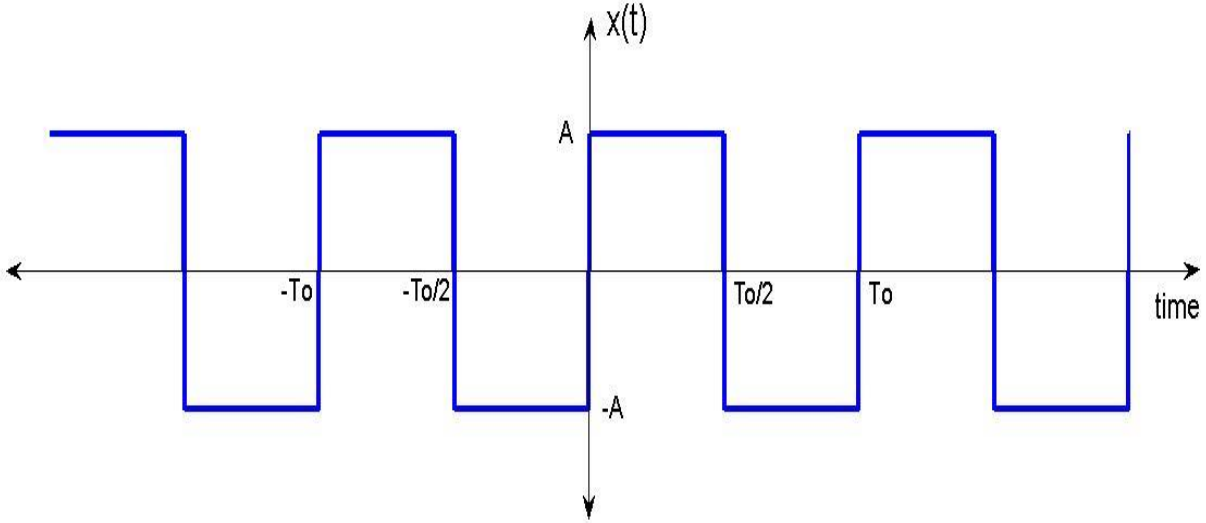
$$x(t) = \frac{A\tau}{T_o} + \sum_{k=1}^{\infty} \frac{2A}{kT_o\omega_o} sin\left(\frac{k\omega_o\tau}{2}\right).cos(k\omega_o t)$$

- Note: Drawing the expanded x(t) in the above equation with very large number k (terms), you will get very close signal to the original signal.

# Example 4:

- The TFS for x(t) shown below is:

$$x(t) = \sum_{\substack{k=1 \\ k\,is\,odd}}^{\infty} \frac{4A}{k\pi}.sin(k\omega_o t)$$






# Example 4:

- You can use the following MATLAB code to draw an approximation of x(t) in time domain with different number of terms, Amplitude, and frequency.
- The code will shown two images; the first one shows the Fourier series signal (Approximated signal) with finite number of terms determined by the user (you), the second shows the exact signal with the approximated to be compared.
- Create a new M-file the type the following commands:

# Example 4:

```
clear all
clc;
A=input('Enter the peak value A= ');
f=input('Enter the frequecy of the signal in Hz f= ');
w=2*pi*f;                    % compute the angular rate
k=input('Enter the number of terms(harmonics) to draw k= ');
x=0;                         % define initial value for x
T=1/f;                       % compute the period
t=-2*T:0.001:2*T;            % draw four cycles of the signal x
for n=1:2:k
    x=x+4*A*sin(n*w*t)/(n*pi); % compute the approximated signal
end
plot(t,x)                    % plot the approximated signal
axis([-2*T 2*T -A-1 A+1]); % limit the x&y axis
grid on;
figure(2);
x1=A*square(w*t);
plot(t,x1,'b',t,x,'r')  % draw the exact and approximated signal
axis([-2*T 2*T -A-1 A+1]);
grid on;
legend('x(t)', 'approximated x(t)');
```

# Properties of TFS:

1. $a_k$ & $b_k$ are real numbers and are found for positive index values k=1,2,3,…
2. Ploting of $a_k$ and $b_k$ against $k\omega_o$ is a one sided spectrum.
3. There is no plot for the phase angle spectrum. But there are two plots; one for $a_k$ and the other for $b_k$.



# Properties of TFS:

4. $a_o$ will be plotted with the $a_k$ spectrum.
5. The following relationships are applied for transformation from one form of FS to another:

$$a_o = C_o$$

$$a_k = 2C_{kR} = 2|C_k| cos\,\theta_k$$

$$b_k = -2C_{kI} = -2|C_k| sin\,\theta_k$$

# 3) Combined Fourier Series (CFS):

- The phase in the TFS is not obvious as the EFS, but TFS is more meaningful.
- A third form of FS is more compact and meaningful than the two previous forms called "Combined Fourier Series-CFS". Which is as follow:

$$x(t) = C_o + \sum_{k=1}^{\infty} d_k \cos(k\omega_o t + \phi_k)$$

# 3) Combined Fourier Series:

- Where:

$$d_k = \sqrt{a_k^2 + b_k^2} = 2|C_k|$$

$$\phi_k = \theta_k = tan^{-1}\left(\frac{b_k}{a_k}\right)$$

- Note that the spectrum draw is single sided spectrum an has two draws ($d_k$ magnitude and the phase $\phi_k$).

# Example 5:

- Draw the EFS, TFS & CFS Spectrum for:

$$f(t) = \sin(2\pi t) + 4\cos(6\pi t) + \sin(6\pi t)$$
$$+ 5\sin(10\pi t) + \cos(14\pi t) + 5$$

# Example 6:

- A half wave rectifier system with $1V_p$ & 50Hz sine wave input signal.

A. Draw the TFS spectrum for the input and output signal.

B. Use the MATLAB to draw the approximated time domain signal of the output using the first three and five terms of the TFS.

# Example 7:

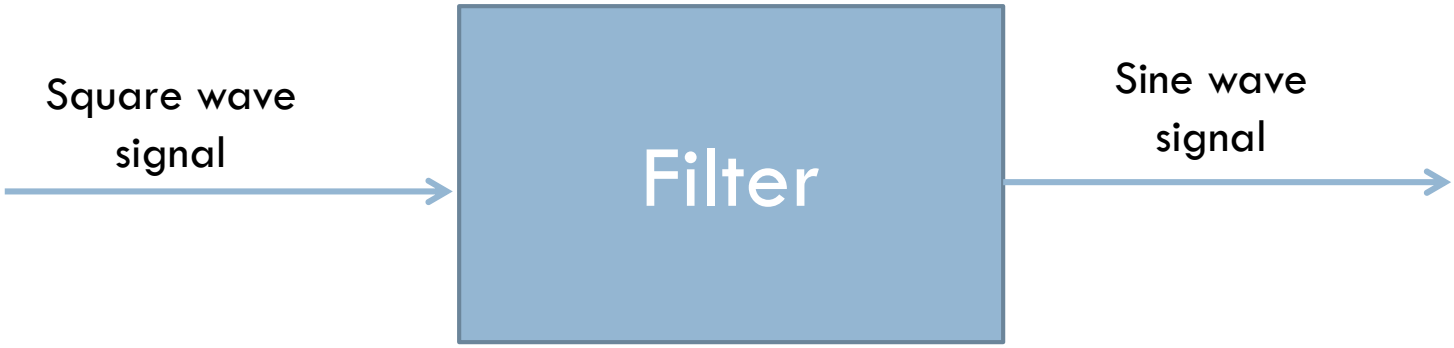
- Find the EFS for:

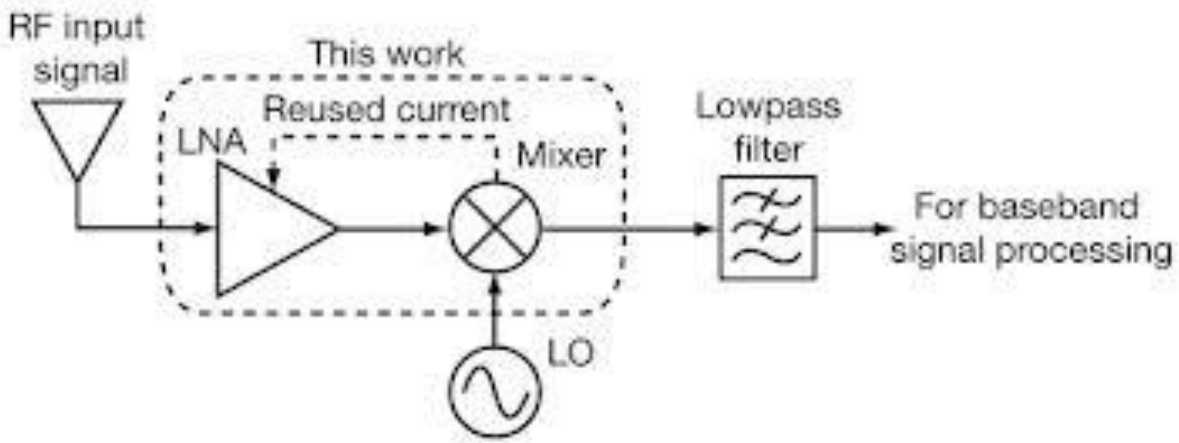
$$x(t) = \sum_{-\infty}^{\infty} \delta(t - nT_o) \qquad where\ n = 0, \pm 1, \pm 2, \ldots$$

# Filters (Electrical Filters):

- Filters in the Electrical Engineering is a frequency selective device.
- It is one of the most basic operation in any signal processing in the telecommunication world.
- Filters is the process by which the relative amplitude of frequency components of a signal are changed or perhaps some frequency components are suppressed (Attenuated).

# Filters (Electrical Filters):



# Exercise 2:

- If the system in “Example 6” become a full wave rectifier:

A. Draw the TFS spectrum of the output signal.

B. Use the MATLAB to draw the approximated time domain signal of the output using the first three terms and five of the TFS.

# Exercise 3:

A. For the signal x(t) shown in "Example 3" Find the TFS spectrum if:

$$1)\quad T_o = 2 \;\;\&\;\; \tau = 1.$$

$$2)\quad T_o = 2 \;\;\&\;\; \tau = 0.5.$$

$$3)\quad T_o = 2 \;\;\&\;\; \tau = 0.2.$$

B. Use the MATLAB to draw the approximated time domain signal using the first three terms and five for each expanded signal x(t) in "A".

# Exercise 4:

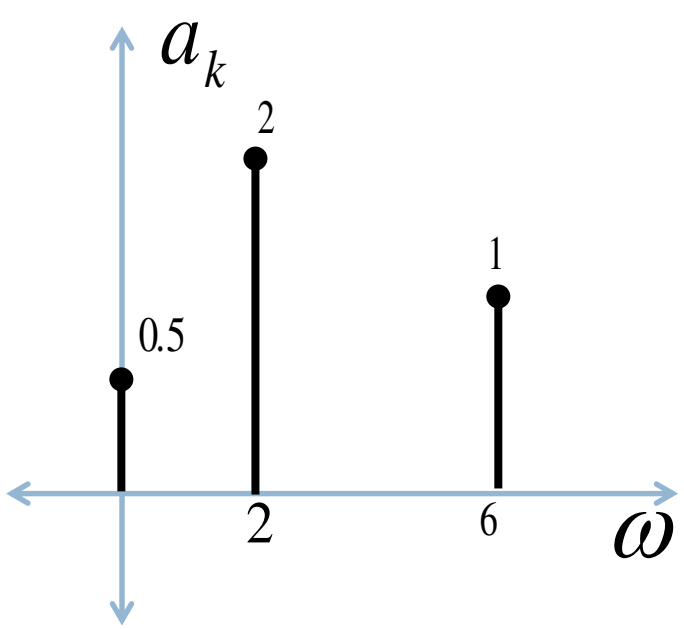
- Consider the following signal:

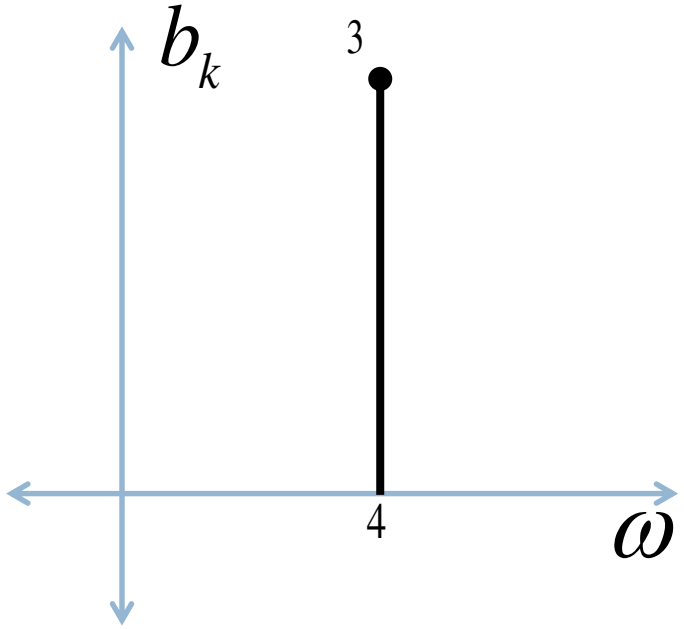
$$x(t) = 1 + 2\sin\left(\frac{7\pi}{3}t\right) - 3\cos\left(\frac{7\pi}{3}t\right) + \sin(7\pi t) + \cos\left(\frac{28\pi}{3}t\right)$$

Find the EFS & CFS, Draw the magnitude and phase spectrum for each of them.

# Exercise 5:

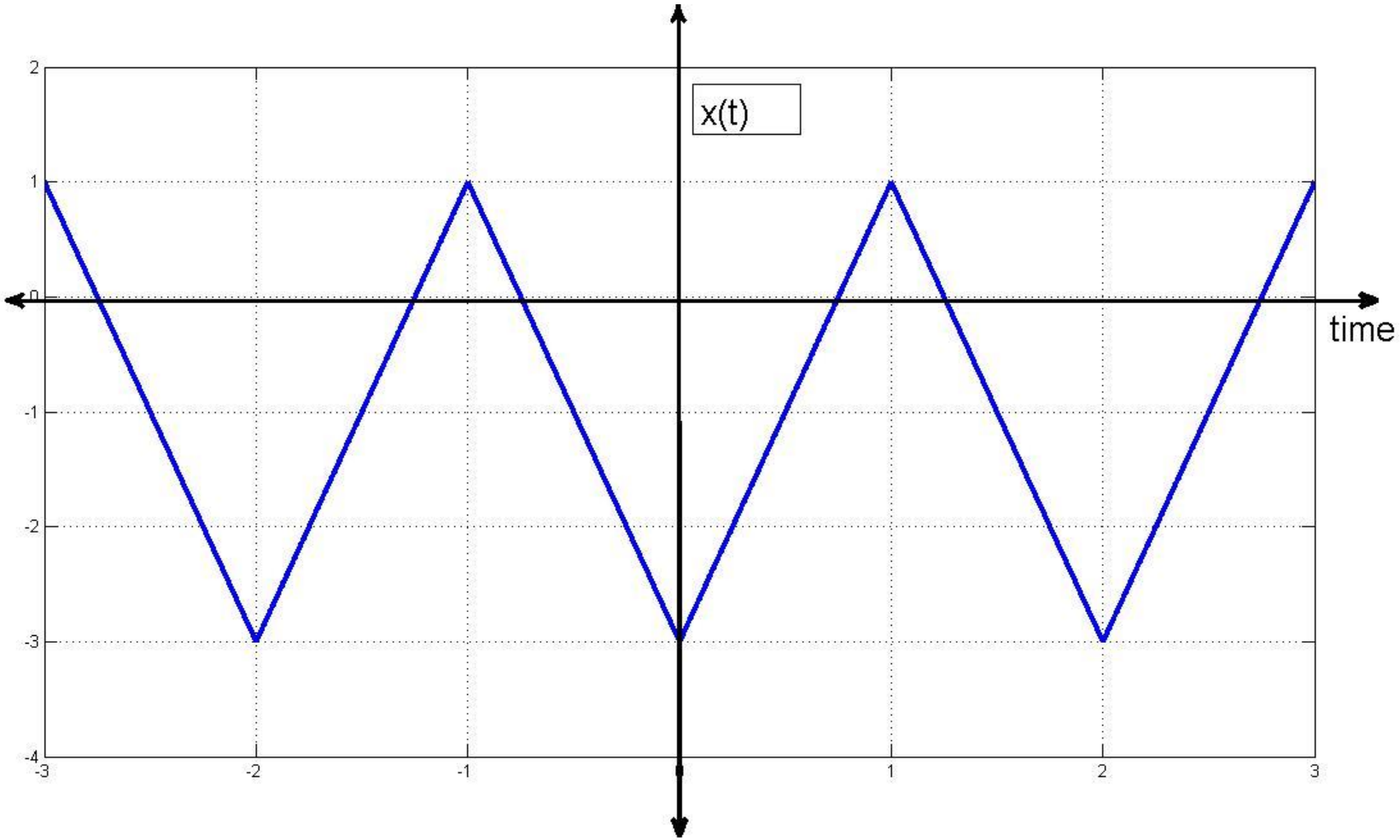
- Write down the expression of x(t) knowing that the spectrum shown below describe x(t) in frequency domain:



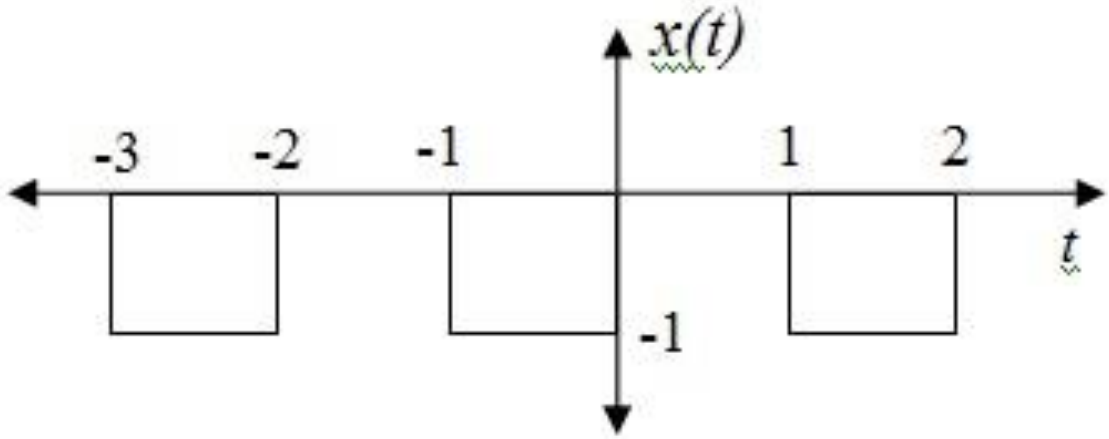
# Exercise 6:

- Find the TFS for the f(t) shown below:



# Exercise 7:

- Find the Exponential Fourier Series (E.F.S) of the Signal $x(t)$ shown below. Draw the magnitude & phase Spectrum of the first four harmonics ($C_{-4}$ to $C_4$).

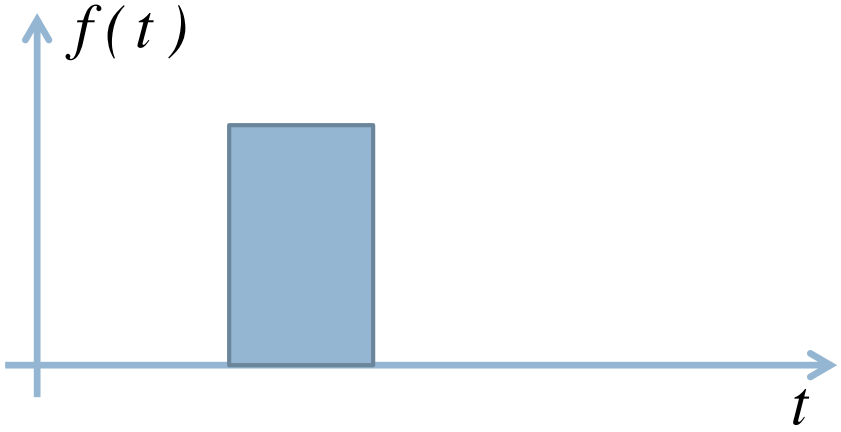


# CHAPTER 4:
# THE FOURIER TRANSFORM

## 4.2: Continuous- Time Fourier Transform

# Introduction:

- The Fourier series (F.T) can be extended to non-periodic waveforms such as a single pulse or a single transient by making $T \rightarrow \infty$.



# Introduction:

- F.T is a mathematical transformation employed to transform signals between time domain and frequency domain, which has many applications in physics and engineering.
- It is reversible, being able to transform from either domain to the other. The term itself refers to both the transform operation and to the function it produces.

# Objectives:

- After completing this session, you will be able to:
  1. Understand the Fourier Transform relationship.
  2. Understand Inverse Fourier Transform relationship
  3. Find the Fourier transform using the integration for some elementary and common signals.
  4. Drawing Non-periodic signals in frequency domain

# Continuous-Time Fourier Transform (F.T):

- The F.T notation is:

$$x(t) \Leftrightarrow X(\omega)$$

$$or \qquad f(t) \Leftrightarrow F(\omega)$$

# Continuous-Time Fourier Transform (F.T):

- The most-common pairs of $f(t)$ and $F(\omega)$ are often given in tables for easy reference.
- The Fourier transform has the useful property that many relationships and operations over the original $f(t)$ correspond to simpler relationships and operations over its image $F(\omega)$.

# Continuous-Time Fourier Transform (F.T):

- To transform f(t) from time to frequency domain:

$$F(\omega) = \int_{-\infty}^{\infty} f(t).e^{-j\omega t}.dt$$

- The Inverse F.T is given by:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega).e^{j\omega t}.d\omega$$

# Example 1:

- The F.T of the signal

$$f(t) = e^{-\alpha t}.u(t) \qquad \alpha > 0$$

Is:

$$F(\omega) = \frac{1}{\alpha + j\omega}$$



# Example 2:

- If:

$$x(t) = e^{-\alpha|t|} \qquad \alpha > 0$$

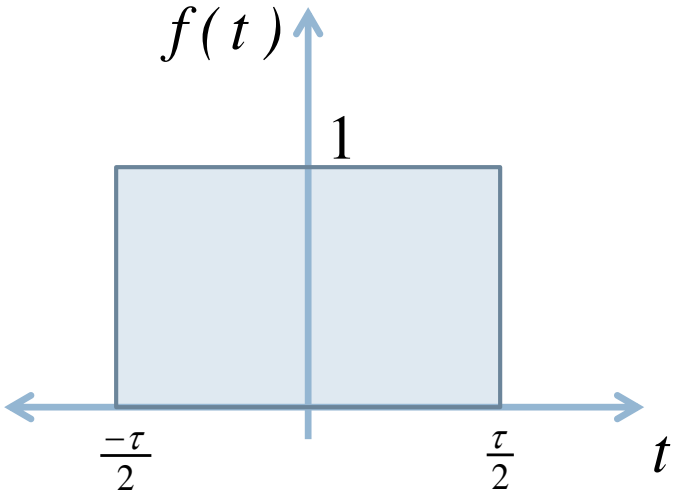
then:

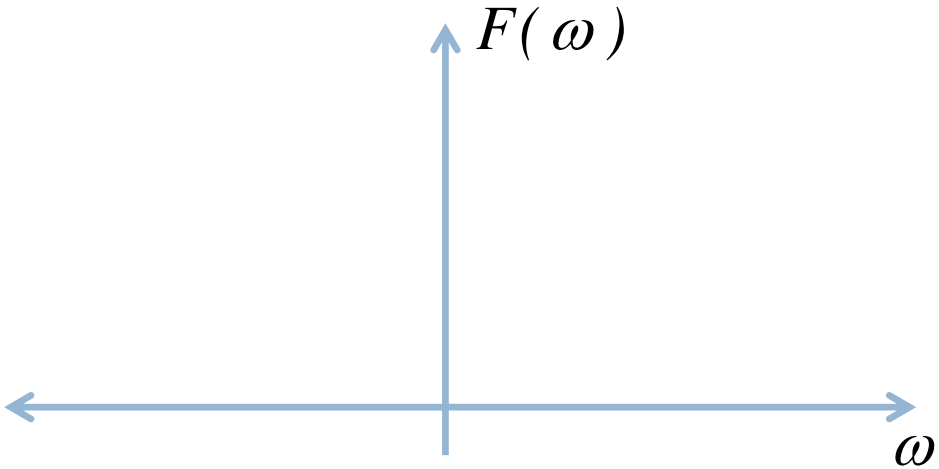
$$X(\omega) = \frac{1}{\alpha - j\omega} + \frac{1}{\alpha + j\omega}$$

# Example 3:

- The F.T for the signal f(t) shown is:

$$F(\omega) = \tau \,\mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right) = \frac{2\sin(\omega\tau/2)}{\omega}$$

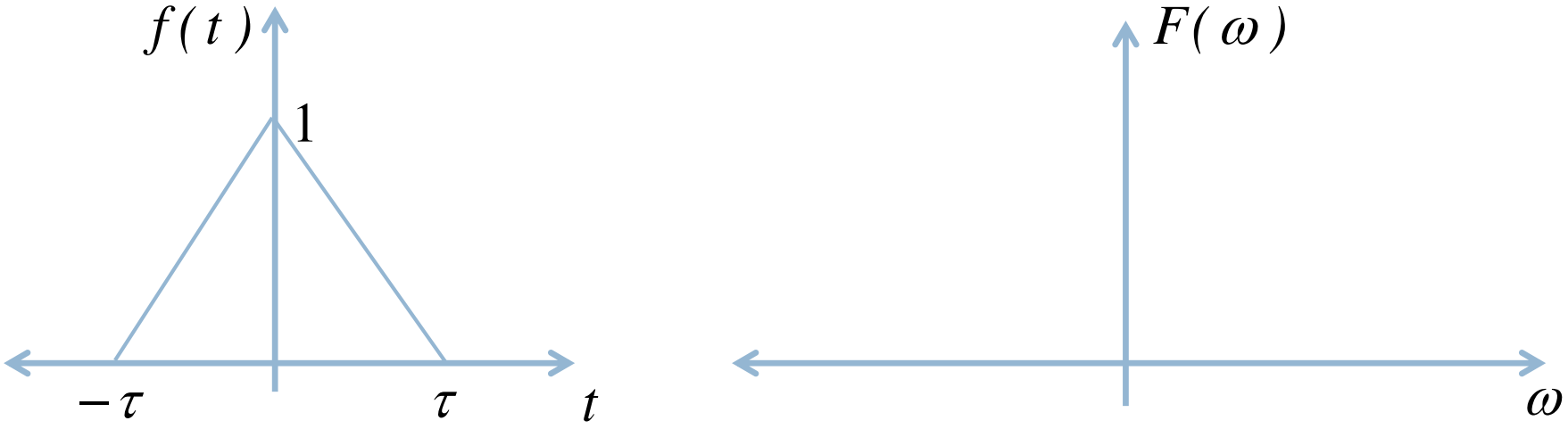


# Example 4:

- The F.T for the signal f(t) shown is:

$$F(\omega) = \tau \,\mathrm{sinc}^2\left(\frac{\omega\tau}{2}\right)$$



# Example 5:

- If:

$$x(t) = \delta(t)$$

then:

$$X(\omega) = 1$$

$$\Rightarrow$$

$$\delta(t) \leftrightarrow 1$$

# Example 6:

- If:

$$x(t) = 1$$

then:

$$X(\omega) = 2\pi\delta(\omega)$$

$$\Rightarrow$$

$$1 \leftrightarrow 2\pi\delta(\omega)$$

# Example 7:

- If:

$$x(t) = e^{j\omega_o t}$$

from example 6 we can conclude that:

$$X(\omega) = 2\pi\delta(\omega - \omega_o)$$

$$\Rightarrow$$

$$e^{j\omega_o t} \leftrightarrow 2\pi\delta(\omega - \omega_o)$$

# Example 8:

- The Fourier Transform for :

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_o)$$

Is:

$$X(\omega) = 2\pi \sum_{n=-\infty}^{\infty} C_n . \delta(\omega - n\omega_o)$$

$$= \frac{2\pi}{T_o} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_o)$$

# CHAPTER 4
# THE FOURIER TRANSFORM PROPERTIES

4.3: Properties of Fourier Transform

# 1) Linearity:

- If

$$x_1(t) \Leftrightarrow X_1(\omega)$$

$$x_2(t) \Leftrightarrow X_2(\omega)$$

then

$$a\,x_1(t) + b\,x_2(t) \Leftrightarrow a\,X_1(\omega) + b\,X_2(\omega)$$

# Example 1:

- The F.T for:

$$f(t) = \cos \omega_o t$$




Is:

$$F(\omega) = \pi[\delta(\omega - \omega_o) + \delta(\omega + \omega_o)]$$

# Exercise 1:

- Find the F.T for:

$$f(t) = \sin \omega_o t$$

# 2) Symmetry

- If x(t) is a real-valued time signal, then:

$$X(-\omega) = X^{*}(\omega)$$

- So, we can conclude:

$$|X(\omega)| = |X(-\omega)| \quad \textit{Magnitude plot is even}$$

$$\phi(\omega) = -\phi(-\omega) \quad \textit{phase Plot is odd}$$

# 3) Time Shifting:

- If:

$$x(t) \Leftrightarrow X(\omega)$$

then:

$$x(t-t_o) \Leftrightarrow e^{-j\omega t_o} X(\omega)$$

Similarly:

$$e^{j\omega_o t}.x(t) \Leftrightarrow X(\omega-\omega_o)$$

# 4) Time scaling:

- If:

$$x(t) \Leftrightarrow X(\omega)$$

- Then:

$$x(\alpha t) \Leftrightarrow \frac{1}{|\alpha|} X\left(\frac{\omega}{\alpha}\right)$$

# 5) Differentiation in the Time domain:

- If:

$$f(t) \Leftrightarrow F(w)$$

- Then:

$$\frac{df(t)}{dt} \Leftrightarrow j\omega F(\omega)$$

- In general:

$$\frac{d^n f(t)}{dt^n} \Leftrightarrow (j\omega)^n F(\omega)$$

# 6) Integration in Time domain:

- If:

$$f(t) \Leftrightarrow F(w)$$

- Then:

$$\int_0^t f(\tau).d\tau \Leftrightarrow \frac{1}{j\omega} F(\omega)$$

- In general:

$$\int_0^t f(\tau).d\tau \Leftrightarrow \pi F(0)\delta(\omega) + \frac{1}{j\omega} F(\omega)$$

# 7) Energy of Aperiodic Signals

- The energy In time domain is:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 .dt$$

- The energy in frequency domain (Parseval's Theorem) is equal to:

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |x(\omega)|^2 .d\omega$$

- In this case:

$$E_t = E_f$$

# 7) Energy of Aperiodic Signals

- In reality, for specific bandwidth we have:

$$\Delta E = \frac{1}{2\pi}\int_{\omega_1}^{\omega_2} |x(\omega)|^2 .d\omega$$

- Due to the similarity:

$$E = \frac{1}{\pi}\int_{0}^{\infty} |x(\omega)|^2 .d\omega$$

# 9) Convolution

- If

$$x(t) \Leftrightarrow X(\omega)$$

$$h(t) \Leftrightarrow H(\omega)$$

- Where $x(t)$ is the input signal to the LTI system $h(t)$.
- Then:

$$x(t) * h(t) \Leftrightarrow X(\omega).H(\omega)$$

# 10) Multiplication by *t* :

- If

$$f(t) \Leftrightarrow F(\omega)$$

- Then

$$(-jt)^n f(t) \Leftrightarrow \frac{d^n F(\omega)}{d\omega^n}$$

- or:

$$t^n f(t) \Leftrightarrow (-j)^n \frac{d^n F(\omega)}{d\omega^n}$$

# 11) Modulation

- If:

$$x(t) \Leftrightarrow X(\omega)$$

$$m(t) \Leftrightarrow M(\omega)$$

- Then:

$$x(t).m(t) \Leftrightarrow \frac{1}{2\pi}[X(\omega) * M(\omega)]$$

# Chapter 5

The Laplace Transform

# 5.1: Introduction:

- The **Laplace transform** is a widely used integral transform in mathematics with many applications in physics and engineering. It is a linear operator of a function $f(t)$ with a real argument $t$ ($t \geq 0$) that transforms $f(t)$ to a function $F(s)$ with complex argument $s$, given by the integral:
- Its called the UNILATERAL Laplace Transform.

$$F(s) = \int_0^{\infty} f(t).e^{-st}.dt$$

# Introduction

- The most-common pairs of f (t) and F (s) are often given in tables for easy reference. The Laplace transform has the useful property that many relationships and operations over the original f (t) correspond to simpler relationships and operations over its image F (s).
- The Laplace transform is related to the Fourier Transform, but whereas the Fourier transform expresses a function or signal as a series of modes of vibration (frequencies), the Laplace transform resolves a function into its moments.

# Introduction:

- Like the Fourier transform, the Laplace transform is used for solving differential, integral equations and initial value problems. In physics and engineering it is used for analysis of linear time-invariant systems such as Electrical circuits, harmonics oscillators, and mechanical systems.
- In such analyses, the Laplace transform is often interpreted as a transformation from the time-domain, in which inputs and outputs are functions of time, to the frequency-domain, where the same inputs and outputs are functions of complex angular frequency, in radians per unit time.
- Given a simple mathematical or functional description of an input or output to a system, the Laplace transform provides an alternative functional description that often simplifies the process of analyzing the behavior of the system, or in synthesizing a new system based on a set of specifications.

# 5.2:The Laplace Transform (L.T):

- The Laplace Transforms Notations are:

$$F(s)$$

$$\ell\{f(t)\}$$

- And it is given by:

$$F(s)=\int_{-\infty}^{\infty} f(t).e^{-st}.dt$$

- Its called the BILATERAL Laplace Transform, Results a function of ***s*** variable.

# The Laplace Transform (L.T):

- Where ***s*** is a complex frequency (measured by $sec^{-1}$:

$$s = \sigma + j\omega$$

$\sigma$ *is the real part*

& $\omega$ *is the imaginary part*

# The Inverse Laplace Transform (I.L.T):

- The Inverse Laplace transform has the notation $f(t) = \ell^{-1}[F(s)]$.
- And its given by:

$$f(t) = \frac{1}{2\pi j} \int_{\sigma - j\omega}^{\sigma + j\omega} X(s).e^{-st}.ds$$

- Usually we use the partial fraction technique to find the Inverse L.T as the L.T function always in the form:

$$F(s) = \frac{N(s)}{D(s)}$$

## Example

let's find Laplace transform of $f(t) = e^t$:

$$F(s) = \int_0^\infty e^t\, e^{-st}\, dt = \int_0^\infty e^{(1-s)t}\, dt = \frac{1}{1-s} e^{(1-s)t} \Big|_0^\infty = \frac{1}{s-1}$$

# Region of Convergence (ROC):

- Whether the Laplace transform X(s) of a signal x(t) exists or not depends on the complex variable s as well as the signal itself. All complex values of s for which the integral in the definition converges form a region of convergence (ROC) in the s-plane. X(s) exists if and only if the argument s is inside the ROC. As the imaginary part $\omega = Im[\ s\ ]$ of the complex variable $s = \sigma + j\omega$ has no effect in terms of the convergence, the ROC is determined solely by the real part $\sigma = Re[\ s\ ] = \Re s$

The ROC can also provide us with information about whether x(t) is Fourier transformable or not. Since the Fourier Transform is obtained from the Bilateral Laplace transform by setting $\sigma$=0, the ROC in this case is a single line (the j$\omega$ axis). Therefore, if the ROC for X(s) includes the j$\omega$ axis. x(t) is Fourier Transformable, and X($\omega$) can be obtained by replacing s in X(s) by j$\omega$.

.

# Example 1:

- The L.T of the signal:

$$x(t) = e^{-at}.u(t)$$

is:

$$X(s) = \frac{1}{s+a}$$

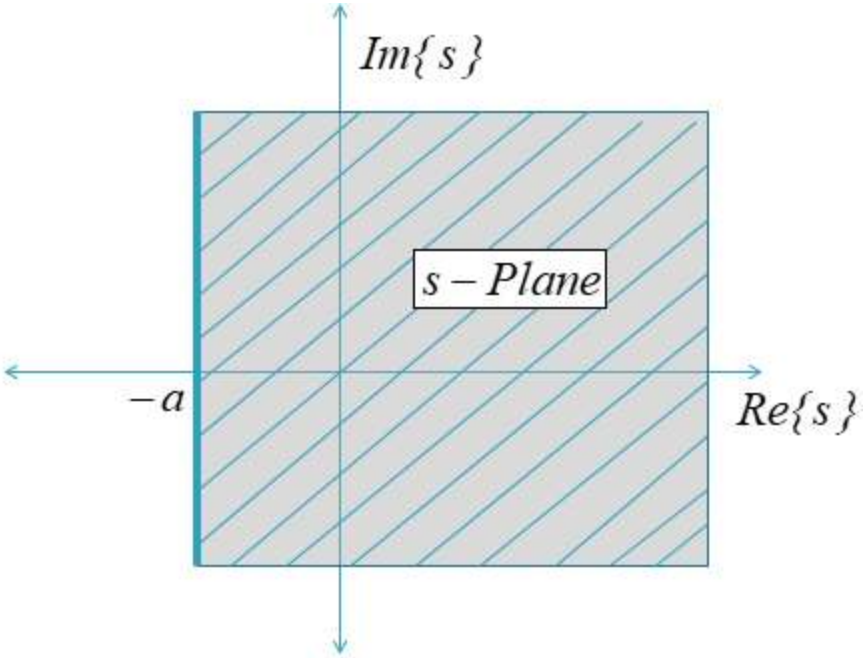


# Region of Convergence (ROC):

- For the signal in “Example 1” has a L.T only if $\sigma > -a$

So, we say that the ROC of x(t) in Example 1, is defined by

$$Re\{ s \} > -a$$



# Example 2:

- The L.T for

$$x(t) = -e^{-at}.u(-t)$$

is also:

$$X(s) = \frac{1}{s+a}$$



And the ROC is defined for

$$Re\{s\} < -a$$

# Example 3:

- The L.T for:

$$x(t) = 3.e^{-2t}.u(t) + 4.e^{t}.u(-t)$$

- Is:

$$X(s) = \frac{3}{s+2} - \frac{4}{s-1}$$

- And the ROC is:

$$-2 < Re\{s\} < 1$$



# Laplace Transform

5.4: Properties of L.T:

# 1) Linearity:

- If

$$x_1(t) \Leftrightarrow X_1(s)$$

$$x_2(t) \Leftrightarrow X_2(s)$$

then

$$a\,x_1(t) + b\,x_2(t) \Leftrightarrow a\,X_1(s) + b\,X_2(s)$$

# Example 1:

- The L.T for:

$$f(t) = (A + Be^{-bt}).u(t)$$

Is:

$$F(s) = \frac{A}{s} + \frac{B}{s+b}$$



# 2) Time Shifting:

- If:

$$x(t) \Leftrightarrow X(s)$$

then:

$$x(t - t_o) \Leftrightarrow e^{-t_o s} X(s)$$

# Example 2:

- Find:

$$\ell\{ u(t) - u(t-2a) \}$$

Sol:

$$\ell\{ u(t) - u(t-2a) \} = \frac{1}{s} - \frac{e^{-2as}}{s}$$

# 3) Shifting in s-Domain

- If:

$$x(t) \Leftrightarrow X(s)$$

- Then:

$$e^{as}.x(t) \Leftrightarrow X(s-a)$$

# Example 3:

- The L.T for:

$$x(t) = e^{-at}.u(t)$$

Is

$$X(s) = \frac{1}{s+a}$$

# 4) Time scaling:

- If:

$$x(t) \Leftrightarrow X(s)$$

- Then:

$$x(\alpha t) \Leftrightarrow \frac{1}{|\alpha|} X\left(\frac{s}{\alpha}\right)$$

# 5) Differentiation in the Time domain:

- If: $$f(t) \Leftrightarrow F(s)$$

- Then:

$$\frac{df(t)}{dt} \Leftrightarrow sF(s) - f(0)$$

- In general:

$$\frac{d^n f(t)}{dt^n} \Leftrightarrow s^n F(s) - s^{n-1} f(0) - \ldots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

# Example 4:

- If $f(t)=t^2.u(t)$, find the L.T for $f^{(2)}(t)$.

Sol:

$$\mathrm{l}\{f''(t)\}=s^2F(s)-f'(0)-s\,f(0)$$

$$=s^2F(s)=\frac{2}{s}$$

Where $F(s)=\frac{2}{s^3}$

# Note:

- One of the most important application of L.T is in solving the differential equations with specified initial conditions and analyzing the RLC circuits. Property “5” can help to do so as in the Examples we are going to see later on in this section.

# Some examples:

1) Let $f(t) = 1, t \geq 0$

$$\mathcal{L}\{1\} = F(s) = \int_0^\infty e^{-st}dt = \lim_{A\to\infty}\int_0^A e^{-st}dt = \lim_{A\to\infty}\frac{-e^{-st}}{s}\Big|_0^A = \lim_{A\to\infty}\left[\frac{1}{s} - \frac{e^{-As}}{s}\right] = \frac{1}{s}, \text{ for all } s > 0$$

---

2) Let $f(t) = t, t \geq 0$

$$\mathcal{L}\{t\} = F(s) = \int_0^\infty e^{-st}t\,dt = \lim_{A\to\infty}\int_0^A e^{-st}t\,dt = \lim_{A\to\infty}\frac{-e^{-st}}{s^2}(st+1)\Big|_0^A =$$

$$= \lim_{A\to\infty}\left[\frac{1}{s^2} - \frac{e^{-As}}{s^2}(sA+1)\right] = \frac{1}{s^2}, \text{ for all } s > 0$$

---

3) Let $f(t) = e^{at}$, $t \geq 0$

$$\mathcal{L}\{e^{at}\} = F(s) = \int_0^{\infty} e^{-st} e^{at} dt = \lim_{A\to\infty} \int_0^{A} e^{-(s-a)t} dt =$$

$$\lim_{A\to\infty} \frac{-e^{-(s-a)t}}{s-a}\Big|_0^A = \lim_{A\to\infty}\left[\frac{1}{s-a} - \frac{e^{-(s-a)A}}{s-a}\right] = \frac{1}{s-a}, \text{ for all } s > a.$$

4) Let $f(t) = \sin at$, $t \geq 0$

$$\mathcal{L}\{\sin at\} = F(s) = \int_0^{\infty} e^{-st} \sin at \, dt = \lim_{A\to\infty} \int_0^{A} e^{-st} \sin at \, dt =$$

$$\lim_{A\to\infty}\left[-\frac{e^{-st}\cos at}{a}\Big|_0^A - \frac{s}{a}\int_0^A e^{-st}\cos at \, dt\right] = \frac{1}{a} - \frac{s}{a}\int_0^{\infty} e^{-st}\cos at \, dt =$$

$$\frac{1}{a} - \lim_{A\to\infty}\left[-\frac{e^{-st}\sin at}{a}\Big|_0^A + \frac{s^2}{a^2}\int_0^A e^{-st}\sin at \, dt\right] = \frac{1}{a} - \frac{s^2}{a^2}F(s)$$

then,

$$F(s) = \frac{1}{a} - \frac{s^2}{a^2}F(s), \quad F(s)\left(\frac{a^2+s^2}{a^2}\right) = \frac{1}{a}, \quad F(s) = \frac{a}{s^2+a^2}, \; s \geq 0$$

# 6) Integration in Time domain:

- If:

$$f(t) \Leftrightarrow F(s)$$

- Then:

$$\ell\{\int_0^t f(t).dt\} \Leftrightarrow \frac{1}{s} F(s)$$

# 7) Multiplication by $t$ or differentiation in s Domain:

- If

$$f(t) \Leftrightarrow F(s)$$

- Then

$$-t\, f(t) \Leftrightarrow \frac{dF(s)}{ds}$$

- In general:

$$(-t)^n f(t) \Leftrightarrow \frac{d^n F(s)}{ds^n}$$

# 8) Divided by $t$ :

- If:

$$f(t) \Leftrightarrow F(s)$$

- Then:

$$\frac{f(t)}{t} \Leftrightarrow \int_{s}^{\infty} F(s).ds$$

# 9) Modulation

- If:

$$x(t) \Leftrightarrow X(s)$$

- Then:

$$x(t).cos(\omega_o t) \Leftrightarrow \frac{1}{2}[X(s+j\omega_o)+X(s-j\omega_o)]$$

$$x(t).sin(\omega_o t) \Leftrightarrow \frac{j}{2}[X(s+j\omega_o)+X(s-j\omega_o)]$$

# 10) Convolution

- If

$$x(t) \Leftrightarrow X(s)$$

$$h(t) \Leftrightarrow H(s)$$

- Where *x(t)* is the input signal to the LTI system *h(t)*.
- Then:

$$x(t) * h(t) \Leftrightarrow X(s).H(s)$$

# 11) Initial Value theorem:

$$x(0^+) = \lim_{s \to \infty} sX(s)$$

$$x^{(N)}(0^+) = \lim_{s \to \infty} s^{N+1} X(s)$$

# e.g:

- If

$$X(s) = \frac{cs + d}{(s - a)(s - b)} \qquad a \neq b$$

$$\Rightarrow X(0^+) = lim \frac{s(cs + d)}{(s - a)(s - b)} = c$$

# 12) Final Value Theorem

- The final value theorem allows us to compute the limit of signal x(t) as $t \to \infty$ from its Laplace transform as follow:

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} s\,X(s)$$

- Condition: the point s=0 must be in ROC of X(s).

# e.g:

- The input $x(t) = A.u(t)$ is applied to a automatic positioning-control system whose transfer function is:

$$H(s) = \frac{c}{s(s+b)+c}$$

- The final value of the output y(t) is obtained as:

$$Y(s) = \frac{cA}{s^2(s+b)+cs}$$

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} \frac{cA}{s(s+c)+c} = A$$

# Properties of Laplace Transform

Some of the Laplace transformation properties are:

If $f_1(t) \leftrightarrow F_1(s)$ and [note: $\leftrightarrow$ implies Laplace Transform]

$f_2(t) \leftrightarrow F_2(s)$, then

| | |
|---|---|
| Linearity Property | $A\, f_1(t) + B\, f_2(t) \leftrightarrow A\, F_1(s) + B\, F_2(s)$ |
| Frequency Shifting Property | $e^{s_0 t}\, f(t)) \leftrightarrow F(s - s_0)$ |
| Integration | $\int_0^t f(\lambda)\, d\lambda \leftrightarrow 1/s\, F(s)$ |
| Multiplication by Time | $T\, f(t) \leftrightarrow (-d\, F(s)/ds)$ |
| Complex Shift Property | $f(t)\, e^{-at} \leftrightarrow F(s + a)$ |
| Time Reversal Property | $f(-t) \leftrightarrow F(-s)$ |
| Time Scaling Property | $f(t/a) \leftrightarrow a\, F(as)$ |